Ministry Approved
2013

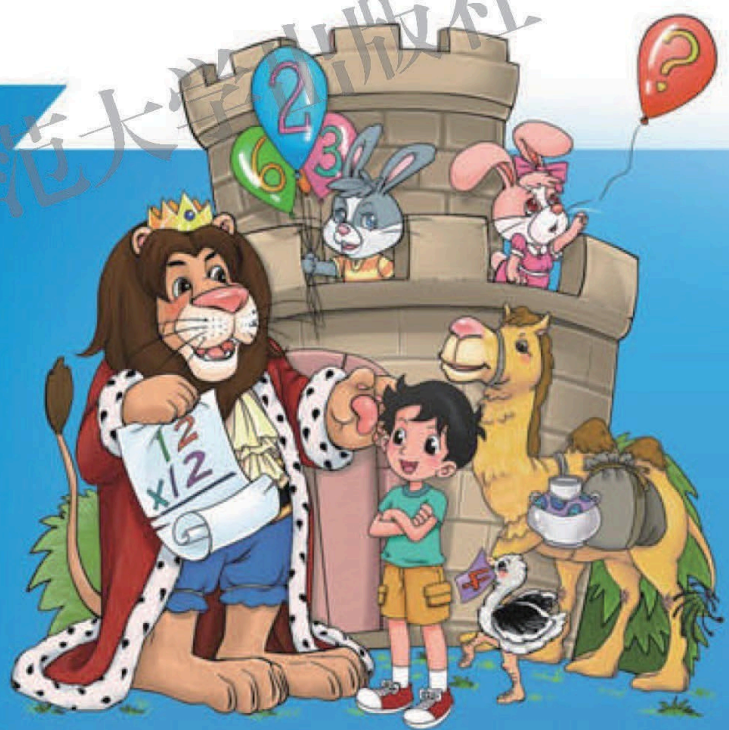
Compulsory Education Textbook

Mathematics

SHUXUE

Grade 4 Volume 1

北京师范大学出版社



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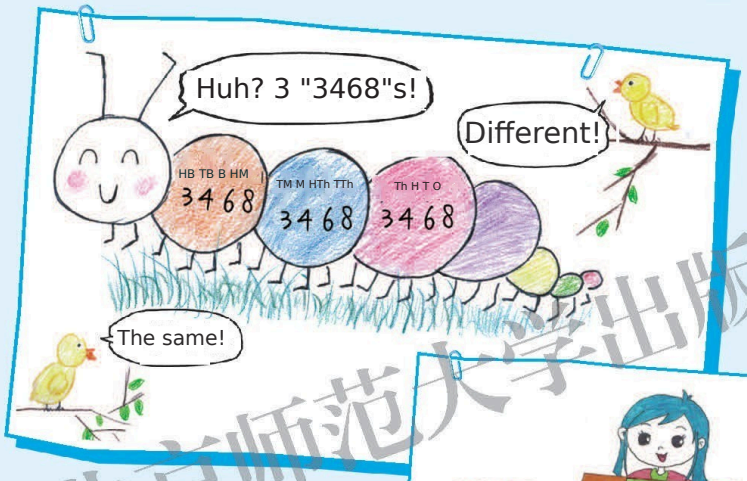
Dear children,

In your three years of mathematics so far, what knowledge and methods have you learned? What have you discovered?

Wise Old Man

I find that much knowledge and many methods in mathematics are connected and related.

Taoqi



When learning mathematics, I like to think for myself first.

Xiaoxiao

Talking with others also helps me learn.

Clever Dog

Children, talk about how you learn mathematics and share it with your classmates.

Your friends, the editors

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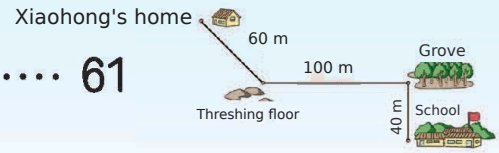

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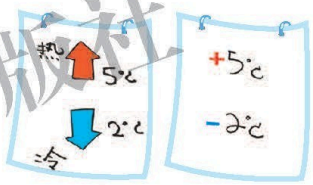

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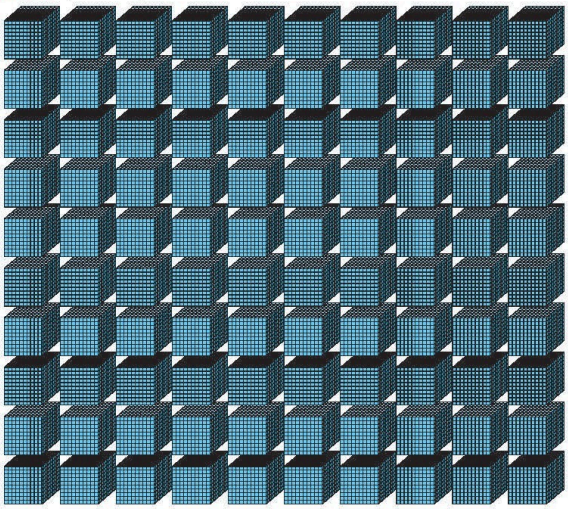
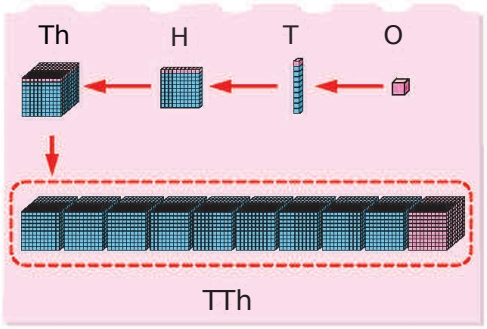
Unit 1 Understanding Larger Numbers

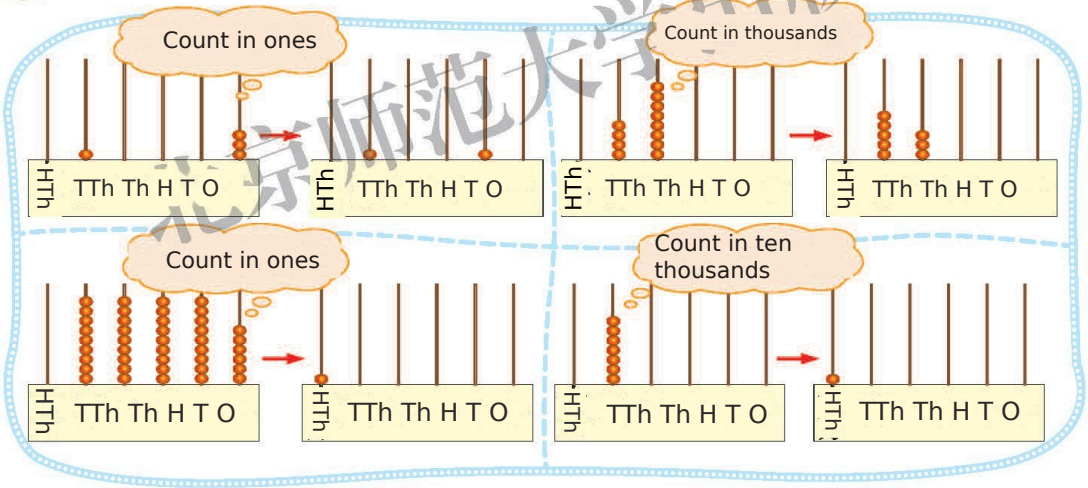
Count

Count and learn.





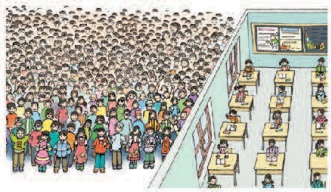
Move the beads and count.



How big is ten thousand? And one hundred thousand?

Running 25 laps around a 400-metre track is ten thousand metres.

2500 classes have about one hundred thousand students.



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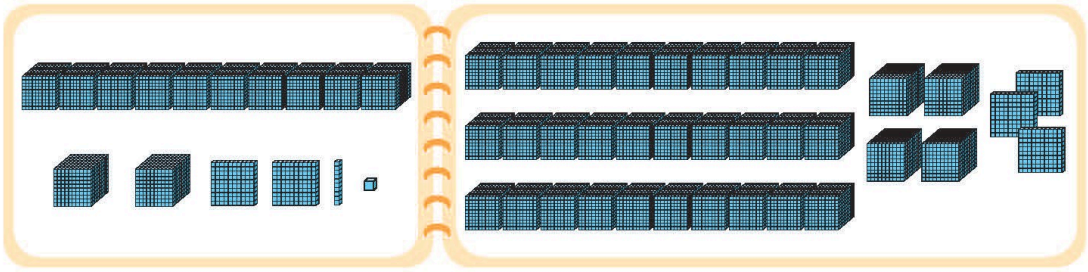
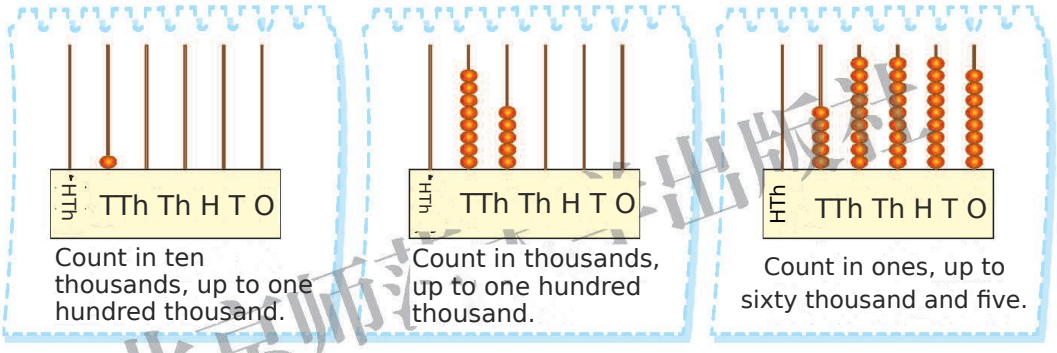
Practice

1. Say how many small cubes there are in total.



2. Move the beads on the counting frame as required and count.



Count in ten thousands, up to one hundred thousand.

Count in thousands, up to one hundred thousand.

Count in ones, up to sixty thousand and five.

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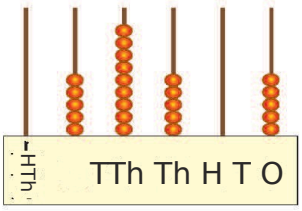
3. Read it and discuss with your partner: what does the "5" in each place represent?

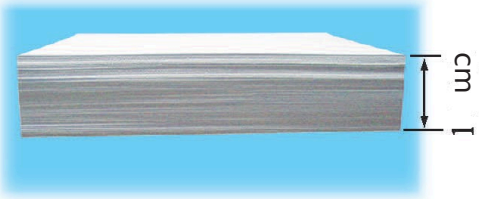


The "5" in the ones place means 5 ( ), the "5" in the hundreds place means ( ) ( ), and the "5" in the ten-thousands place means ( ) ( ).

4. A stack of 100 sheets of paper is about 1 cm high.

(1) Work out: about how high is a stack of ten thousand sheets? And one hundred thousand sheets?

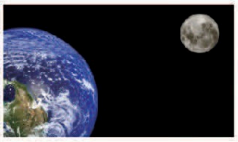
(2) Use familiar everyday objects to describe how high a stack of one hundred thousand sheets of paper is.



Larger Numbers

Talk about large numbers in daily life

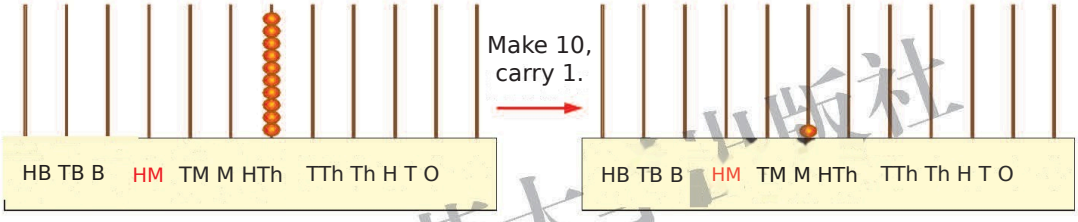
A person has about one hundred and twenty thousand hairs.

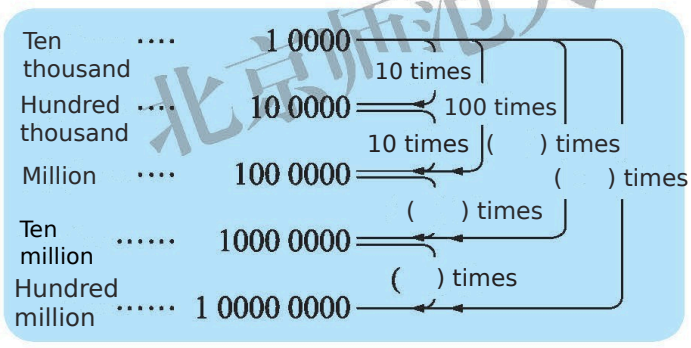


The average distance from the Moon to the Earth is about three hundred and eighty-four thousand km.

According to the 2010 population statistics, the total world population is about six billion nine hundred million.

Count and think: how are two neighbouring counting units related?





10 hundred thousands make one million, 10 millions make ten million, 10 ten millions make one hundred million...

Below is the place-value chart. Read it and discuss: what do you notice?

Period	·····	Hundred millions				Ten thousands				Ones			
Place	·····	HB	TB	B	HM	TM	M	HTh	TTh	Th	H	T	O
Counting unit	·····	HB	TB	B	HM	TM	M	HTh	TTh	Th	H	T	one

Each period has 4 counting units, like "ones, tens, hundreds, thousands".

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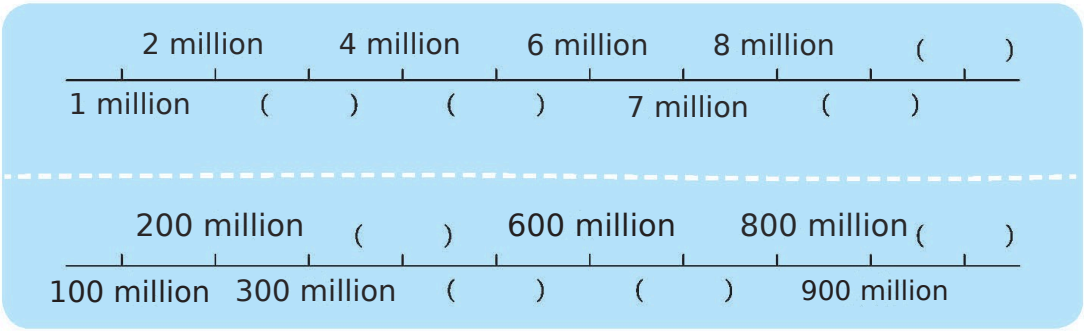
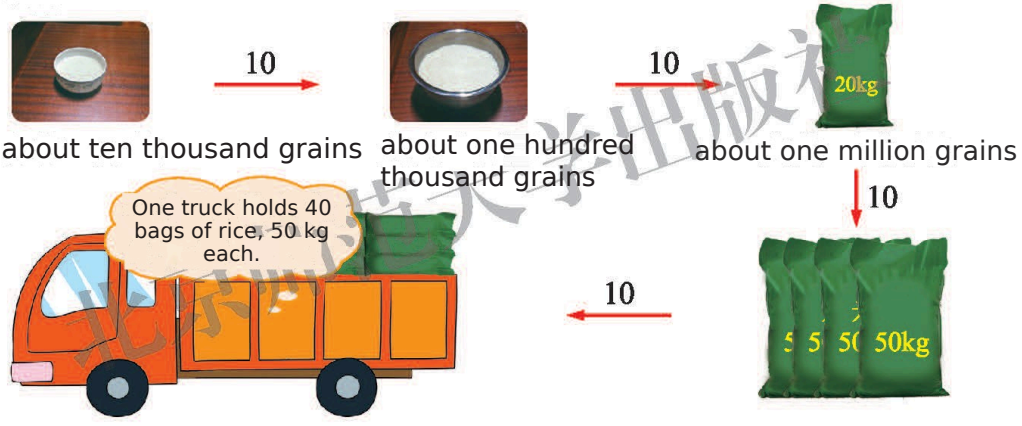
Practice

1. Count and fill in.



2. Look and talk: about how much rice is one million grains? Ten million grains? One hundred million grains?



3. Write the numbers.

(1) The number made up of 1 hundred million, 4 hundred thousands and 5 ones.

(2) The number made up of 6 millions, 3 ten thousands, 6 hundreds and 3 ones.

HM	TM	M	HTh	TTh	Th	H	T	O

4. Show 2345678 on the counting frame and talk about how many millions, hundred thousands, ten thousands, thousands, hundreds, tens and ones make up this number.

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Census

In 2010 China carried out its sixth national census; the total population was 1370536875 people. Below are the census results for some provinces, cities and special administrative regions. (unit: people)

Beijing: 19612368

Anhui: 59500510

Hong Kong: 7097600

Guangdong: one hundred and four million, three hundred and three thousand, one hundred and thirty-two

Taiwan: twenty-three million, one hundred and sixty-two thousand, one hundred and twenty-three

Macao: five hundred and fifty-two thousand, three hundred

Can you read the total population of the whole country? And of Beijing, Anhui and Hong Kong?

First group the digits: one billion...

Hundred millions	Ten thousands	Ones
13	7053	6875

TM	M	HTh	TTh	Th	H	T	O
5	9	5	0	0	5	1	0

fifty-nine million, five hundred thousand, five hundred and ten

Can you write the population of Macao, Taiwan and Guangdong?

.....	Hundred millions				Ten thousands				Ones			
.....	HB	TB	B	HM	TM	M	HTh	TTh	Th	H	T	O
five hundred and fifty-two thousand, three hundred							5	5	2	3	0	0
twenty-three million, one hundred and sixty-two thousand, one hundred and twenty-three												
one hundred and four million, three hundred and three thousand, one hundred and thirty-two												

Discuss: how do you read and write numbers?

Practice

1. Answer these questions.

In 2012, Canada had about 33490000 people.

In 2012, Saudi Arabia had about twenty-eight million six hundred and ninety thousand people.

In 2012, the UK had about 61110000 people.

In 2012, Brazil had about one hundred and ninety-eight million seven hundred and forty thousand people.

(1) Read the populations of Canada and the UK, and tell your partner how you read them.

(2) In the place-value chart below, write the populations of Saudi Arabia and Brazil.

HM	TM	M	HTh	TTh	Th	H	T	O

2. Read the numbers first, then draw lines to match.

96720000 6080070 12000000080 400900

1200000080 9672000 60800070 4000900

ninety-six million, seven hundred and twenty thousand

sixty million, eight hundred thousand and seventy

twelve billion and eighty

four hundred thousand, nine hundred

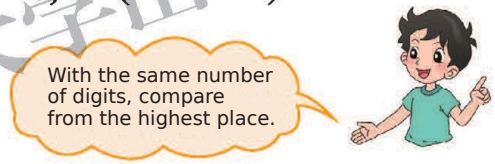
3. Collect some information about large numbers from newspapers, magazines and other media, and share it with the class.

Land Area

Below are the areas of China's 5 largest provinces and autonomous regions (unit: km²).

Arrange the areas of Sichuan Province, the Tibet Autonomous Region and the Xinjiang Uygur Autonomous Region from largest to smallest, and explain how you compared them.

( ) > ( ) > ( )



Arrange the areas of the 5 provinces and autonomous regions on the map above from largest to smallest, and tell your partner how you compared them.

seven-digit numbers

six-digit numbers

( ) > ( ) > ( ) > ( ) > ( )

Rewrite these numbers in units of "ten thousand" or "hundred million".

For convenience, we sometimes need to rewrite numbers that are exact multiples of ten thousand or hundred million in units of "ten thousand" or "hundred million".

The area of the Xinjiang Uygur Autonomous Region is about:

166 0000 km² = 166 ten thousand km²

The total population of China is about:

14 00000000 people = 14 hundred million people

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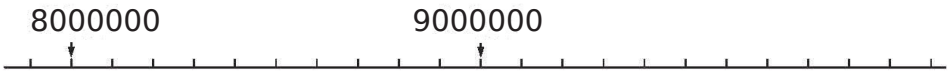
Practice

1. Mark the following numbers on the line and arrange them from smallest to largest.

9600000　8500000　10000000　7900000



(　　) < (　　) < (　　) < (　　)

2. Read the numbers in the table below and arrange them from smallest to largest.

Table of National Land Resources

Type	Cultivated land	Forest land	Pasture	Other farmland
Area (km^2)	1259300	2307200	2635200	256500

(　　) < (　　) < (　　) < (　　)

3. Rewrite each of the following numbers in units of "ten thousand".

20500000　　8000000　　120000

4. The table below gives an overview of China's investment in landscaping and urban sanitation. Read the data in the table and rewrite the numbers in units of "hundred million".

Year	Item	Investment (¥)	Investment (¥100 million)
2000	Landscaping	14300000000	
	Urban sanitation	8400000000	
2005	Landscaping	41100000000	
	Urban sanitation	14800000000	
2010	Landscaping	229700000000	
	Urban sanitation	30200000000	

5. Join the dots below in order, from the smallest number to the largest.

73000

72 ten thousand

one hundred and forty million

983000

1000 ten thousand

nine million, seven hundred and two thousand

9720000

102000000

one hundred million and four thousand

Approximate Numbers

On 1/10/2009, the celebration of the 60th anniversary of National Day began with a 60-gun salute. The national flag guard marched 169 steps, standing for the 169 extraordinary years since the Opium War of 1840. The parade and march-past together took nearly 66 minutes, with 56 formations and echelons, and about 200000 people were reviewed. The huge Chinese painting "This Land So Rich in Beauty" has a canvas area of nearly 20000 square metres.

In the data above, which are exact numbers? Which are approximate numbers?

In "56 formations and echelons", "56" is an exact number.

In "about 200000 people", "200000" is an approximate number.

The actual area of the huge Chinese painting "This Land So Rich in Beauty" is 18000 square metres, but a report said "nearly 20000 square metres". How was "20000" obtained?

From the figure below, 18000 is closer to 20000.

Rounding 18000 to the nearest ten thousand gives 20000.

$18000 \approx 20000$

"≈" is the sign for "is approximately equal to".

The exact number of people in the National Day parade was 233482. Find roughly where this number is on the line below. How was "about 200000 people" obtained?

233482 is closer to 200000. In fact, it was rounded to the nearest hundred thousand.

Which place to round to depends on...

Practice

1. Read the data below. Which are exact numbers? Which are approximate numbers?

In 2010, 190 countries and 56 international organisations took part in the Shanghai World Expo. The solar panels on the roof of the Theme Pavilion covered more than 30000 square metres. By 31/10/2010, there had been more than 73080000 visits, an average of about 400000 a day.

2. Mount Hua is one of China's famous Five Great Mountains. It is about 2155 m above sea level. Mark it on the line below. About how many metres is that, to the nearest hundred?

2100 m 2150 m 2200 m

3. Mark on the lines, then fill in the blanks.

Round the population of Shanghai to the nearest ten thousand:

23019148

23010000 23020000

23019148 people ≈ ( ) people

Round the population of Shandong Province to the nearest hundred million:

90000000 100000000

95793065 people ≈ ( ) people

4. Complete the table.

	Nearest ten	Nearest hundred	Nearest thousand	Nearest ten thousand
25763				
29957				
30168				

Round to the nearest ten thousand...

5. Do a survey first, then answer the questions below.

(1) How many students are in your grade? What is that to the nearest ten?

(2) How many students are in your school? What is that to the nearest hundred?

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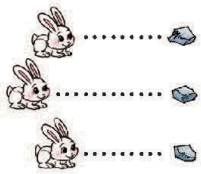
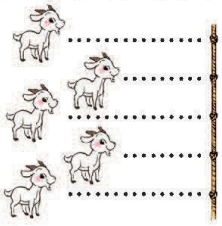
Counting with Knotted Ropes

Do you know how people in ancient times counted? Read on.

In ancient times, people counted the animals they caught with stones, knotted ropes or notches.



Counting with stones

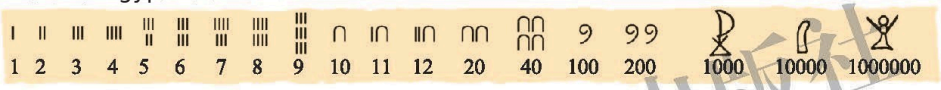


Counting with knots

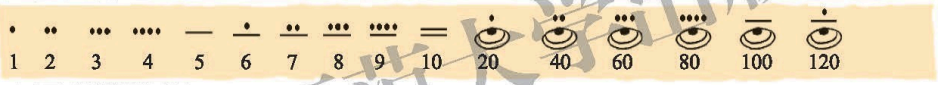
Counting with notches

Later people gradually invented some counting symbols. Five thousand years ago, people began to use various symbols to represent numbers.

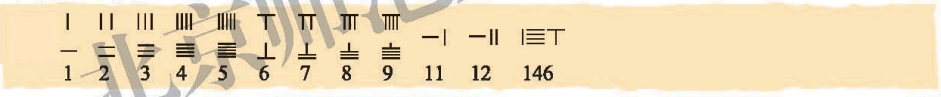
Ancient Egyptian numerals



Mayan numerals



Chinese rod numerals (top row: vertical form; bottom row: horizontal form)



0 1 2 3 4 5 6 7 8 9

The 10 digits from 0 to 9 that we use today can represent any number. These digits are called Hindu-Arabic numerals.

The numbers 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, ... that represent the number of objects are all natural numbers. When there are no objects at all, we use 0. 0 is also a natural number.

What do you know about natural numbers? Talk about it with your partner.

0, 1, 2, 3, ⋯

Each number is always 1 more than the one before it.

Natural numbers are counted in base ten: each counting unit is ten times the next smaller one.

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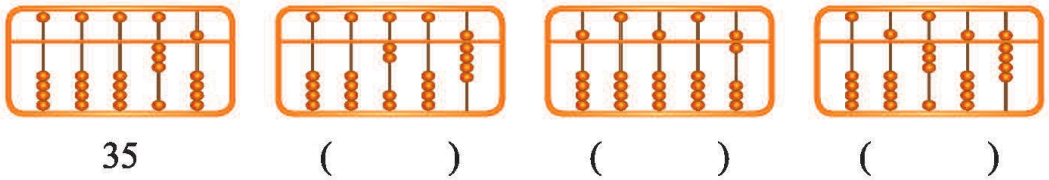
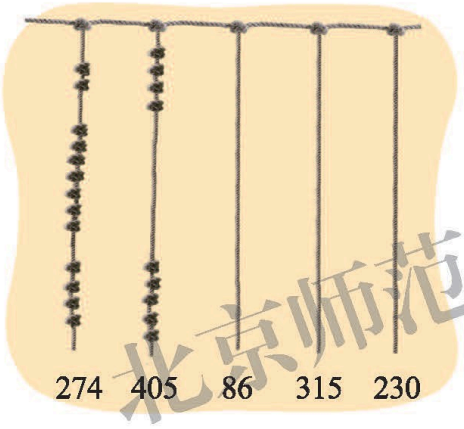
Practice

1. Look and fill in the appropriate numbers in the brackets.



35 ( ) ( ) ( )

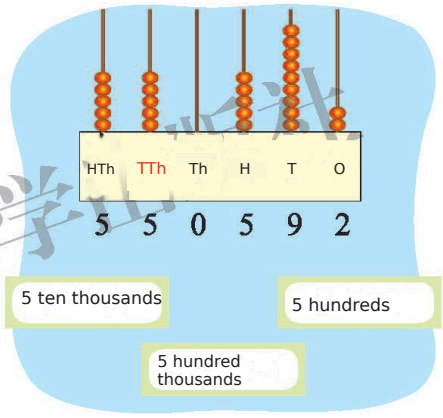
2. Can you understand how the numbers below are represented? Can you represent the other numbers?



274 405 86 315 230

3. What is the difference in meaning between the 5 beads in the hundred thousands, ten thousands and hundreds places?

Match.



4. Find the pattern and fill in.

(1) 2002, 3003, 4004, _____, _____, 7007, _____, _____。

(2) 6060, 6040, 6020, _____, _____, 5960, _____, _____。

With decimal counting, just a few symbols can represent any natural number – a great invention in the history of mathematics. Over long practice, people developed different number systems as they were needed. Today the binary system is widely used in computers, and the sexagesimal (base-60) system is used for time and angles, such as 1 hour =60 minutes and 1 minute =60 seconds.

What other number systems have you seen in daily life? Want to learn more about number systems? Look it up!

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Exercise 1

1. First read or write the numbers on the lines below, then say how many digits each has and what the highest place is.

The Palace Museum in Beijing is the largest palace complex in the world, with an area of about 720000 square metres.

The National Library of China covers about two hundred and fifty thousand square metres. By the end of 2012 its collection had reached thirty-one million one hundred and ninety thousand items, fifth among the world's national libraries.

2. I say, you write; I write, you read.

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3. The total output of some fruits in China in 2009 was as follows:

31680000 tonnes

14260000 tonnes

7940000 tonnes

8830000 tonnes

First rewrite the numbers above in units of "ten thousand", then arrange them in order from largest to smallest.

( ) > ( ) > ( ) > ( )

4. Make a place-value chart, write these numbers in it and read them.

957182	629351428	805506300	28001009

5. The table shows the average distances of the eight planets from the Sun. Read the numbers and rewrite them in units of ten thousand.

Planet	Distance/km	Distance/10000 km
Mercury	57910000	5791
Venus	108200000	
Earth	149600000	
Mars	227940000	
Jupiter	778330000	
Saturn	1426980000	
Uranus	2870990000	
Neptune	4497070000	

(1) Which planet has the smallest average distance from the Sun? Which has the largest?

(2) What is Jupiter's average distance from the Sun in kilometres? Do its two "7"s mean the same thing? Talk about it.

6. Use the five digits 6, 6, 6, 0 and 0 to write these numbers.

(1) Largest 5-digit number.

(2) Number closest to 60000.

(3) Number with 0 in the thousands place.

7. Say to which place each of the following approximate numbers was rounded.

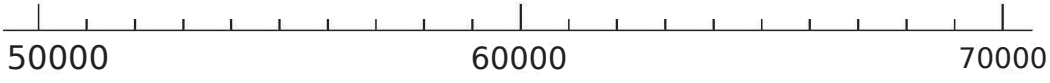
Price: ¥518, about ¥500.

Per hour: 78 km, about 80 km.

The Beijing-Kowloon Railway runs from Beijing in the north to Shenzhen in the south, linking to Kowloon, Hong Kong.

Main line length: 2397 km, about 2000 km.

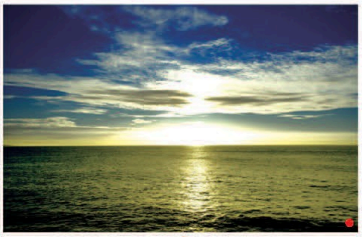
8. If a report gives a figure as "about 60000", what could the actual figure be? Draw its range.

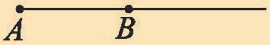


Unit 2 Lines and Angles

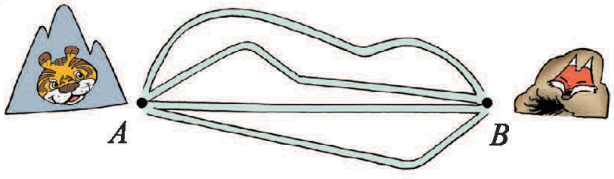
Recognising Lines

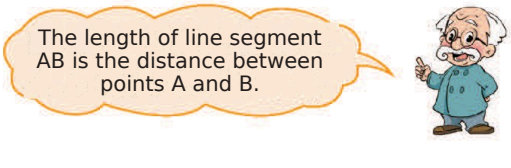
Look and say.



Line segment	Ray	Straight line
(1) A line segment has two endpoints. (2) It has a definite length.	(1) A ray has one endpoint. (2) A ray can extend infinitely in one direction.	(1) A line has no endpoints. (2) A line can extend infinitely in both directions.
 Read as: line segment AB (or BA)	 Read as: ray AB.	 Read as: straight line AB (or BA)

Discuss: what routes are there from Tiger Mountain to Fox Cave? Trace the shortest route. What do you find?





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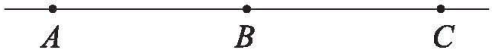
Practice

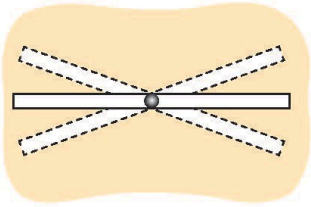
1. Find and talk about things in everyday life that look like line segments.

2. Talk about the lines, rays and line segments in the figure.

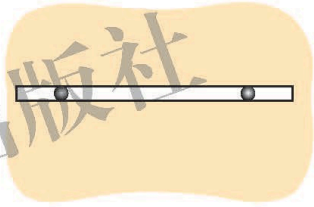


3. Do it and discuss.

(1) Pin a cardboard strip to a board with one thumbtack. Can the strip rotate? Through one point, how many straight lines can you draw?



(2) Pin a cardboard strip to a board with two thumbtacks. Can the strip rotate? Through two points, how many straight lines can you draw?



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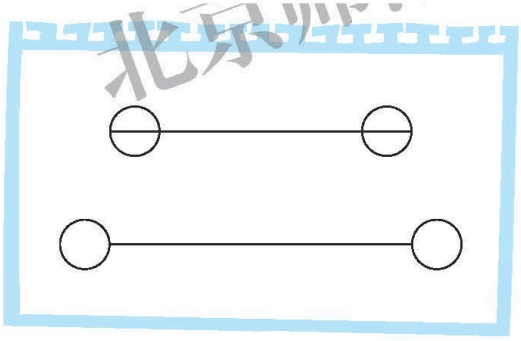
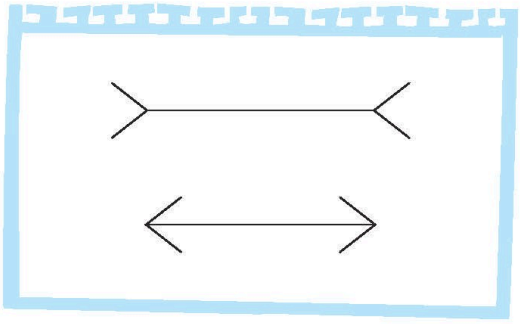
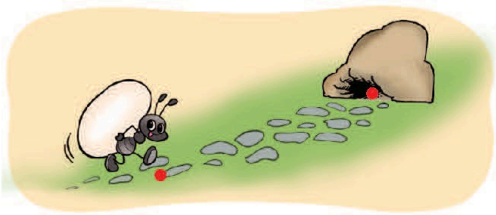
4. Is each pair of line segments the same length? Look and measure.



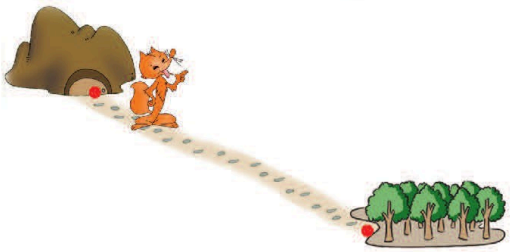


5. How far is the ant from the hole?

Draw and measure.

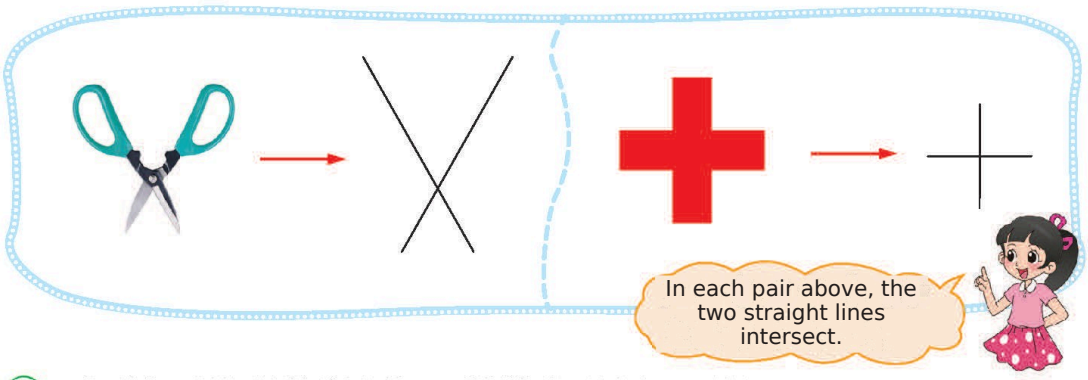


6. What is the distance from the fox's home to the woods? Draw and measure.

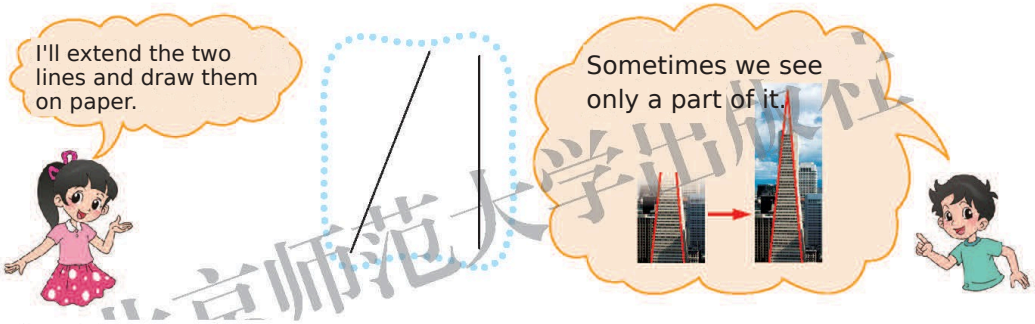


Intersecting and Perpendicular

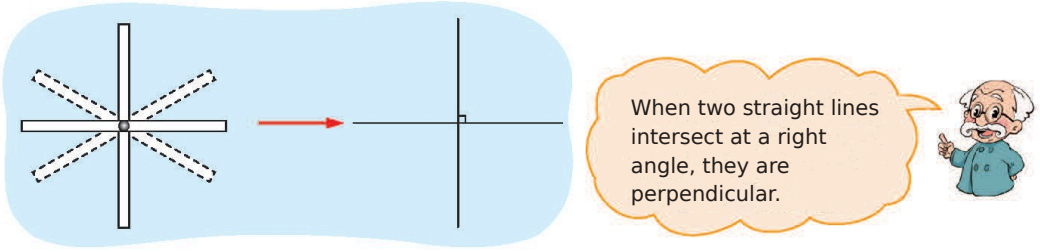
Look and say.



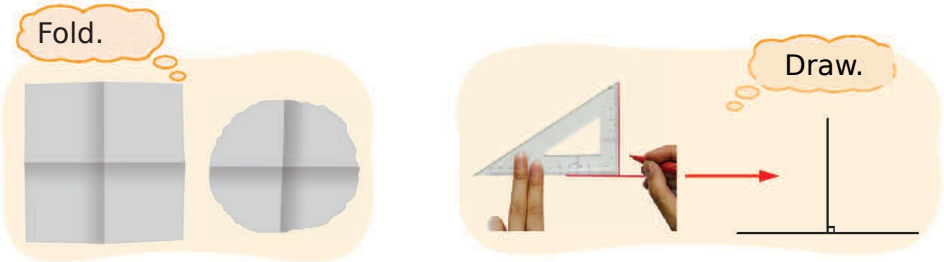
Do the two lines below intersect? Explain your thinking.



Do it and recognise.



Find a way to make a pair of perpendicular lines.



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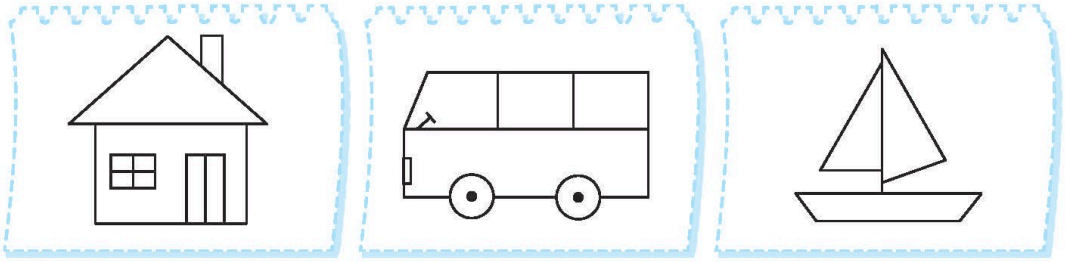
Practice

1. Find two pairs of intersecting or perpendicular line segments in each figure.

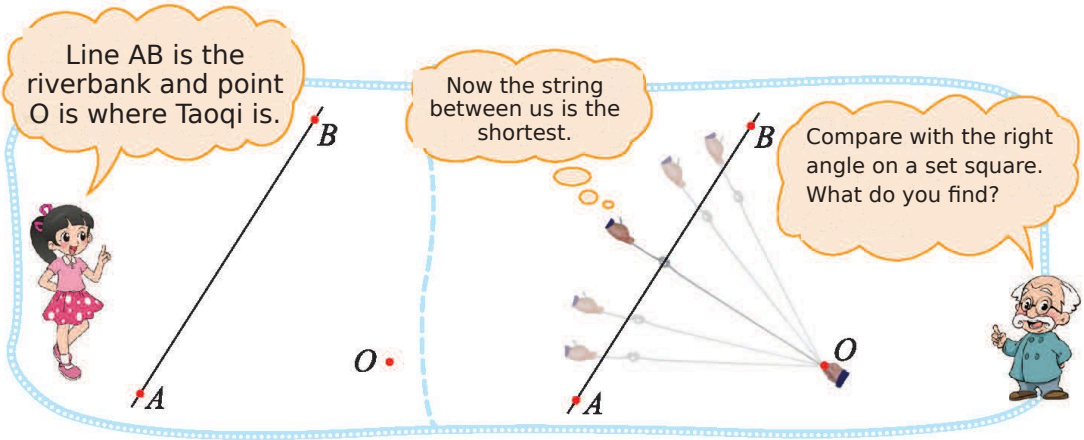


2. Which two sides of each figure are perpendicular? Check with a set square.

3. Draw a line perpendicular to each line below.

4. Mini experiment.

Try it in pairs.



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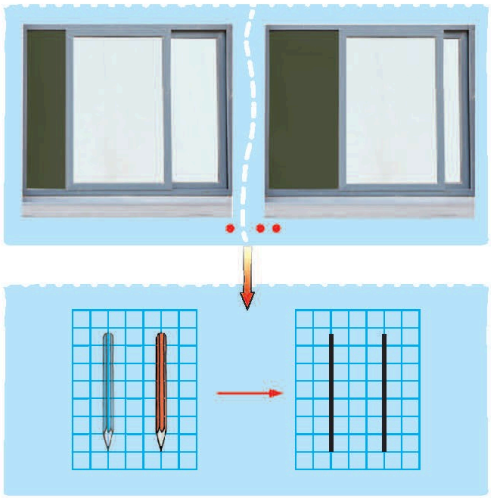
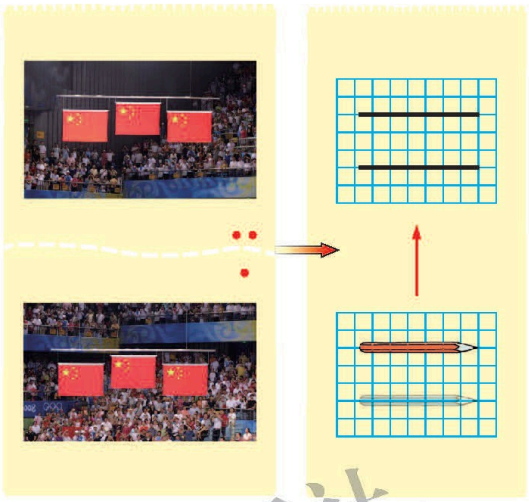
Translation and Parallel Lines

Talk about translations in everyday life. Slide a pencil on squared paper and draw.



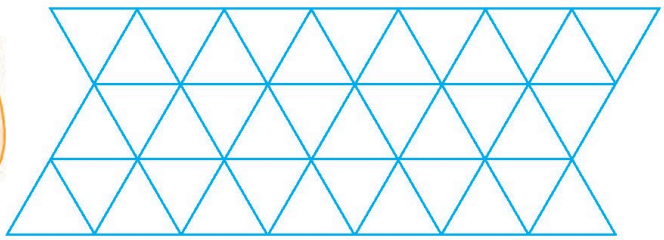


Two straight lines like these are parallel.

The pencils stay the same number of squares apart after moving.

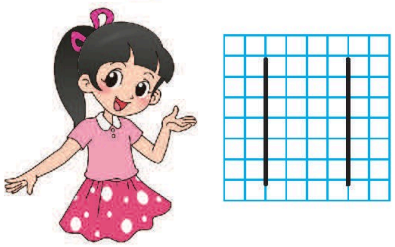
Find two pairs of parallel line segments and trace them in different colours.

These two lines are also parallel.

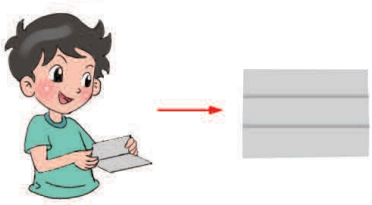


Can you find a way to make a pair of parallel lines?

I use squared paper.

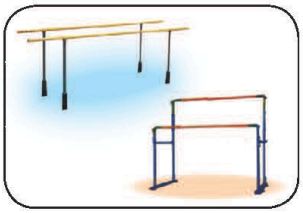


I fold paper.

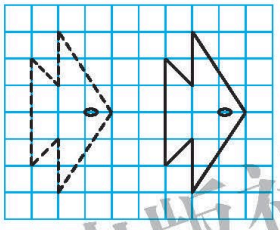


Practice

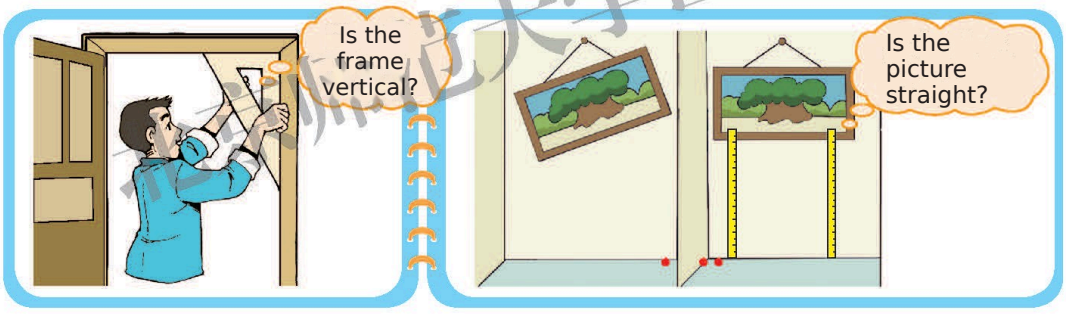
1. Find and talk about parallel lines in everyday life.



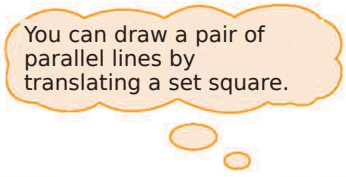
2. From each figure below, find two pairs of parallel line segments and trace them in different colours.

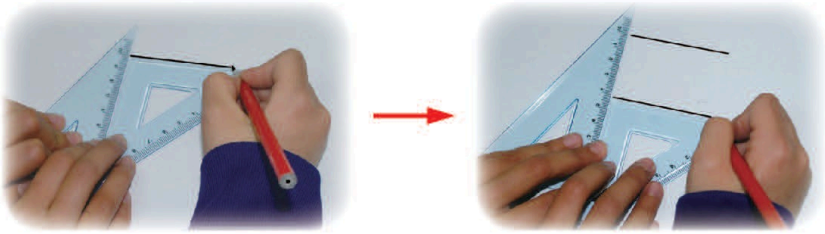


3. Can you use what you have learned to explain why?



4. You can also make a pair of parallel lines this way. Try it.





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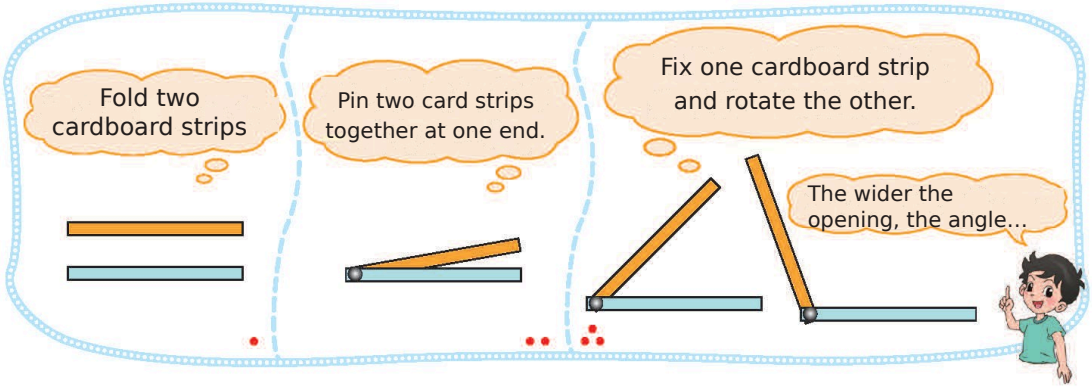
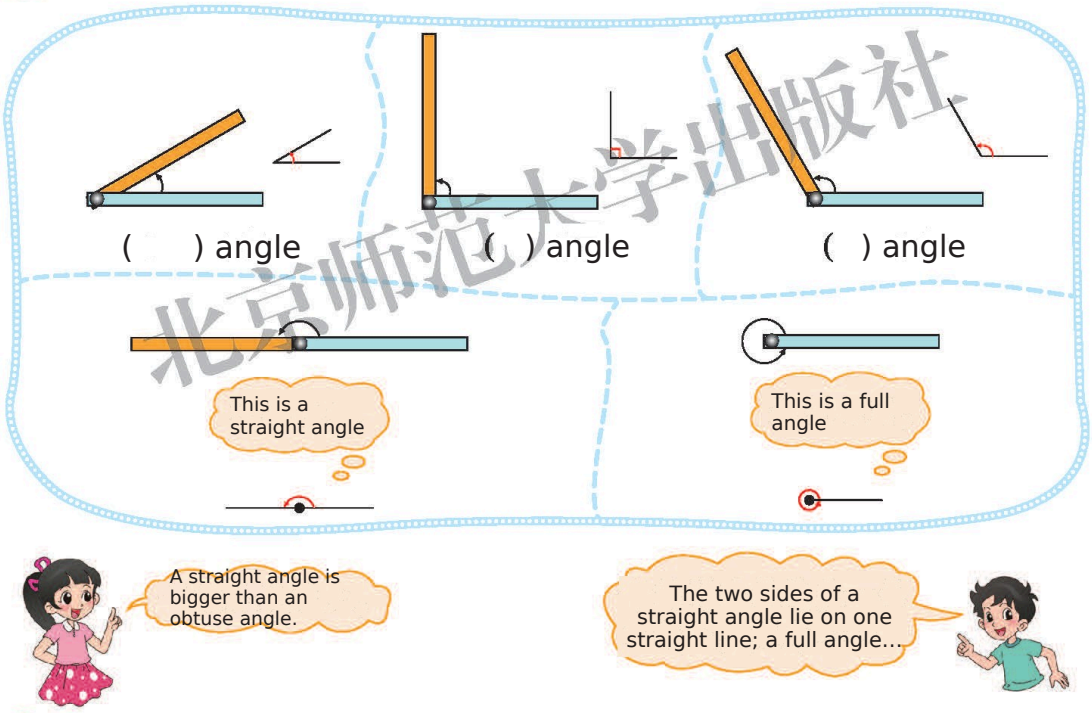
Rotation and Angles

Can you make a "moving angle"? Try it and think.



Watch the angles made as the strip turns. Name them and say what you notice.



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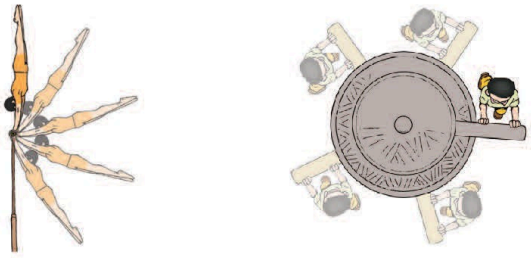
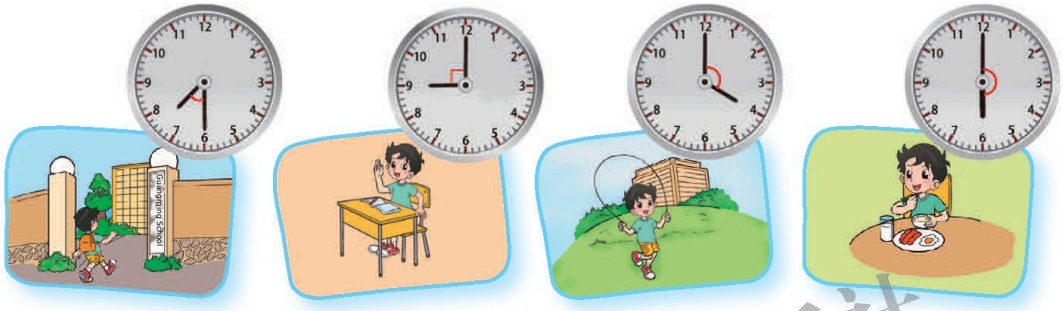
Find and describe straight angles and full angles in daily life.



Practice

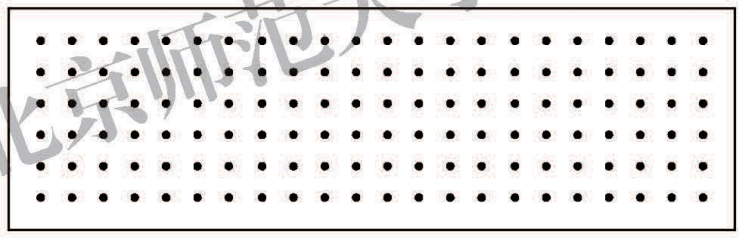
1. On the clock, turn the hands to make an acute angle, a right angle, an obtuse angle, a straight angle and a full angle. Do it with a partner.

2. Look at each clock face. Name the angle made by the hour and minute hands.



( ) ( ) ( ) ( )

3. On the dot grid, draw an acute angle, a right angle, an obtuse angle and a straight angle.

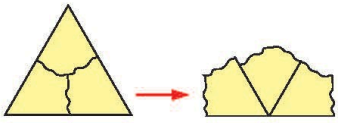


4. Fold a piece of paper. Compare with your deskmate and talk: are the angles you folded the same size?

(1) Fold a right angle.

(2) Fold an obtuse angle.

5. Cut out a paper triangle and a paper parallelogram. Tear off the angles of the triangle and fit them together. Talk with your partner: what do you find? What if you tear off the angles of the parallelogram and fit them together?



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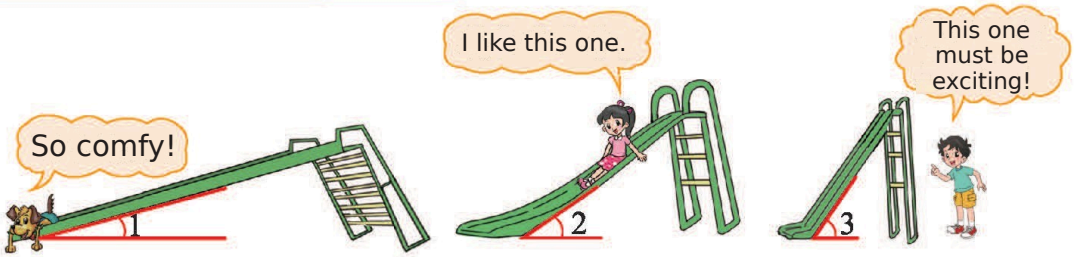
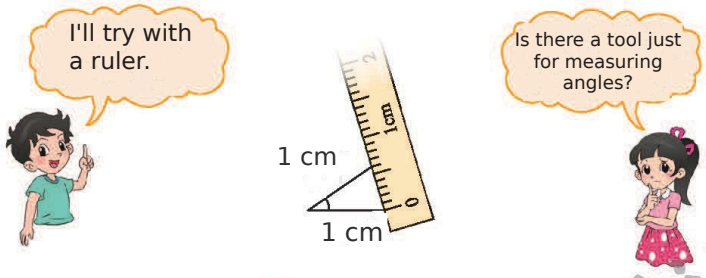
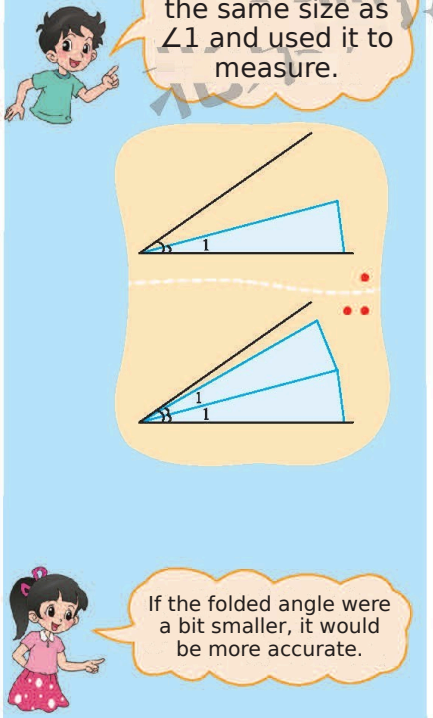
Measuring Angles (1)



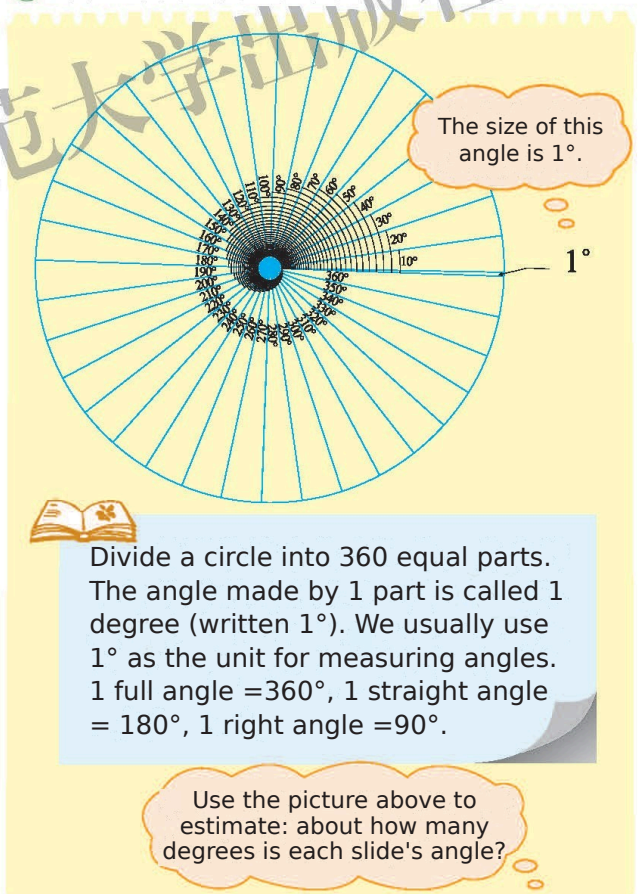
How can we measure the sizes of the three angles? Discuss.



Taoqi measures ∠2 like this. Try it yourself.



Think and identify



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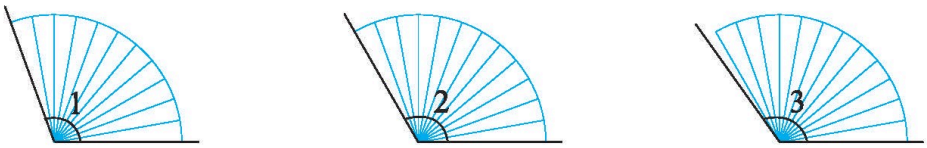
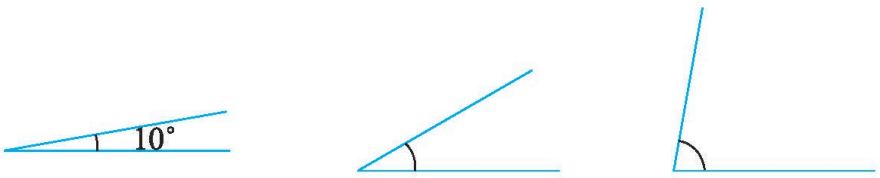
Practice

1. Count and talk: compare the sizes of ∠1, ∠2 and ∠3 below.

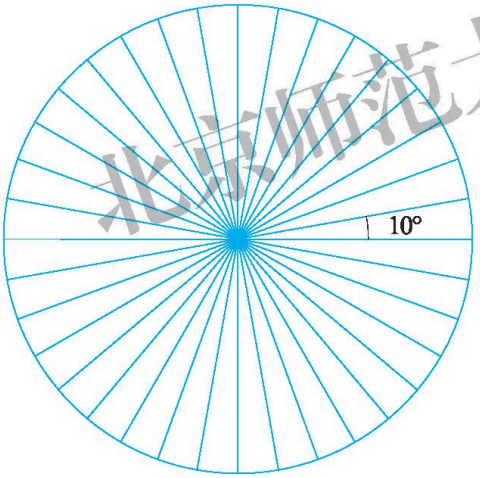


2. Estimate: how many degrees is each angle below?



( ) ( )

3.

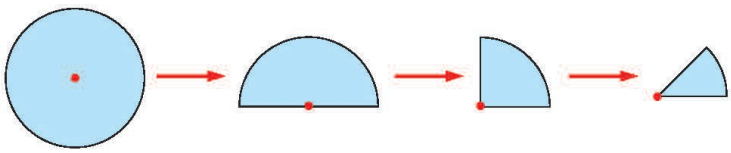


(1) Can you mark angles of 30°, 60°, 90°, 120°, 180°, and 270° on the figure?

(2) Can you find two 50° angles in the figure? Mark them.

(3) Use Figure 1 in the appendix to cut out an angle you want.

4. Fold a circular piece of paper in half three times. Think and fill in.



360°	( )	( )	( )
(full angle)	( )	( )	( )

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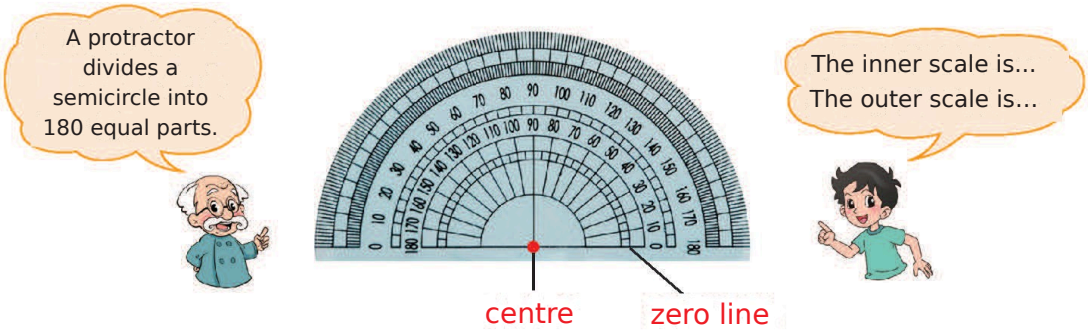
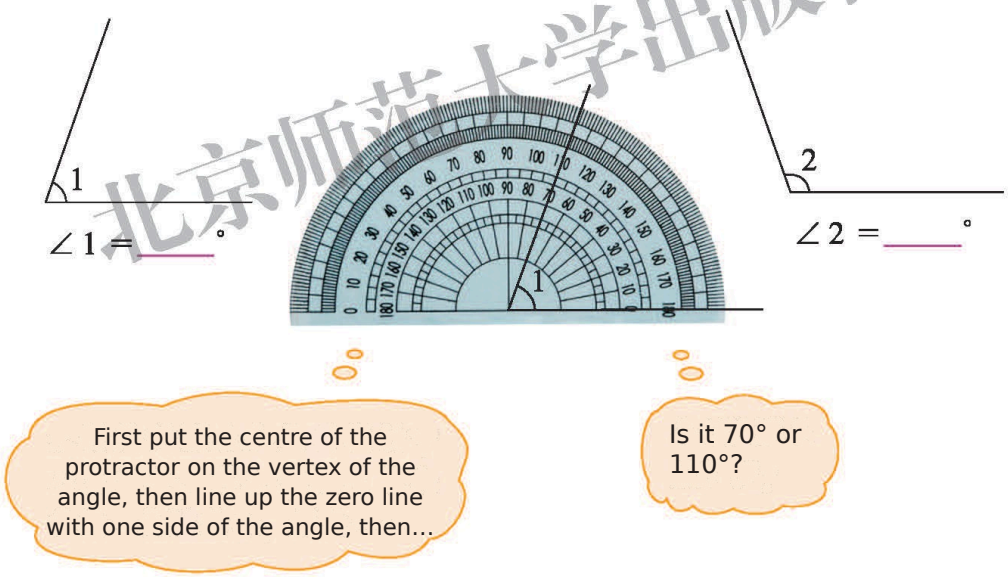
Measuring Angles (2)

A protractor is usually used to measure the size of an angle. Look at the protractor, get to know it and talk about it.

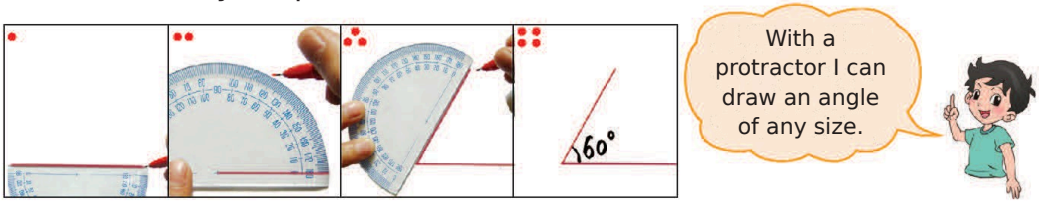


Can you find 50° and 140° angles on the protractor? Draw them on the pictures above and discuss with your partner.

Estimate, then measure the two angles below. How did you measure?



How do we draw an angle of 60°? Think, draw and discuss your method with your partner.



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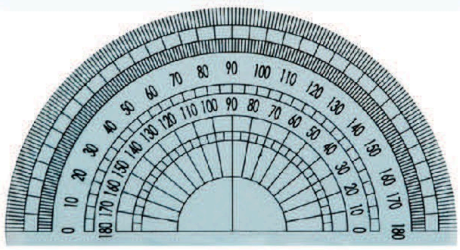
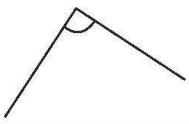
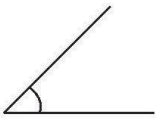
Practice

1. Find angles of 70° and 130° on the protractor and draw them on the figure on the right.



2. Use a protractor to measure each angle of the set square.

3. Estimate each angle below, then measure it. Was your estimate close?

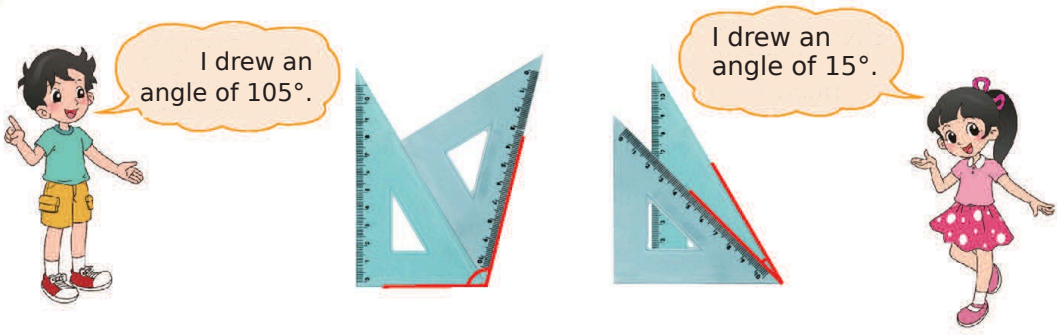
			
Estimate			
Measure			

4. Measure angles in daily life.

5. Draw the following angles.

35° 115° 80° 160°

6. What different angles can you draw with a pair of set squares?



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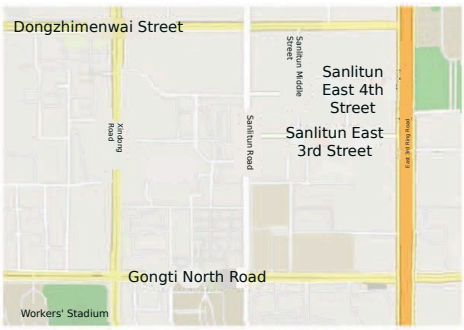
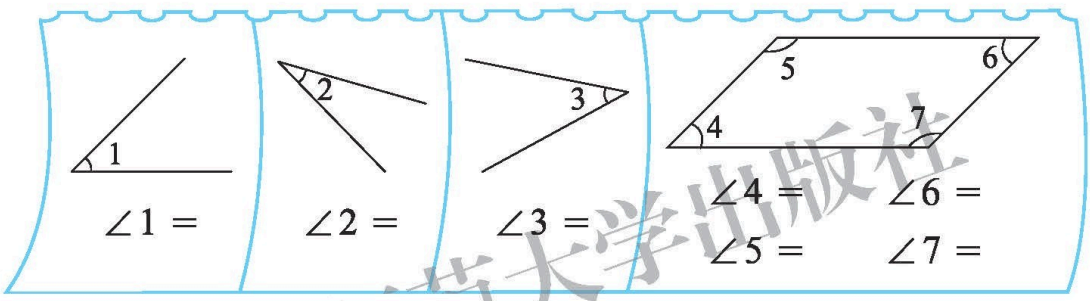
Exercise 2

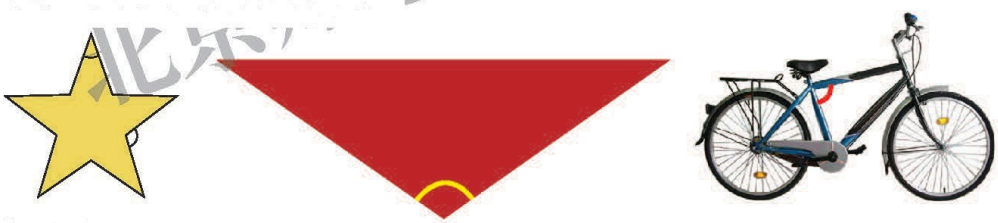
1. The picture on the right is part of a map of Beijing. Find two pairs of parallel roads and two pairs of perpendicular roads.



2. Estimate first, then measure.



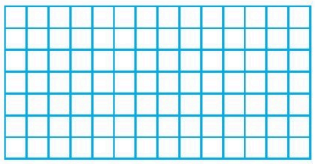
3. Measure each angle below.



4. Draw.

(1) Draw a line perpendicular to segment AB.

(2) Draw two parallel lines on the grid.



5. Cut out Figure 2 from the appendix, then fold it and try.

(1) Find a way to fold this paper into a rectangle, then cut it out.

(2) Fold the rectangle to make angles of 45° and 135°.

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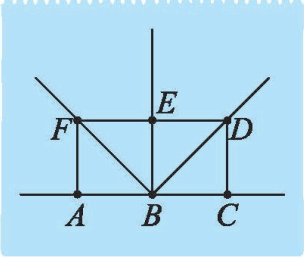
6. Use a protractor to draw the following angles.

45° 70° 140° 170°

7. Find and discuss.



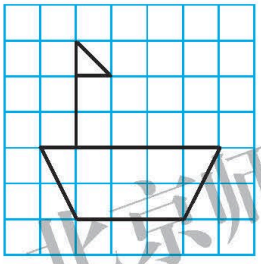
(1) Find a line segment, a line and a ray in the figure.

(2) In the figure, find a pair of parallel lines and a pair of perpendicular lines.

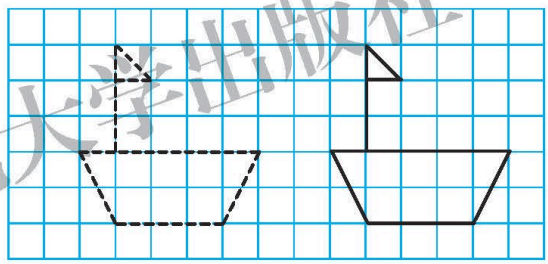
(3) Find two acute angles, two right angles, two obtuse angles and two straight angles in the figure.

8. Trace and find.

(1) Trace the parallel and perpendicular lines in the boat picture.



(2) Find the parallel lines in the two pictures, before and after the translation.



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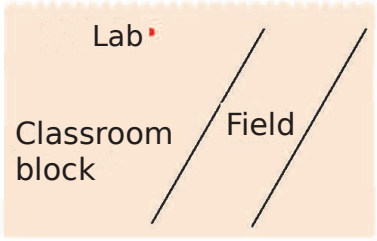
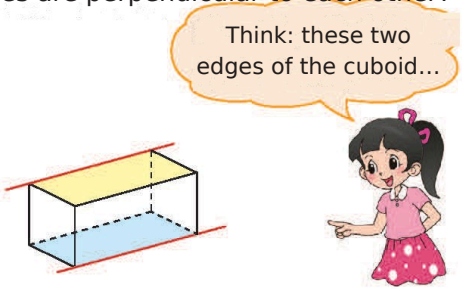
9. (1) Draw the shortest route from the lab to the classroom block.

(2) Draw the shortest route from the lab to the field.



10. Get a paper box shaped like a cuboid or a cube. Look: which edges of the cuboid or cube are parallel to each other? Which edges are perpendicular to each other?



Unit 3 Multiplication

Satellite Orbit Time

The first satellite China launched took 114 minutes for 1 orbit of the Earth.

Estimate: how long do 21 orbits of the Earth take? Discuss with a partner.

110×20=2200 (min), more than 2200 min.

120×20=2400 (min), about 2400 min.

Calculate: how long does it take to orbit the Earth 21 times? Explain how you worked it out.

□○□=□( )

114×20=2280
114×1=114
2280+114=2394

114×21
=114×7×3
=798×3
=2394

Do you understand how they calculated it? Discuss with your partner.

114×21

X	100	10	4
20	2000	200	80
1	100	10	4

2280
+ 114
2394

114
×21
114 …… 114×1
228 …… 114×20
2394

Answer: _____。

Calculate and discuss: what should you watch out for in column multiplication?

135
×74
_____

Step 2 works out 135×70……

Pay attention to carrying when calculating.

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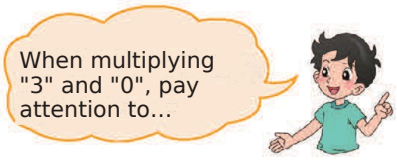
Try It

Try calculating 408×23 and say what to pay attention to.

$$\begin{array}{r} 408 \\ \times\ 23 \\ \hline 1224 \\ 816 \\ \hline 9384 \end{array}$$



How could you work these out? Discuss your ideas with a partner.

54×312

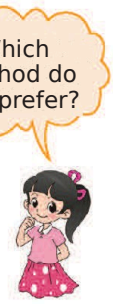
$$\begin{array}{r} 54 \\ \times\ 312 \\ \hline 108 \\ 54 \\ 162 \\ \hline 16848 \end{array}$$

$$\begin{array}{r} 312 \\ \times\ 54 \\ \hline 1248 \\ 1560 \\ \hline 16848 \end{array}$$

210×47

$$\begin{array}{r} 210 \\ \times 47 \\ \hline 1470 \\ 840 \\ \hline 9870 \end{array}$$

$$\begin{array}{r} 210 \\ \times 47 \\ \hline 147 \\ 84 \\ \hline 9870 \end{array}$$



Practice

1. In July 2011, Beijing started the 114 telephone booking service: patients can book an appointment at a chosen hospital by dialling 114. In August 2011, Aunt Guo worked 23 days in total and answered an average of 144 calls a day.

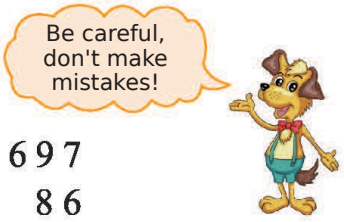
(1) How many calls did Aunt Guo answer in August 2011? Estimate, then calculate.

(2) Explain your working to a partner.

2.

At the 60th National Day parade, each soldier marching past Tian'anmen took 128 steps, each 75 cm long. How many centimetres did each soldier march in total? How many metres is that?

3. Calculate, and explain your working to your partner.



$$\begin{array}{r} 213 \\ \times\ 32 \\ \hline \end{array} \qquad \begin{array}{r} 138 \\ \times\ 54 \\ \hline \end{array} \qquad \begin{array}{r} 126 \\ \times\ 25 \\ \hline \end{array} \qquad \begin{array}{r} 697 \\ \times\ 86 \\ \hline \end{array}$$

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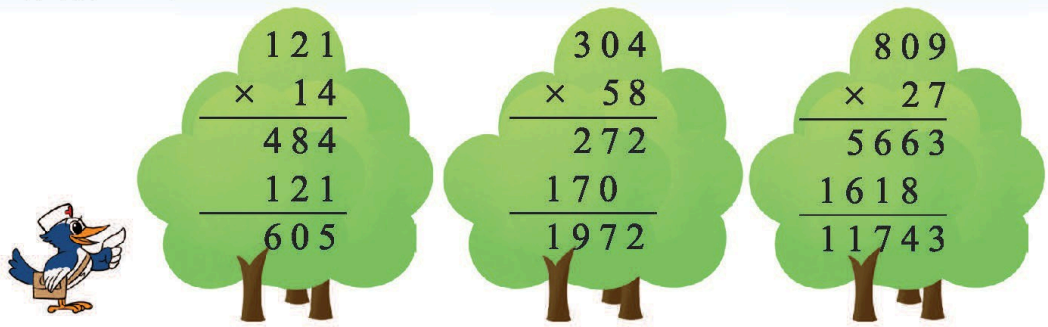
4. Forest Doctor.



5. Calculate, and explain your working to your partner.

$$\begin{array}{r} 203 \\ \times\ 32 \\ \hline \end{array} \qquad \begin{array}{r} 632 \\ \times\ 50 \\ \hline \end{array} \qquad \begin{array}{r} 609 \\ \times\ 57 \\ \hline \end{array} \qquad \begin{array}{r} 530 \\ \times\ 87 \\ \hline \end{array}$$

6. Wrapping one small gift box takes 31 cm of ribbon. There are 320 such boxes. Estimate: is 90 m of ribbon enough?

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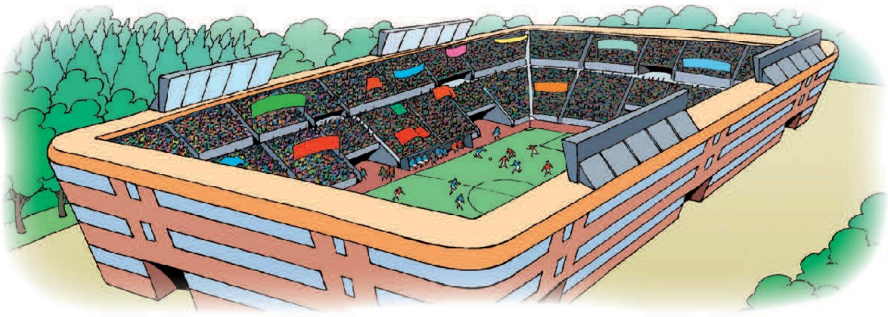
7.

912×25	473×28	26×130
108×24	82×403	460×23

8. (1) A shop buys 80 recorders wholesale from a factory at ¥140 each. How much must the shop pay the factory?

(2) After selling 70 recorders, the shop began to sell them at a lower price. If all of these recorders are sold, do you think the shop will make a profit or a loss? Discuss with your partner.

How Many Spectators

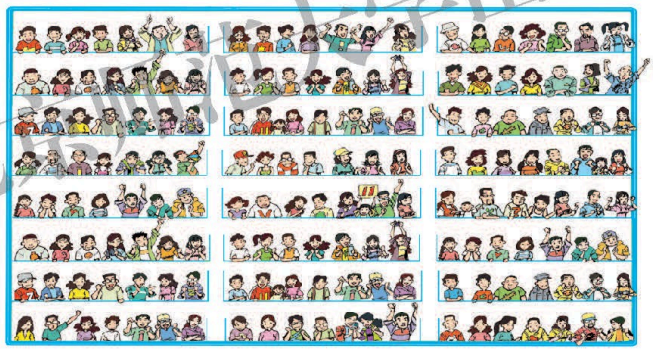


How could you estimate how many spectators this stadium can hold?

Count about how many spectators are in one row and how many rows there are, then multiply.

You can first estimate about how many spectators are in one stand.

Below is one of the stands. About how many spectators are in it?



Divide the seats into 3 roughly equal parts, then count how many spectators are in 1 part...

You can also divide the seats into...

This stadium has 28 stands in total. If each stand has about the same number of spectators, about how many spectators can this stadium hold?

Multiply the rough number of spectators in one stand by 28.

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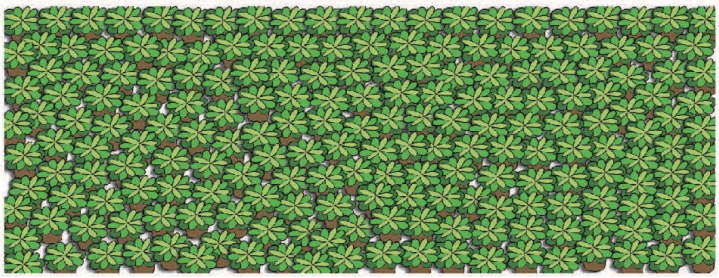
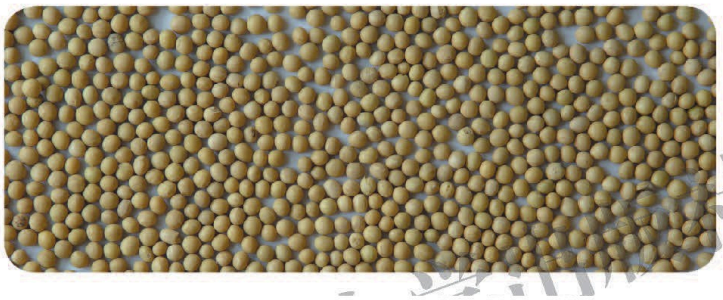
Practice

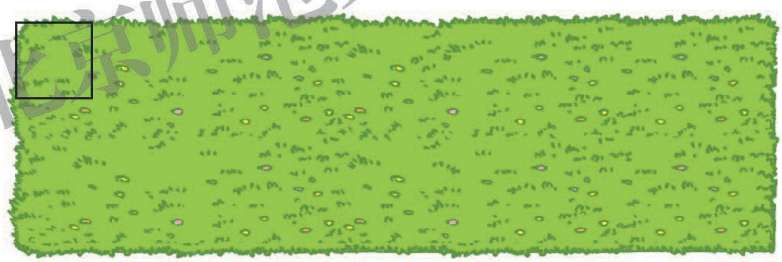
1. Estimate how many seedlings Grandpa Wang grew. Share your method with a partner.



2. Estimate how many soybeans there are. How did you do it? Share with your group or class.



3. Below is a picture of a lawn. If the box in it stands for 1 square metre, estimate the lawn's area.



4. The table below shows the daily sales (in yuan) of Dongfang Newsstand from 1 to 10 October. Can you estimate the total sales for these ten days?

Date	1	2	3	4	5	6	7	8	9	10
Sales	206	201	206	204	205	198	196	198	195	203

5. Choose a newspaper and estimate the number of words on one page. Use words, pictures or number sentences to explain how you estimated.

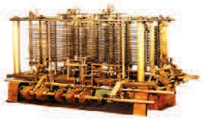
The newspaper I chose has pictures. This is how I dealt with them...

The headline is in big print. I dealt with it like this...

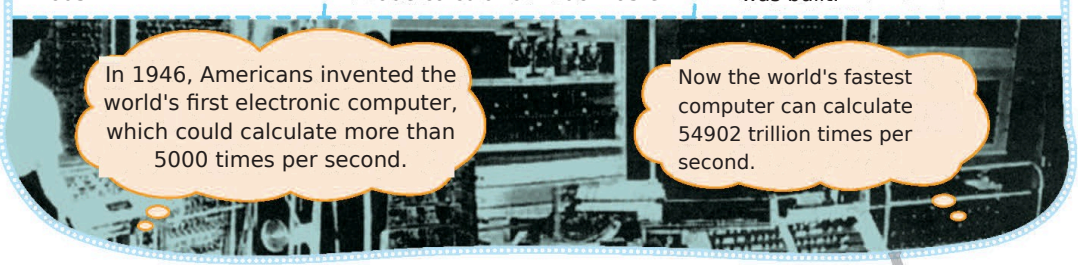
Amazing Calculating Tools

More than two thousand years ago, Chinese people calculated with counting rods.

More than one thousand years ago, Chinese people began to use the abacus widely, which made calculation much faster.



In 1822, the first computer that could be controlled automatically by a program was built.



In 1946, Americans invented the world's first electronic computer, which could calculate more than 5000 times per second.

Now the world's fastest computer can calculate 54902 trillion times per second.

A calculator is a calculating tool we often use in daily life. What do you already know about calculators?

I can work out 25×4: enter 25, then ×, then 4, then =. The number on the screen is the answer.

Wrong key? Press CE to clear it ……

Set each other some calculations and work them out together with a calculator.

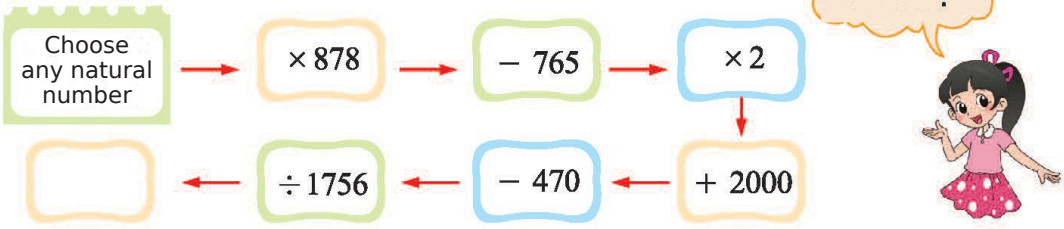
I estimate the result in case I press a wrong key.

245 × 789

21 ÷ 7

I can do this one in my head. No calculator needed!

Work it out with a calculator.



Practice

1. Work these out with a calculator.

$876+569$	$416000\div128$	782×534
$756+685-298$	$3363\times21\div57$	$438\times36-1734$

2. What can go in the brackets? Estimate, then check with a calculator to find it.

Number sentence	$88\times(\quad)$	$73\times(\quad)$
Product range	3000~3400	4000~4200

3. Read and calculate.

You often find these appliances still plugged in when they are not being used.

A TV on standby uses about 48 kilowatt-hours of electricity a year. That doesn't seem like much! Get your calculator ready. Let's start!

There are about 410000000 families in the country. If each family has 1 television, about how much electricity do these families waste in a year?

If each family uses 100 kilowatt-hours a month, how much is that in a year?

How many families could the wasted power supply for a year?

Unplug them when they are not in use!

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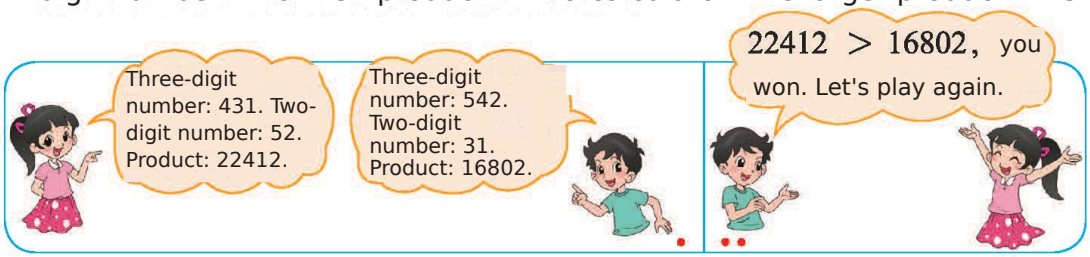
4. Maths game.

Use the five digits 1, 2, 3, 4, 5 to make a two-digit number and a three-digit number. Find their product with a calculator. The larger product wins.



(1) How many games did you and your partner play? How many did you win?

(2) Share your "secret" for winning with your partner.

Fun Number Sentences

There are many interesting number sentences in arithmetic. Taoqi and Xiaoxiao are going to explore the patterns behind them. Would you like to go with them? Don't forget to bring your calculator.

Calculate, then look carefully and say what you notice.

$1 \times 1 = 1$

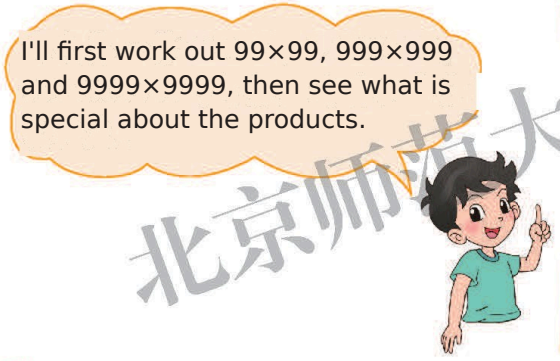
$11 \times 11 =$ _____

$111 \times 111 =$ _____

$1111 \times 1111 = ?$

$11111 \times 11111 = ?$

Without calculating, can you write the products of 99999×99999 and 999999×999999?



$99 \times 99 =$ _____

$999 \times 999 =$ _____

$9999 \times 9999 =$ _____

$99999 \times 99999 = ?$

$999999 \times 999999 = ?$

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Look at the number sentences and answers below. What is special about them? Can you write a few more like these? Check your results with a calculator.

$1 \times 9 + 2 = 11$

$12 \times 9 + 3 = 111$

$123 \times 9 + 4 = 1111$

$1234 \times 9 + 5 =$ _____

$12345 \times 9 + \square =$ _____

$123456 \times \square + \square =$ _____

How do you feel about what you learned today? Tell your partner.

Practice

1. The strange number 142857.

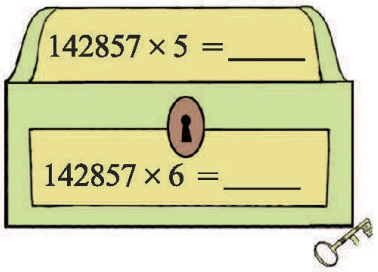
Work out the number sentences below and look for a pattern in the answers. Can you directly write the answers on the treasure box and find the password?

$142857 \times 1 =$ _____

$142857 \times 2 =$ _____

$142857 \times 3 =$ _____

$142857 \times 4 =$ _____



2. Find the mysterious four-digit number.

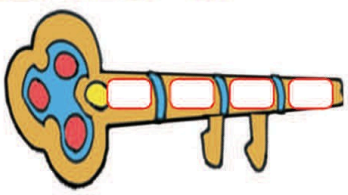
From the ten digits 0~9, choose any four and make the largest and smallest numbers. For example, with 2, 9, 8, 4, the largest number is 9842 and the smallest is 2489.

Then find the difference of the two numbers, make the largest and smallest numbers from the four digits of the result, and subtract again...

$$\begin{array}{r} 9842 \\ -\ 2489 \\ \hline 7353 \end{array} \qquad \begin{array}{r} 7533 \\ -\ 3357 \\ \hline 4176 \end{array}$$

If you keep repeating this process, can you find a mysterious four-digit number?

Write the mysterious four-digit number on the key below and open the gate of Wisdom Castle!

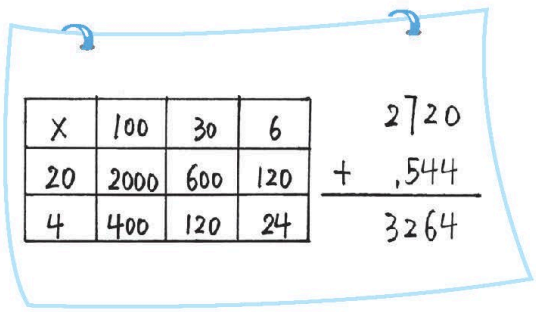


Exercise 3

1. When Xiaoxiao gives a speech she speaks 215 words per minute on average. In the "National Day Speech Contest" she must speak for 15 minutes. She has prepared a 3000-word speech. Is that enough?

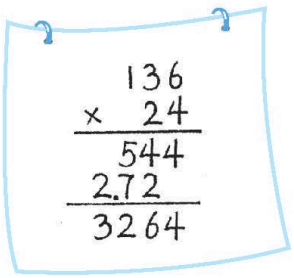
Estimate first, then calculate. Explain your working to a partner.

2. This is how Taoqi and Xiaoxiao each calculated 136×24.



×	100	30	6
20	2000	600	120
4	400	120	24

2720
+ 544
3264



136
× 24
544
272
3264

Tell your partner what each step means.

3. Without calculating, decide if each is right or wrong.

$58 \times 18 = 4534$ ( )　　$88 \times 34 = 318$ ( )

$150 \times 40 = 600$ ( )　　$350 \times 70 = 2450$ ( )

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4.

48×23	72×124	102×15	56×456
603×34	25×112	460×18	35×440

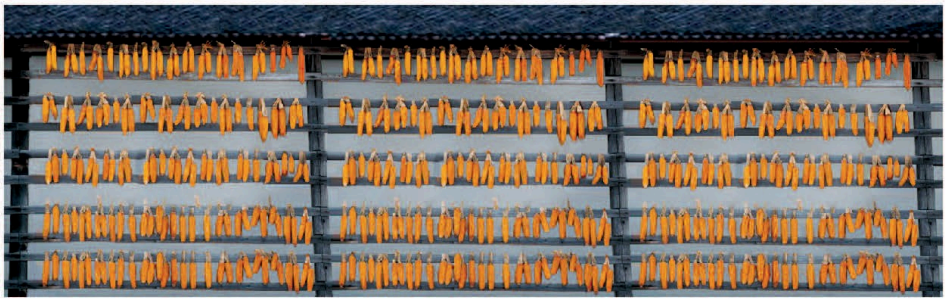
5. The table shows the daily sales of the Little Doctor stationery shop from 1 to 7 September. (Unit: yuan)

Date	1	2	3	4	5	6	7
Sales	293	285	292	297	302	316	312

(1) Estimate: about how much are the total sales for these 7 days?

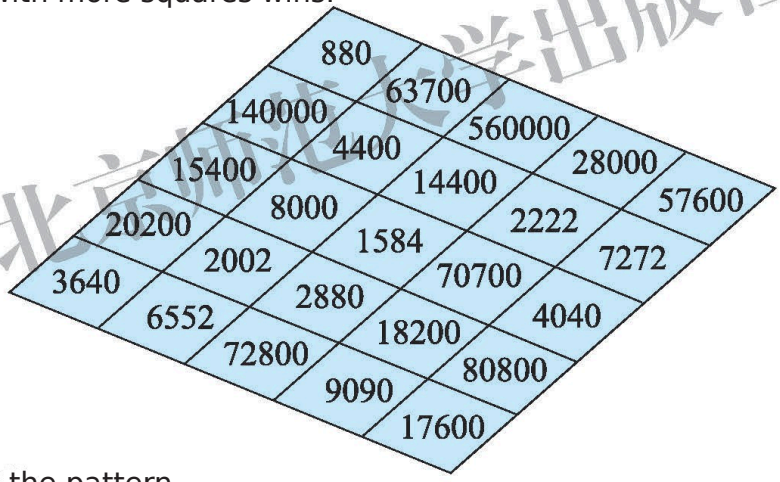
(2) At this rate, estimate the shop's total sales for the whole of September.

6. About how many corn cobs are there? Explain your method to a partner.



7. Maths game: Capture the Squares. From 22, 40, 72, 91, 101, 200, 700 and 800, choose any two numbers and multiply them.

If the product matches a number on the board, mark it; then no one else can take that number. The first player whose squares form a straight line wins. If all the squares are taken and no straight line is formed, the player with more squares wins.



8. Explore the pattern.

(1) Use the grid to explain how each three-digit number below is made. Work them out. What do you notice?

6	1	8
7	5	3
2	9	4

$618 \times 618 + 753 \times 753 + 294 \times 294$

$816 \times 816 + 357 \times 357 + 492 \times 492$

(2) Use the three digits in each column to make three-digit numbers and write number sentences like those above. Is the pattern the same?

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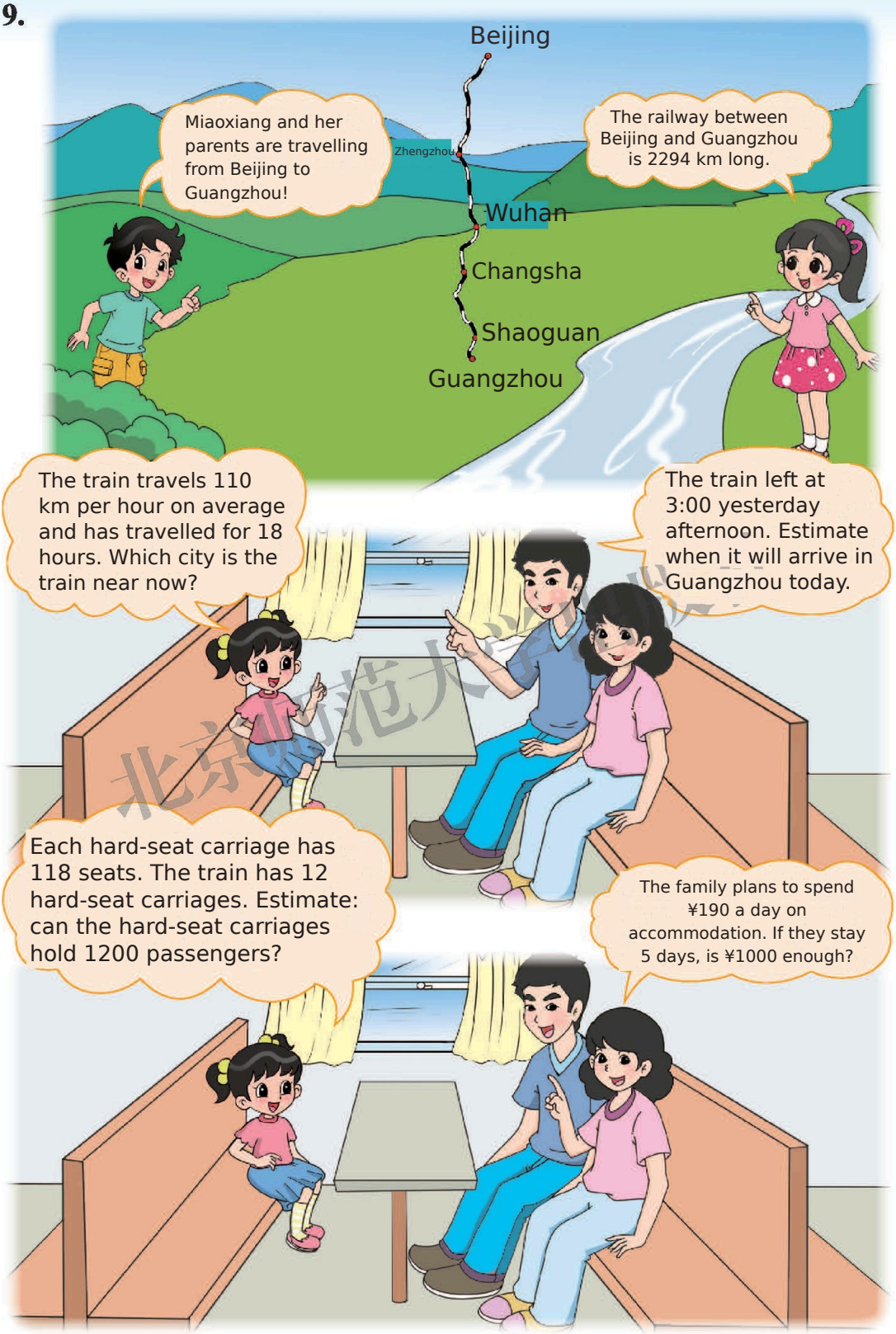
9.


Beijing
Zhengzhou
Wuhan
Changsha
Shaoguan
Guangzhou
Miaoxiang and her parents are travelling from Beijing to Guangzhou!
The railway between Beijing and Guangzhou is 2294 km long.
The train travels 110 km per hour on average and has travelled for 18 hours. Which city is the train near now?
The train left at 3:00 yesterday afternoon. Estimate when it will arrive in Guangzhou today.
Each hard-seat carriage has 118 seats. The train has 12 hard-seat carriages. Estimate: can the hard-seat carriages hold 1200 passengers?
The family plans to spend ¥190 a day on accommodation. If they stay 5 days, is ¥1000 enough?

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Review

What I Learned

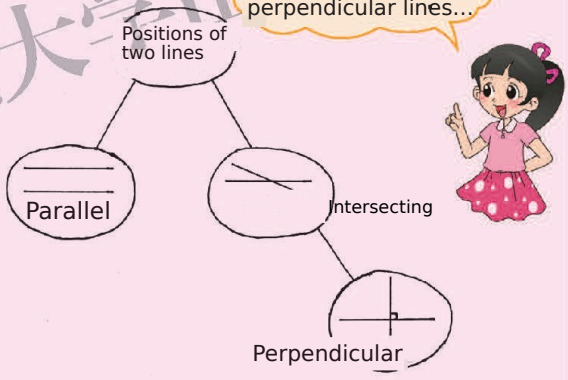
I know many large numbers, like China's population: 1370536875. To read, write, compare or rewrite one, first split it into classes: hundred millions, ten thousands, ones.

13 7053 6875 1 billion 370 million 536 thousand 875

To the nearest ten thousand: about 137054 ten-thousands. To the nearest hundred million: about 14 hundred million. A place-value chart also helps us read and write large numbers.

.....	B	H M	T M	M	H Th	T Th	Th	H	T	O

Acute		< 90°
Right		= 90°
Obtuse		> 90° and < 180°
Straight		= 180°
Full		= 360°



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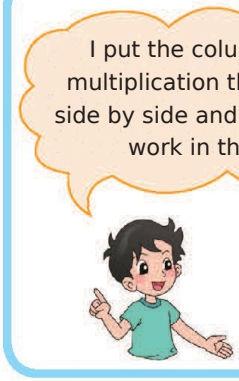


$$\begin{array}{r} 23 \\ \times\ 5 \\ \hline 115 \end{array} \qquad \begin{array}{r} 23 \\ \times\ 35 \\ \hline 115 \\ 69 \\ \hline 805 \end{array} \qquad \begin{array}{r} 23 \\ \times\ 135 \\ \hline 115 \\ 69 \\ 230 \\ \hline 3105 \end{array}$$

My Progress

I found that numbers within ten thousand and numbers above ten thousand are linked in how they are read, written and compared...

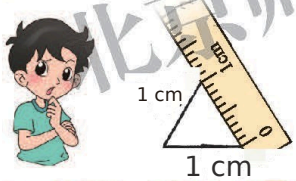
The activity that impressed me most was when the teacher took us through the process of making a protractor...

I have a new understanding of what an angle is...

I can use a calculator to explore patterns in maths...

My Questions

I think the size of this angle is 1 cm. Is that right?



I know there are many units of length and area. Besides the "degree", are there other units for angles?

I wonder why a calculator can always calculate so fast. How does it work inside?

When we use multiplication to solve real-life problems, which problems can be solved by estimating?

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My Maths Diary

Write a maths diary about what you have learned in maths so far.

Consolidate and Apply

1. Read the data in the table and answer the questions below.

Children's Books Printed in Some Provinces and Municipalities Unit: copies

Region	2010	2011	2012
Beijing	12260000	16170000	30930000
Shanghai	25490000	31740000	39430000
Shandong	10020000	6290000	24060000
Guizhou	640000	550000	3190000
Gansu	650000	760000	2850000

(1) In 2010, the total number of children's books printed in Beijing was ( ) ten thousand copies; in 2011, the total number printed in Gansu was ( ) ten thousand copies.

(2) For 2010 and then for 2012, compare the total numbers of children's books printed in the provinces and municipalities, and arrange them in order from smallest to largest.

(3) What do you notice from the data above?

2. Round each number in the table below to the nearest ten thousand and write the approximate number.

Number of Pupils in Primary Schools, 2007–2012

Year	Number of pupils	Rounded (10,000s)
2007	105640027	
2008	103315122	
2009	100714661	
2010	99407043	
2011	99263674	
2012	96958985	

3. Can you read the numbers below? Write the numbers given in words in figures.

Our country's largest outdoor activity camp for young people covers 3735200 square metres, of which 1334000 square metres is water. The state invested nearly ¥1000000000. The camp has eight zones, such as the Courage and Wisdom Zone, the Science Exploration Zone and the Life Practice Zone. One hundred thousand trees and one million two hundred thousand flowers and shrubs were planted, and one hundred and seventy thousand square metres of lawn were laid.

4. Measure each angle below and say what kind of angle it is.

5. Draw the following angles.

85° 40° 140° 170°

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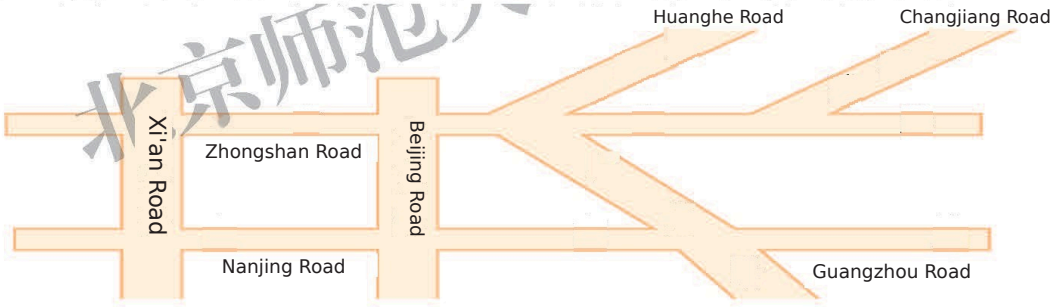
6. Which roads in the map below are parallel to each other? Which are perpendicular?



7.

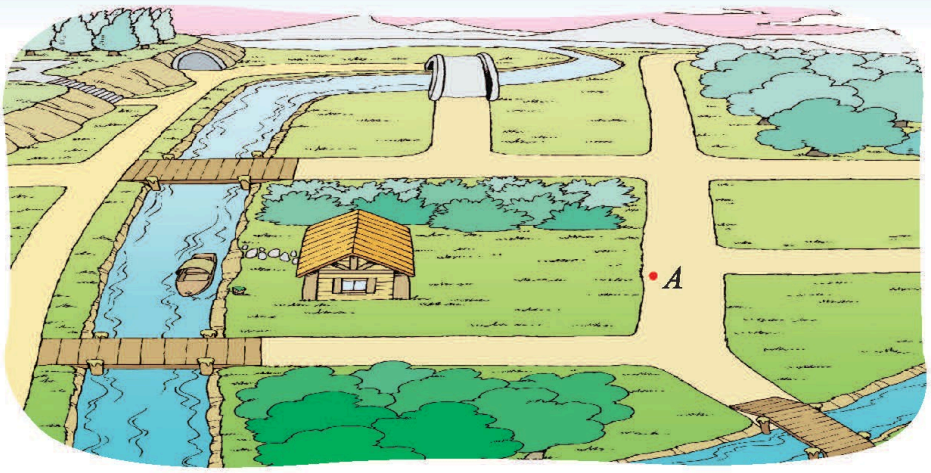
305 × 52	624 × 78	46 × 589	229 × 80
508 × 35	74 × 120	201 × 24	27 × 315

8. The school plans to buy 12 televisions and 40 computers and has ¥250000 in total. Is that enough?

¥1900

¥4800

9.



(1) Find two pairs of parallel or perpendicular line segments in the picture and trace them in colour.

(2) Draw the shortest path from point A to the small house.

10. In the figures below, which two line segments are parallel? Which two are perpendicular?

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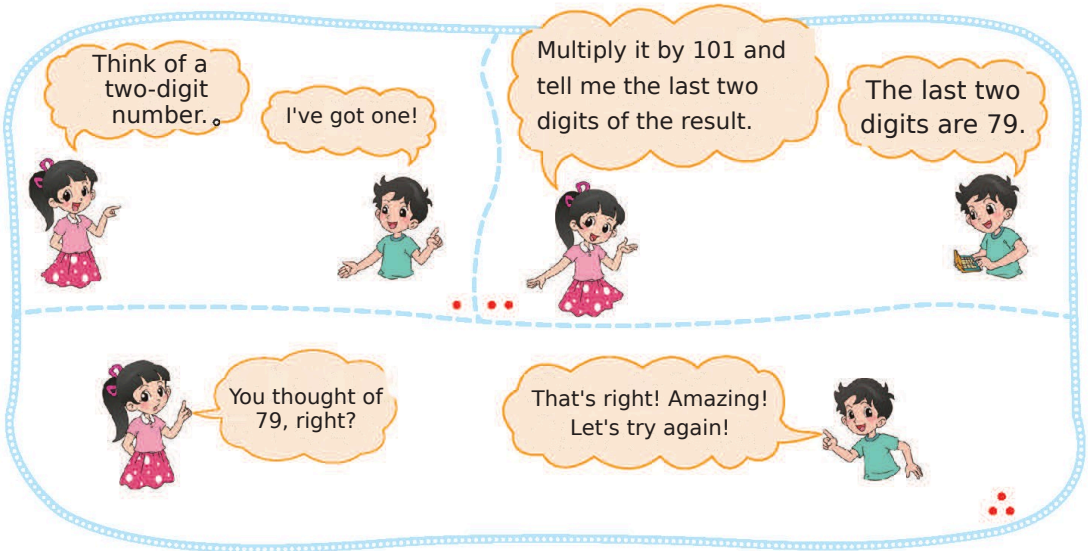
11. Choose a page in your Chinese textbook with a lot of text and estimate how many characters it has.

12. The magical 101.



Unit 4 Laws of Operations

Buying Stationery

How much do 3 calculators and 1 pen cost?

Add the cost of 3 calculators to the cost of 1 pen.

22×3=66 (yuan)

24÷4=6 (yuan)

66+6=72 (yuan)

Is my number sentence correct?

$22\times3+24\div4$

Answer: _____。

How do you calculate Xiaoxiao's number sentence? Discuss your idea with your partner.

$22\times3+24\div4$
$=66+24\div4$
$=66+6$
$=72$ (yuan)

$22\times3+24\div4$
$=66+6$
$=72$ (yuan)

First say the order of operations for each problem below, then calculate.

$35+65\times40\div5$

$12\times(153-83)\div8$

$(96-6)\times(15+9)$

Think: what do you calculate first, and what next? Good habits are important!

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Try It

Can you add brackets to make 9÷3×5-2=1 true?

Round brackets alone won't work!

First 5−2=3, then 3×3=9, finally 9÷9=1.

Use square brackets "[ ]" to help.

$$9 \div [3 \times (5-2)]$$
$$= 9 \div [3 \times 3]$$
$$= 1$$

Work out the ( ) first, then the [ ].

Can you sum up the order of operations in mixed calculations?

With only addition and subtraction, or only multiplication and division...

When there are both addition/ subtraction and multiplication/ division...

If there are brackets, you should...

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Practice

1. Look at "Buying Stationery" on the previous page. Ask two maths questions and try to solve them.

2. How many more kilometres per hour does the minibus travel than the jeep?

I drove 270 km in 3 hours.

I drove 140 km in 2 hours.

3. First say the order of operations for each problem below, then calculate.

$1000-289+11$	$92 \div 4 \times 25$	$28 \times (207 \div 3)$
$176-176 \div 8$	$97-12 \times 6+43$	$68+360 \div (20-16)$

4. Compare and calculate.

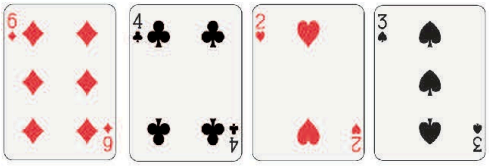
$12 \times (8+4) \div 2$
$12 \times [(8+4) \div 2]$

$234 \div (51-48) \times 3$
$234 \div [(51-48) \times 3]$

5. Game.

(1) How can you use the numbers on the 4 playing cards below to make 24?

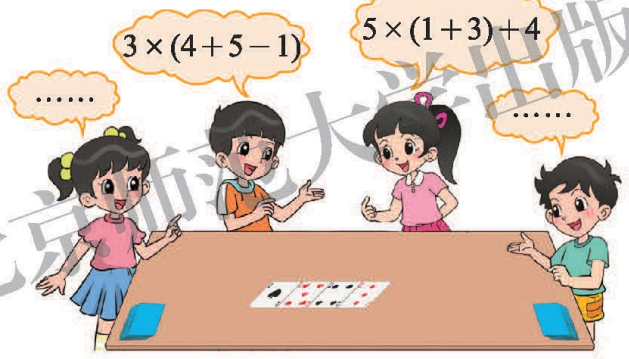
Write at least two ways.



_____

_____

(2) Play with your partner. Each time choose 4 number cards; whoever first works out 24 scores 1 point.



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6.

$[196-(96+4)] \div 6$
$72 \div [(12+24) \div 9]$

$288-[8 \times (26-14)]$
$200 \div [(172-72) \div 25]$

7. At the school sports day, the pupils of Class 4 (1) did very well (see the table below for the number of prize winners). Their class teacher plans to spend ¥350 on prizes (see the picture on page 47 for the prizes and prices). Help the teacher plan what to buy and explain why.

First place	Second place	Third place	Merit award
3 pupils	4 pupils	6 pupils	12 pupils

Commutative Laws of Addition and Multiplication

Look at each set of number sentences below. Write another set like them and say what you notice.

$4 + 6 = 10$
$6 + 4 = 10$
$\longrightarrow 4 + 6 = 6 + 4$

$3 \times 5 = 15$
$5 \times 3 = 15$
$\longrightarrow 3 \times 5 = 5 \times 3$

I find that when two numbers are added...

$62 + 53 = 115$
$53 + 62 = 115$
$\longrightarrow 62 + 53 = 53 + 62$

$7 \times 9 = 63$
$9 \times 7 = 63$
$\longrightarrow 7 \times 9 = 9 \times 7$

I find that when two numbers are multiplied...

Can you explain your discovery using examples from daily life?

The distance from the cinema to the school is the same as from the school to the cinema.

$42 + 35 = 35 + 42$

School — 35 m — Youth Centre — 42 m — Cinema

Across: 6 chairs in each row, 5 rows. Down: 5 chairs in each column, 6 columns. It's really the same.

$6 \times 5 = 5 \times 6$

Use a and b to stand for two numbers. Can you write the rule you found above? Think about it and learn it.

$a + b = b + a$

$a \times b = b \times a$

These two number sentences show the commutative laws of addition and multiplication.

Can you use what you learned today to explain the reasoning behind the calculations below?

$358 + 276 = 634$

$$\begin{array}{r} 358 \\ +\ 276 \\ \hline 634 \end{array}$$

Check:

$$\begin{array}{r} 276 \\ +\ 358 \\ \hline 634 \end{array}$$

$5 \times 107 = 535$

$$\begin{array}{r} 107 \\ \times\ \ 5 \\ \hline 535 \end{array}$$

Practice

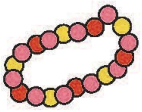
1. Use the example below to explain why the equation is true.

$16+12=12+16$

16 spotted cows

12 brown cows

$20\times6=6\times20$



20 beads on 1 string; make 6 strings.

6 beads on 1 string; make 20 strings.

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2. Use the commutative laws of addition and multiplication to fill in the boxes.

$45+76=\square+45$

$28+13=\square+\square$

$\square+\square=\square+\square$

$45\times102=102\times\square$

$296\times200=\square\times\square$

$\square\times\square=\square\times\square$

3. Calculate, then check using the commutative law of addition or multiplication.

$918+395$ 35×27

4. Do subtraction and division follow the commutative law too? Try some examples.

10−6=4, but I can't work out 6−10, and it's certainly not 4. So...

Associative Law of Addition

Look at the number sentences below. Can you write another set like them? Say what you notice.

$(4+8)+6 = 12+6 = 18$

$4+(8+6) = 4+14 = 18$

$(4+8)+6 = 4+(8+6)$

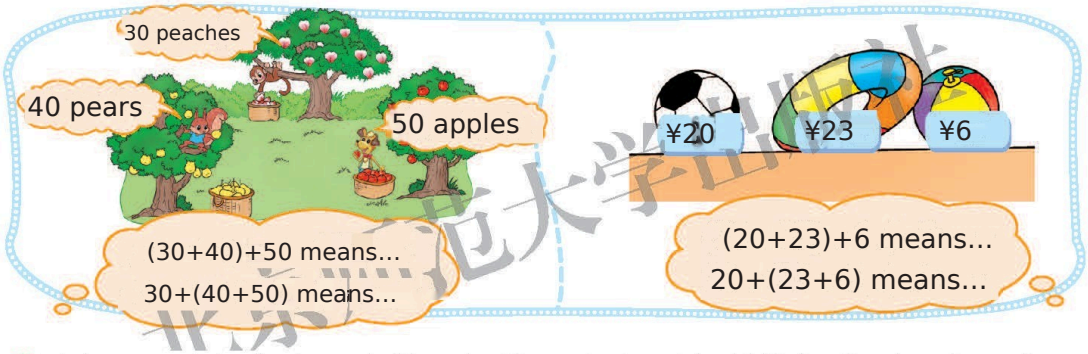
$(19+62)+38 = 81+38 = 119$

$19+(62+38) = 19+100 = 119$

$(19+62)+38 = 19+(62+38)$

When adding three numbers, add the first two first, or the last two first...

Use a real-life example to explain what you found.



Use a, b and c to stand for three numbers. Can you write the rule you found above? Think about it and learn it.

$$(a+b)+c=a+(b+c)$$

This is the associative law of addition.

How can you calculate more simply? Think and work it out.

$57+288+43=\square$

$57+288+43$
$=(57+43)+288$
$=100+288$
$=388$

$57+288+43$
$=288+(57+43)$
$=288+100$
$=388$

The commutative and associative laws of addition can make some additions easier.

Practice

1. Use the example below to explain why the equation is true.

Shopping list

Gloves	¥10
Apples	¥7
Pears	¥13

$10+7+13=10+(7+13)$

2. Use the commutative and associative laws of addition to fill in the boxes.

$25+13+87=25+(\square+87)$

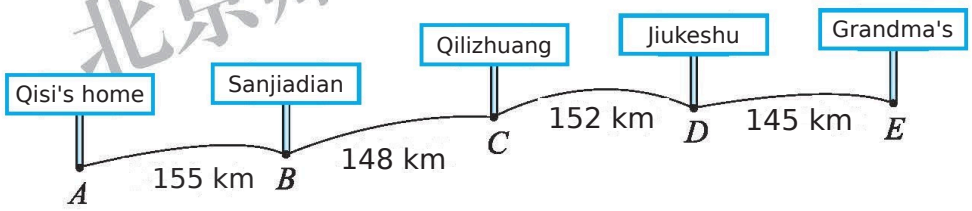
$(51+29)+71=\square+(\square+\square)$

$15+34+85+66=(\square+\square)+(\square+\square)$

3. Look at the addends in each expression and calculate.

$88+156+44$	$28+69+172$	$91+34+109+366$

4. Qisi and his dad take the train from A to E to visit Grandma. How many kilometres does the train travel altogether?



5. Before, when we solved a problem like the one on the right, we could use two different methods, so we know that:

$100-24-36=100-(24+36)$

(1) Follow the example and write two more pairs of number sentences like this.

(2) Try to describe this rule in words or letters.

Mum took ¥100. She spent ¥24 on fish and ¥36 on vegetables. How much does she have left?

$100-24-36$
$=76-36$
$=40$ (yuan)

$100-(24+36)$
$=100-60$
$=40$ (yuan)

Answer: ¥40 is left.

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Associative Law of Multiplication

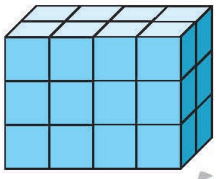
Look at the number sentences below. Can you write another set like them? Say what you notice.

$(2\times4)\times3$ $2\times(4\times3)$
$=8\times3$ $=2\times12$
$=24$ $=24$
$(2\times4)\times3=2\times(4\times3)$

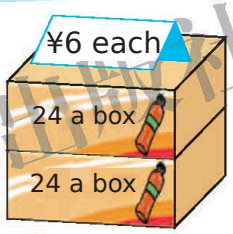
$(7\times4)\times25$ $7\times(4\times25)$
$=28\times25$ $=7\times100$
$=700$ $=700$
$(7\times4)\times25=7\times(4\times25)$

To multiply three numbers, first multiply the first two...

Use a real-life example to explain what you found.



(2×4)×3 means...
2×(4×3) means...



(2×24)×6 means...
2×(24×6) means...

Use a, b and c to stand for three numbers. Can you write the rule you found above? Think about it and learn it.

$(a\times b)\times c=a\times(b\times c)$

This is the associative law.

How can you calculate more simply? Think and work it out.

$125\times9\times8=\square$

First look at the signs and numbers in the number sentence.

You can calculate 125×8 first.

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Practice

1. Use the example below to explain why the equation is true.

$$(40 \times 3) \times 6 = 40 \times (3 \times 6)$$

2. Use the commutative and associative laws of multiplication to fill in the blanks.

$35 \times 2 \times 5 = 35 \times (\square \times \square)$

$(25 \times 60) \times 4 = (\square \times \square) \times \square$

$125 \times 4 \times 25 \times 8 = (\square \times \square) \times (\square \times \square)$

3. Look closely at the number sentences below, then calculate.

$38 \times 25 \times 4$	$125 \times 3 \times 8$	$(13 \times 5) \times 6$

4.

5. This is how Taoqi calculated 24×25.

24 × 25
= 6 × 4 × 25
= 6 × (4 × 25)
= 6 × 100
= 600

(1) Can you understand it? Share your ideas with a partner.

(2) Try to use the commutative and associative laws of multiplication to calculate the following.

64×125	$125 \times 25 \times 32$

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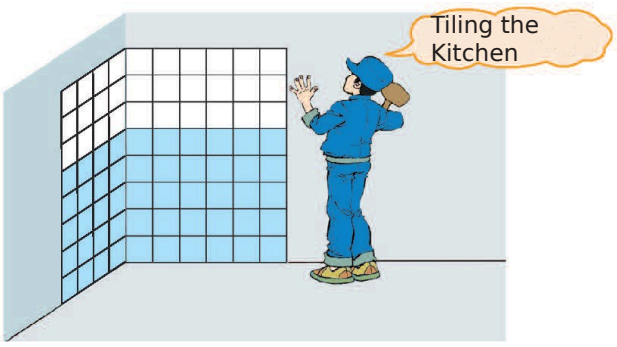
Distributive Law



How many tiles were laid? Explain how you calculated it.

$3\times10+5\times10$ $=30+50$ =80 (tiles)	$(3+5)\times10$ $=8\times10$ =80 (tiles)

$4\times8+6\times8$ $=32+48$ =80 (tiles)	$(4+6)\times8$ $=10\times8$ =80 (tiles)

Answer: _____

Look at the two groups of number sentences above. What do you notice?

I found:
$3\times10+5\times10=(3+5)\times10$。

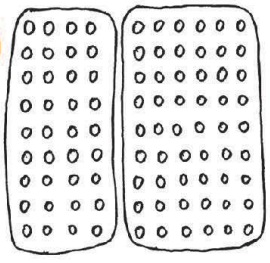
I found:
$4\times8+6\times8=(4+6)\times8$。

Use a, b and c to stand for three numbers. Can you write the rule you found above? Think about it and learn it.

$(a+b)\times c=a\times c+b\times c$

This is the distributive law.

Use $4\times9+6\times9$ to explain why the distributive law is true.



$4\times9+6\times9$

4 nines 6 nines

10 nines in all.

That is $(4+6)\times9$.

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Try It

Look closely at (80+4)×25, then calculate.

You can work out 80×25, then 4×25...

$$(80+4)\times25$$
$$=80\times25+4\times25$$
$$=2000+100$$
$$=2100$$

I use the distributive law.

Look closely at 34×72+34×28, then calculate.

Woof! Both multiplications have 34.

It means 72 groups of 34 plus 28 groups of 34, 100 groups of 34 in total.

$$34\times72+34\times28$$
$$=34\times(72+28)$$
$$=34\times100$$
$$=3400$$

I use the distributive law.

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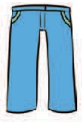
Practice

1. The school is buying outfits for 28 choir members.

¥46



¥54

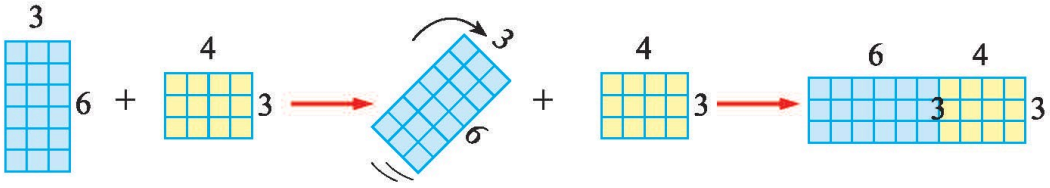
(1) Below are Taoqi's and Xiaoxiao's number sentences. Tell your partner how they were thinking.

$(46+54)\times28$

$46\times28+54\times28$

(2) Work out how much the outfits cost in total.

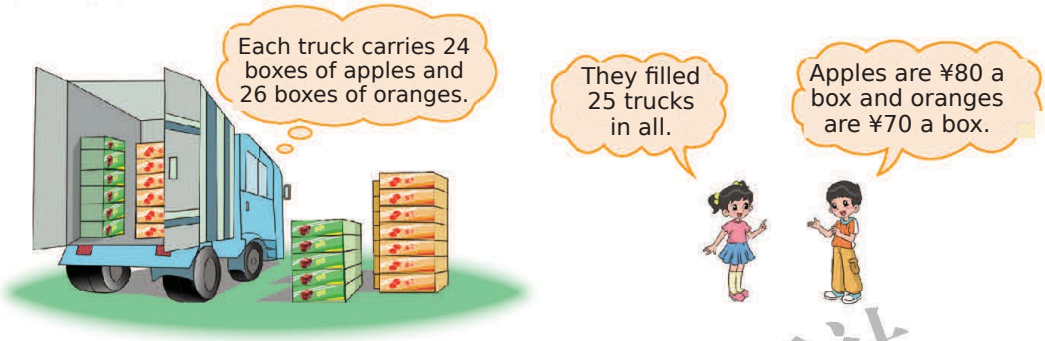
2. Use the diagram to tell your partner why 3×6+4×3=(6+4)×3 is true.



3. Look closely at the number sentences below, then calculate.

$(20+4)\times 25$	$35\times 37+65\times 37$	$19\times 66+81\times 66$
$32\times(200+3)$	$12\times 26+12\times 4$	$125\times(80+8)$
$12\times 65+65\times 88$	$8\times(125+9)$	$39\times 6+39\times 54$

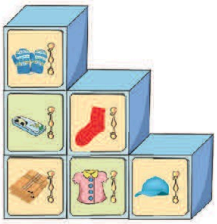
4. The fruit harvest is in.



(1) How many boxes of fruit are there in all?

(2) How much will all this fruit sell for in total?

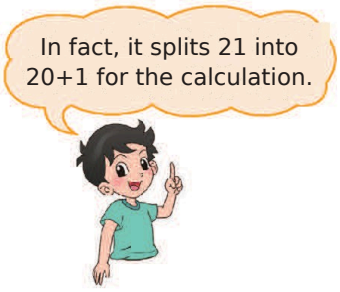
5. Mum ordered Taoqi a set of small cabinets that can be put together in any way. Each small cabinet costs ¥18 and each sticker on a cabinet door costs ¥2. Work out how much the whole set of cabinets costs.



6. We often multiply multi-digit numbers in columns.

(1) Can you use the distributive law to explain why it works?

$$\begin{array}{r} 26 \\ \times\ 21 \\ \hline 26 \\ 52 \\ \hline 546 \end{array}$$



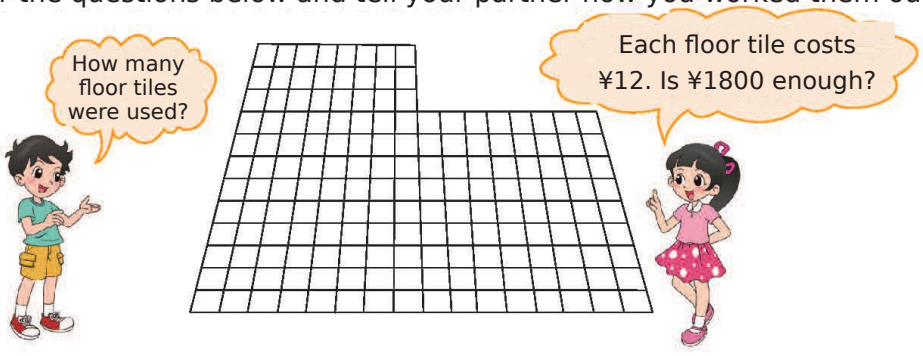
$$\begin{aligned} &26\times 21 \\ &=26\times(1+20) \\ &=26\times 1+26\times 20 \\ &=26+520 \\ &=546 \end{aligned}$$

(2) Try to use the distributive law to calculate the following.

58×11	47×102

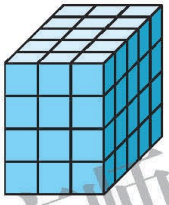
Exercise 4

1. Answer the questions below and tell your partner how you worked them out.



2. Use the example below to explain why the equation holds.

$(3\times5)\times4=3\times(5\times4)$



$3\times8+3\times6=3\times(8+6)$

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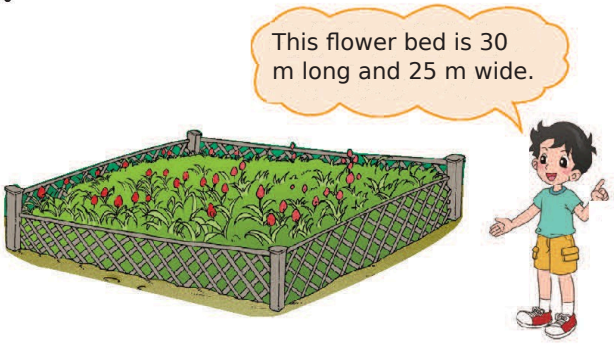
3.

$85\times82+82\times15$	$5\times289\times2$	$(125\times25)\times4$
$(125+17)\times8$	$25\times97+25\times3$	$378+527+73$
$167+289+33$	$58+39+42+61$	$76\times25+25\times24$

4.



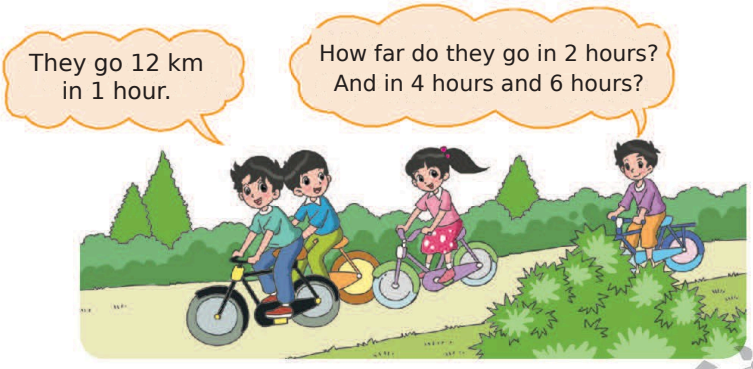
(1) How long is the fence around this flower bed?

(2) If about 40 tulips are planted per square metre, about how many tulips are planted in this flower bed?

5. Match.

$75 \times 2 \times 5$	$38 \times (29 + 1)$
$58 + 147 + 42$	$100 \times 25 + 4 \times 25$
$38 \times 29 + 38$	$58 + 42 + 147$
$(100 + 4) \times 25$	$75 \times (2 \times 5)$

6.



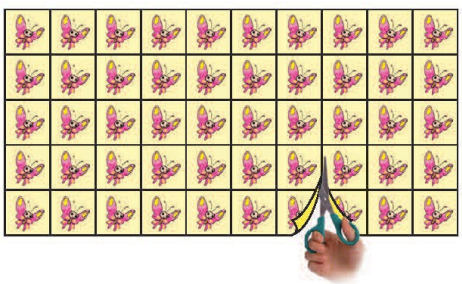
(1) Write number sentences for the questions above. Look at how the multiplier changes in each one. Can you find a pattern in how the product changes?

(2) Using your rule, can you write the answers below directly?

$150 \times 20 = 3000$	$360 \times 24 = 8640$
$150 \times 40 =$	$360 \times 12 =$
$150 \times 60 =$	$360 \times 6 =$

7. (1) $5 \times 10 - 5 \times 3 = 5 \times (10 - 3)$

Is it true? Find a way to check.



(2) Can you understand the reasoning below?

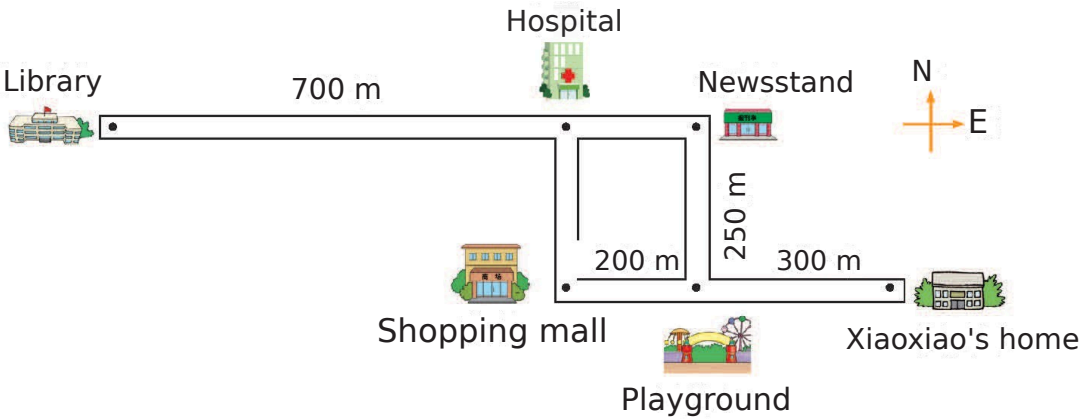
$$\begin{aligned} & 13 \times 99 \\ = & 13 \times (100 - 1) \\ = & 13 \times 100 - 13 \times 1 \\ = & 1287 \end{aligned}$$

(3) Can you make these calculations simpler?

$206 \times 14 - 6 \times 14$ $\qquad$ 72×99

Unit 5 Direction and Position

To the Library



Xiaoxiao goes from home to the library. How can she describe her route clearly?

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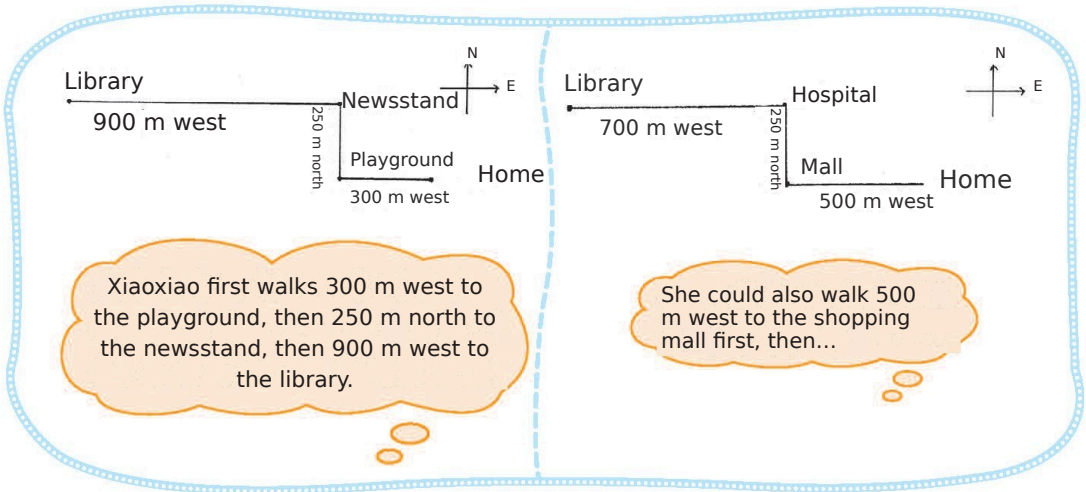
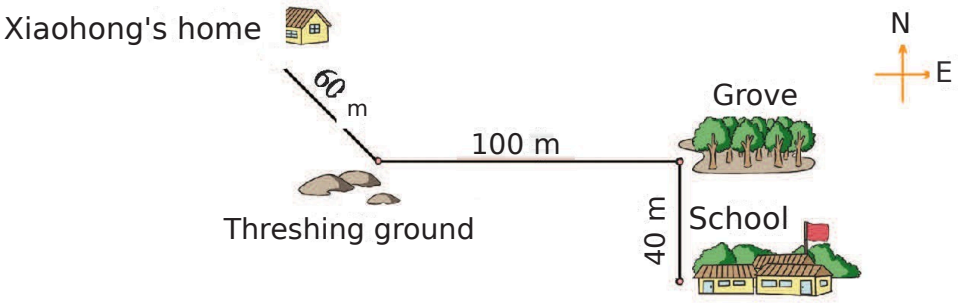
Draw and then describe Xiaoxiao's route from home to the library.



How does Xiaoxiao get home from the library? Draw it and describe it.

Practice

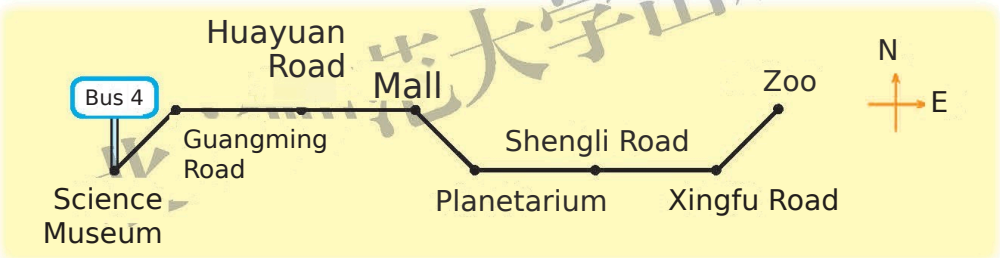
1. Describe Xiaohong's route to school and back, then fill in the blanks.



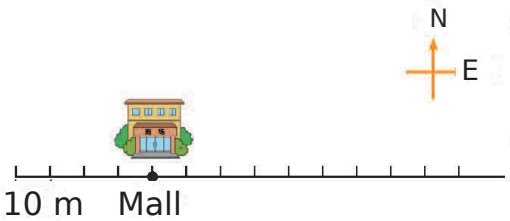
From home, Xiaohong goes ____ for ____ m to the threshing ground, then goes ____ for ____ m to the grove, and then goes ____ for ____ m to school.

From school she goes ____ for ____ m to ____, then goes ____ for ____ m to ____, and then goes ____ for ____ m back home.

2. Describe bus 4's round trip from its first stop to its last stop.



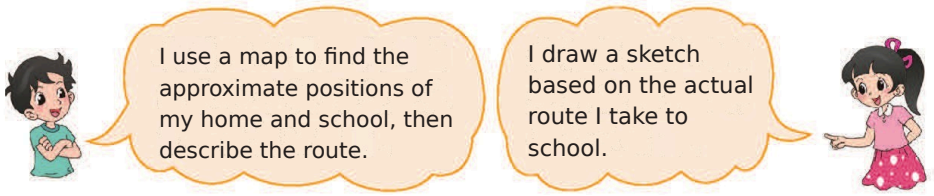
3. Draw as required.



(1) There is a bookshop 60 m east of the mall. Mark its position with △.

(2) There is a fitness square 50 m north of the bookshop. Mark its position with ☆.

4. Hands-on activity. Can you describe the route from your home to school?



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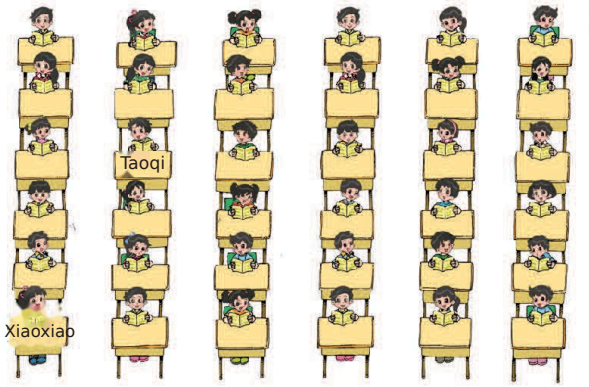
Position

This is the seating chart of Taoqi's class.

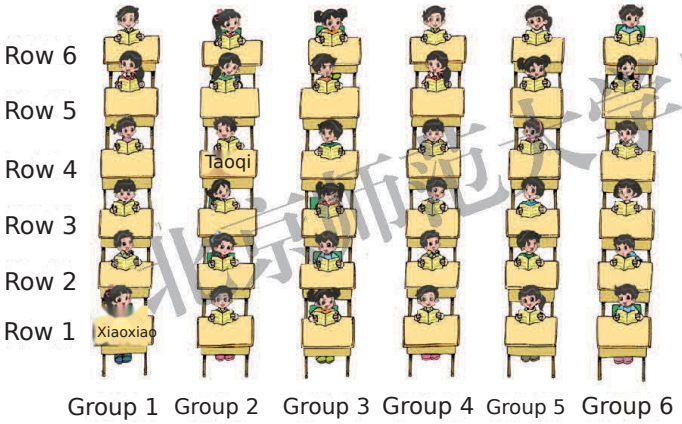


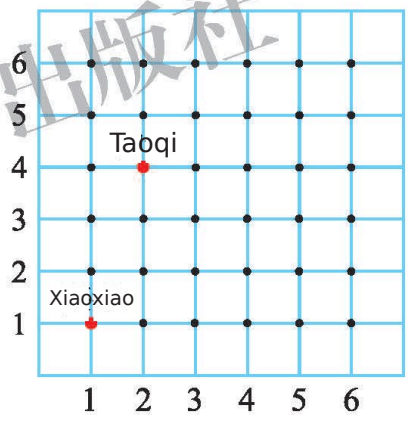
Discuss: which seat is Taoqi in? What about Xiaoxiao?

Taoqi sits in Group 2, 3rd from the back.

Xiaoxiao is in Group 1, 1st from the front.

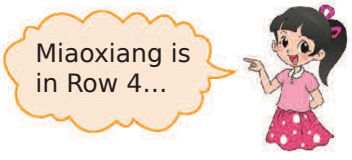
Look and say.





Taoqi is in Group 2, Row 4. This can be written as the number pair (2, 4). How can Xiaoxiao's position be written?

Qisi's and Miaoxiang's positions are written as the number pairs (4, 3) and (1, 4). Can you find their positions in the diagram? Say which group and which row they sit in.



Which group and which row do your 3 good friends sit in? Try to write their positions as number pairs.

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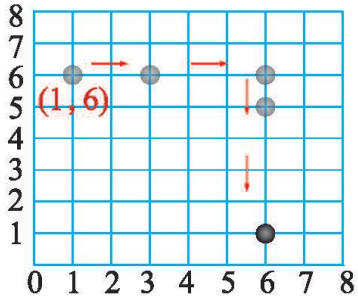
Try It

Use number pairs to write each position of the game piece as it moves.



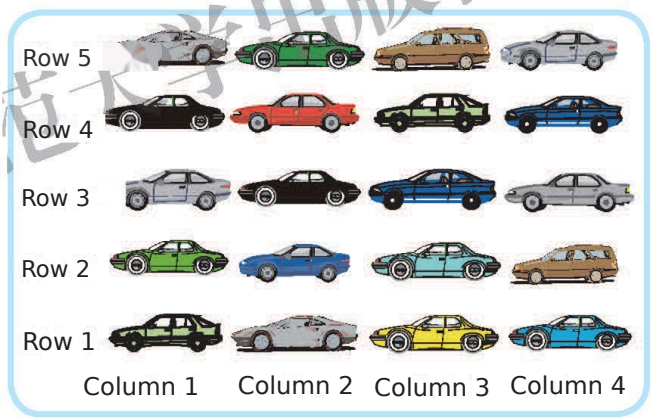
Observe how the number pair changes as the piece moves. What do you find?

The number pairs (1, 6), (3, 6) and (6, 6) are all in Row 6.

When moving vertically, the 1st number in the number pair does not change...

Practice

1. Find and fill in.

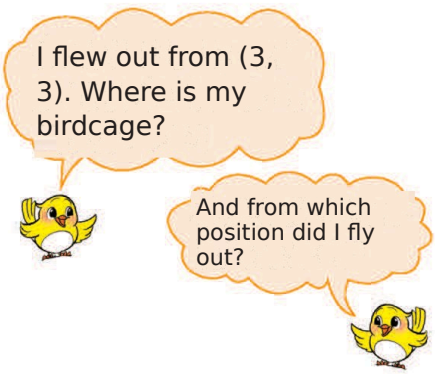


(1) The red car is in column ( ) and row ( ). Its number pair is (,).

(2) The yellow car is in column ( ) and row ( ). Its number pair is (,).

(3) Qisi's family car is at (4, 3) and Miaoxiang's family car is at (2, 2). Mark them on the diagram.

2. Two birds fly out of two birdcages.



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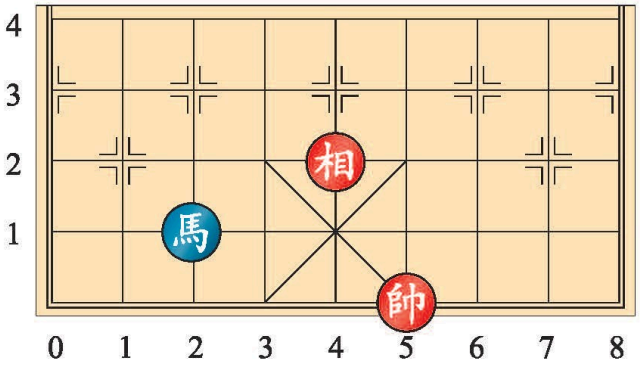
3. “馬” Where can it move next? Write the positions as number pairs.

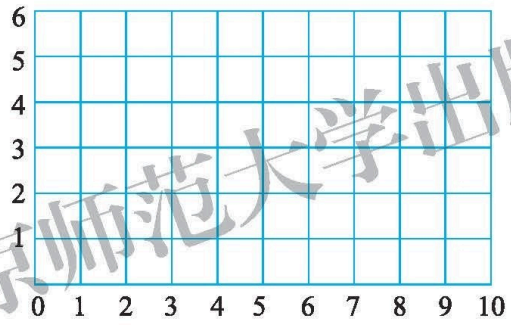
“相” What about it?



“馬” goes in an L

“相” goes 2 points diagonally

4. Draw a rectangle in the grid below, and use number pairs to express the positions of its four vertices. Think: what do you find?



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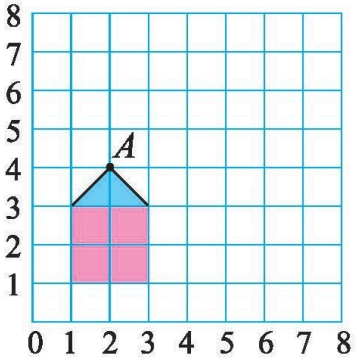
5. Maths game: find the secret point.

Two deskmates play a game on grid paper. One thinks of a secret point and writes down its position; the other starts guessing where it is. If a guess is wrong, the first player says how many squares off it is. Guessing it correctly within three tries (including the third) wins; otherwise you lose.

6. On the grid, draw the little house after it moves up 4 squares and then right 5 squares, and write the new position of its vertex A as a number pair.



Unit 6 Division

Buying Stationery

¥20

¥10

¥30

¥8

How many schoolbags can you buy with ¥80?

□ ÷ □ = □ ( )

8 ¥10 notes; every 2 notes buy 1 schoolbag, so I can buy...

$8 \div 2 = 4$

$80 \div 20 = 4$

$20 \times 4 = 80$

$80 \div 20 = 4$

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Answer: _____。

Can you use the column method?

Why is "4" written in the ones place?

$$\begin{array}{r} 4 \\ 20\overline{)80} \\ \underline{80} \\ 0 \end{array}$$

$4 \times 20 = 80$

How many calculators can you buy with ¥160? How much money is left?

□ ○ □ = □ ( ) ····· □ ( )

$$\begin{array}{r} 5 \\ 30\overline{)160} \\ \underline{150} \\ 10 \end{array}$$

5×30=150, less than 160
6×30=180, more than 160,
so the quotient should be 5.

Answer: _____。

Try It

Can you work out 600÷30? Explain your thinking.

$20 \times 30 = 600$
$600 \div 30 = 20$

$60 \div 30 = 2$
$600 \div 30 = 20$

$60 \div 3 = 20$
$600 \div 30 = 20$

Use the column method to calculate 520÷40 and explain how you did it.

$40\overline{)520}$

When the divisor is a two-digit number, first look at the first two digits of the dividend...

$$\begin{array}{r} 1 \\ 40\overline{)520} \\ 40 \\ \hline 12 \end{array}$$

Why is "1" written in the tens place?

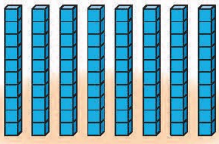
$$\begin{array}{r} 13 \\ 40\overline{)520} \\ 40 \\ \hline 120 \\ 120 \\ \hline 0 \end{array}$$

Why is "3" written in the ones place?

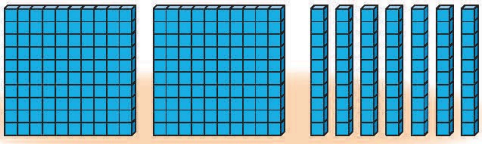
Practice

1. Circle and calculate.

$80 \div 40$



$270 \div 30$



2.

$560 \div 70$	$720 \div 90$	$360 \div 40$	$240 \div 30$
$640 \div 80$	$200 \div 50$	$180 \div 60$	$280 \div 70$

3. What is the largest number that fits in the brackets?

$30 \times (\quad) < 220$	$40 \times (\quad) < 140$	$60 \times (\quad) < 200$
$50 \times (\quad) < 85$	$60 \times (\quad) < 306$	$80 \times (\quad) < 314$

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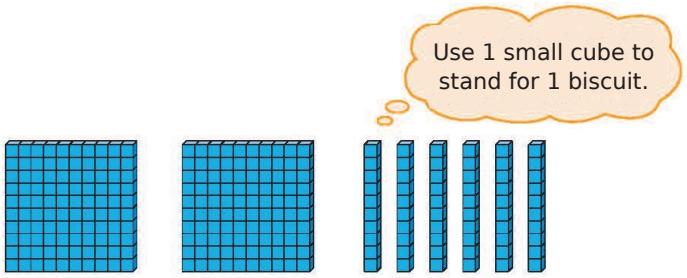
4. Calculate and explain how you did it.

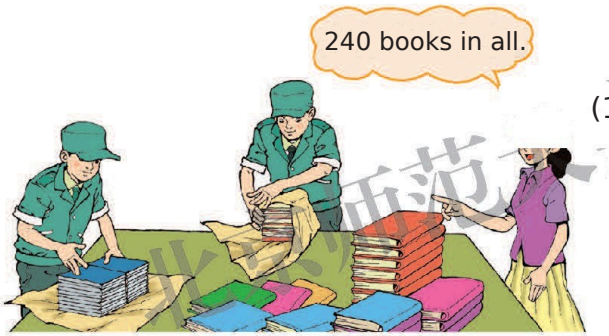
$900 \div 30$ $800 \div 20$ $400 \div 40$ $600 \div 20$

5. 260 biscuits are shared equally among 20 children. How many biscuits does each child get? Use the sharing of the biscuits to explain what each step of the column method means.

$20\overline{)260}$



6. Bundling books.



(1) How many bundles of 40 books?

(2) How many bundles of 50 books? How many books are left?

7. First say how many digits the quotient has, then use the column method.

$928 \div 50$	$168 \div 20$	$780 \div 60$	$460 \div 80$
$297 \div 30$	$570 \div 40$	$620 \div 70$	$350 \div 90$

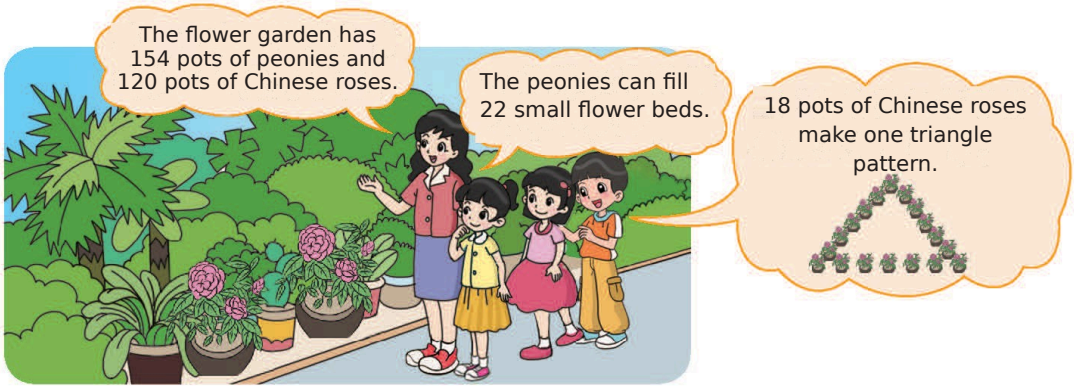
8. Taoqi is reading a 281-page storybook. He closed it by accident, but he remembers that the page numbers of the two consecutive pages he just read add up to 81.

(1) What are the page numbers of the two pages Taoqi just finished reading?

(2) How many pages of this storybook are left unread?

(3) If Taoqi reads 20 pages every day, how many days will it take to finish the remaining pages?

Visiting the Flower Garden



On average, how many pots of peonies are in each small flower bed? Estimate first, then use the column method.

$154 \div 22 = \square$ ( )

"15" is less than "22", so the quotient goes in the ones place and has one digit.

Treat "22" as "20"

$20 \times 7 = 140$,

$20 \times 8 = 160$,

It might be 7 pots.

Think of "22" as "20" to estimate.

$$\begin{array}{r} 7 \\ 22\overline{)154} \\ 154 \\ \hline 0 \end{array}$$

Answer: _____。

How many triangle patterns can the Chinese roses make? How many pots are left? Estimate first, then use the column method.

$120 \div 18 = \square$ ( ) ····· $\square$ ( )

The quotient has one digit!

Treat "18" as "20"

$20 \times 6 = 120$,

It might be 6 patterns.

Think of "18" as "20" to estimate.

$$\begin{array}{r} 6 \\ 18\overline{)120} \\ 108 \\ \hline 12 \end{array}$$

Answer: _____。

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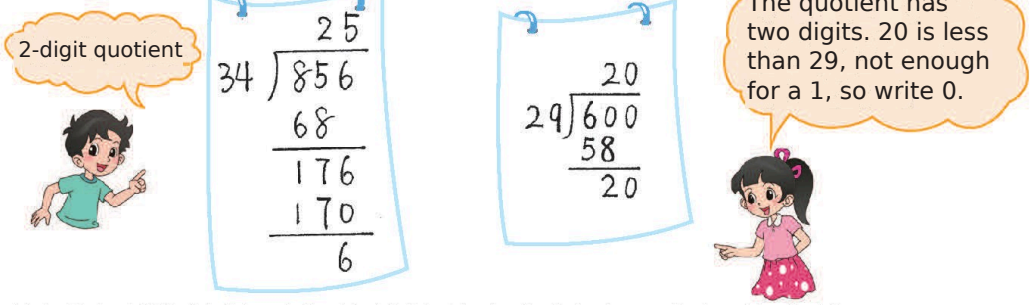
Try It

Think and calculate.

$856 \div 34 = \square \cdots\cdots \square$ $600 \div 29 = \square \cdots\cdots \square$



Use an example to show how to try a quotient when the divisor has two digits, and share with your partner

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Practice

1. First say which ten you round the divisor to when trying the quotient, then calculate.

$31\overline{)70}$ $28\overline{)159}$ $54\overline{)381}$ $47\overline{)351}$

2. Using 177÷36, tell your partner how you try the quotient and calculate.

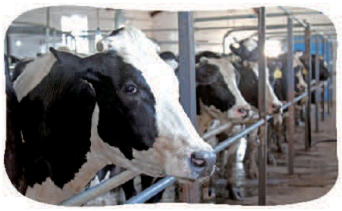
3. At the start of the new term, the school took 216 books from the library to add to the "Little Book Corner" in each of its 24 classes. On average, how many books can each class get?

Book Corner

4. Estimate how many digits each quotient has, tell your partner why, then calculate.

$550 \div 22$	$438 \div 21$	$756 \div 54$
$989 \div 43$	$961 \div 37$	$700 \div 57$

5. A dairy farm has 38 cows. One day in August (month 8) they gave about 800 kg of milk. Do you agree with Taoqi?



That day each cow gave about 20 kg of milk on average.

6. Forest Doctor.

$$\begin{array}{r} 6 \\ 23\overline{)138} \\ 128 \\ \hline 10 \end{array} \qquad \begin{array}{r} 6 \\ 30\overline{)180} \\ 180 \\ \hline 0 \end{array} \qquad \begin{array}{r} 2 \\ 27\overline{)640} \\ 54 \\ \hline 10 \end{array}$$

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7. ☐ Which digits in the box give a one-digit quotient? Which give a two-digit one?

$43\overline{)☐36}$ $54\overline{)☐28}$ $7☐\overline{)711}$ $☐5\overline{)567}$

8. Teacher Li took ¥2000 to buy sports equipment.

¥35

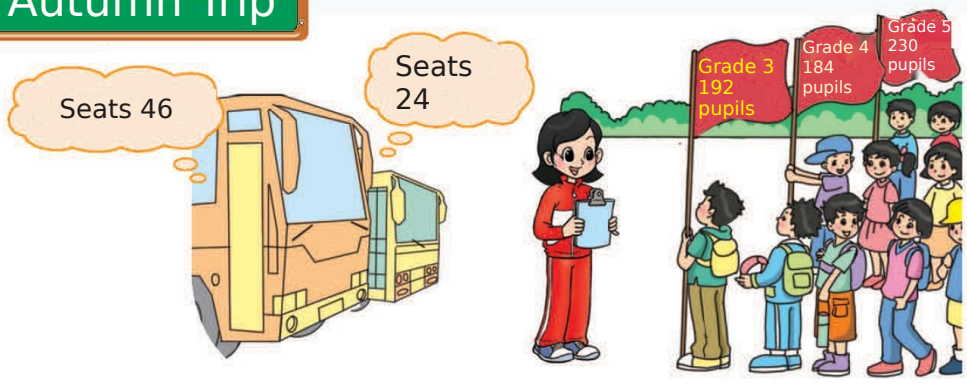
¥13

¥5

(1) Teacher Li bought 10 balls and 200 shuttlecocks. How much did that cost?

(2) How many skipping ropes can be bought with the money left?

Autumn Trip



All Grade 3 pupils take small buses. How many are needed? Estimate, then use the column method.

$192 \div 24 = \square$ ( )

$24 \times 10 = 240$, so fewer than 10 buses

Treat 24 as 20, and the quotient is 9. But...

A quotient of 9 is too big; change it to 8.

$$\begin{array}{r} 9 \\ 24\overline{)192} \\ 216 \end{array} \qquad \begin{array}{r} 8 \\ 24\overline{)192} \\ 192 \\ \hline 0 \end{array}$$

Answer: _____

All Grade 4 pupils go in large buses. How many buses are needed?

$184 \div 46 = \square$ ( )

Treat 46 as 50, $50 \times 3 = 150$, so try 3...

$$\begin{array}{r} 3 \\ 46\overline{)184} \\ 138 \\ \hline 46 \end{array} \qquad \begin{array}{r} 4 \\ 46\overline{)184} \\ 184 \\ \hline 0 \end{array}$$

Your remainder is the same as the divisor, so the quotient 3 is too small. It should be 4.

Answer: _____

What puzzled you when trying a quotient? How did you solve it? Discuss with your partner.

When I try a quotient the way we learned, it is sometimes too big or too small

If the quotient is too big, make it smaller; if it is too small, make it bigger.

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Try It

Below is how Taoqi worked out 624÷13. Look and discuss.

$$\begin{array}{r} 10 \\ \uparrow \\ 13\overline{)624} \\ 65 \end{array}$$

$$\begin{array}{r} 4 \\ 13\overline{)624} \\ 52 \\ \hline 104 \end{array}$$

$$\begin{array}{r} 10 \\ \uparrow \\ 49 \\ 13\overline{)624} \\ 52 \\ \hline 104 \\ 117 \end{array}$$

$$\begin{array}{r} 48 \\ 13\overline{)624} \\ 52 \\ \hline 104 \\ 104 \\ \hline 0 \end{array}$$

Oh no, the quotient was too big both times!

When trying a quotient, you often can't get it right first time. Just adjust it as you go!

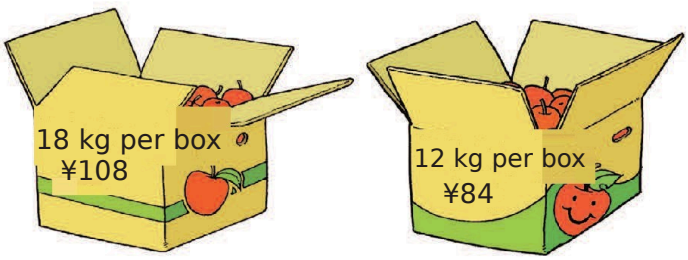
Taoqi asks, "When trying a quotient, when might it be too small? When might it be too big?" First discuss in groups, then use examples to explain your ideas.

Practice

1. Fill in each ◯ with >, < or = without calculating.

263 ÷ 27 ◯ 10　　602 ÷ 31 ◯ 20　　487 ÷ 18 ◯ 30

2. Which apples are cheaper?



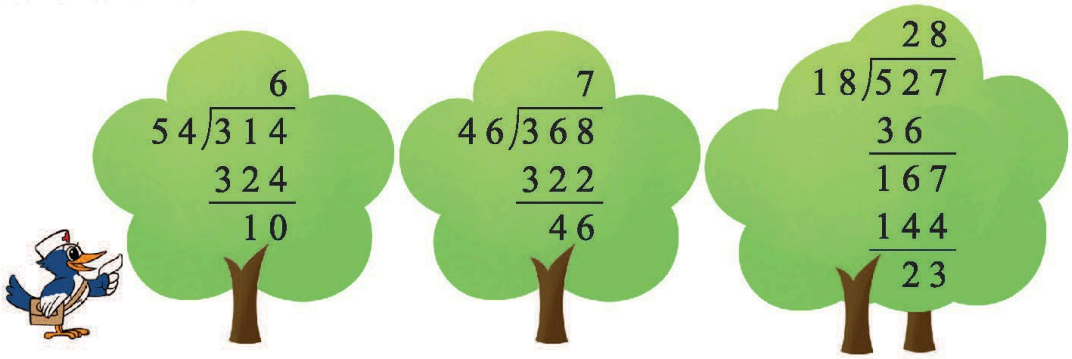
3.

361 ÷ 54	448 ÷ 89	342 ÷ 43	216 ÷ 27

4. Estimate first, then calculate, and tell your partner how you worked it out.

$492 \div 63$	$233 \div 34$	$351 \div 58$
$840 \div 35$	$630 \div 31$	$961 \div 19$

5. Forest Doctor.



6.

(1) How old is the teacher?

(2) Xiaoming is 13. How many times as old as Xiaoming is the ginkgo tree?

(3) The museum is 8 m tall. How many times as tall as the museum is the tree?

7. The magazine I Am a Little Host was launched in August 2012 (month 8). The table shows how many emails its editors received from readers in the first 4 months.

Month	8	9	10	11
Emails	89	713	868	979

(1) Estimate: about how many emails a day were received on average in month 9?

(2) Calculate: how many emails a day were received on average in month 10?

(3) How many times as many emails were received in month 11 as in month 8?

(4) Ask another maths question and try to answer it.

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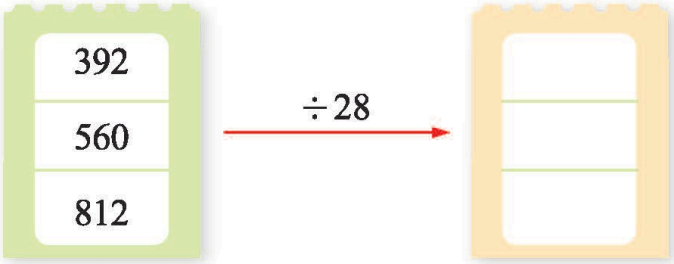
Exercise 5

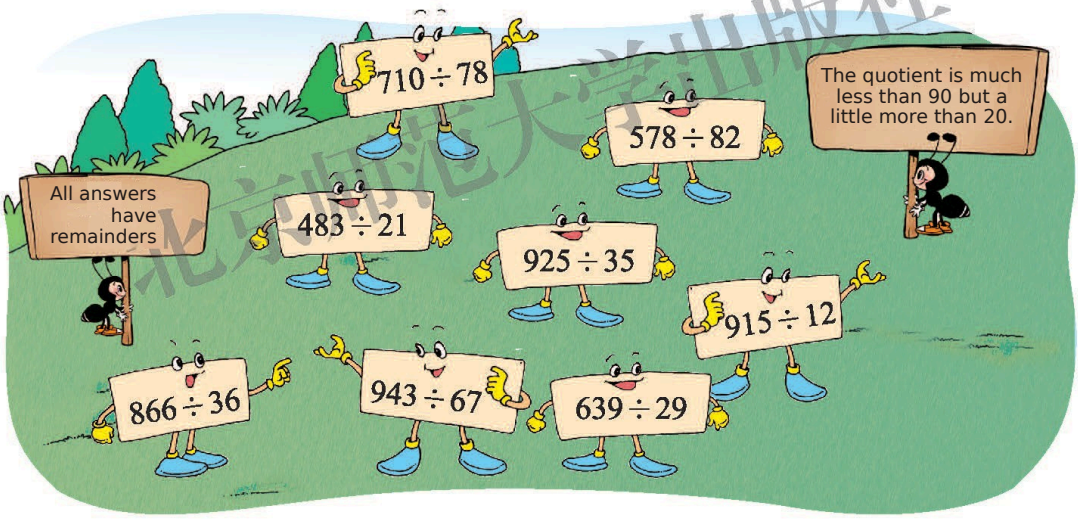
1. First estimate how many digits the quotient has, then calculate.

$333 \div 37$	$372 \div 45$	$294 \div 29$
$328 \div 42$	$395 \div 56$	$765 \div 74$

2. Fill in the boxes, then tell your partner how you tried the quotients.



3. Find the ones that match each clue, and share with your partner.



4. Can she finish reading it before school starts? Estimate, then calculate.

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5. Estimate which is nearest your age. Mark its ☐ with ✓.

550 months	550 hours	550 weeks	550 minutes
☐	☐	☐	☐

6. First estimate, then calculate.

$784 \div 49$	$966 \div 23$	$923 \div 88$
205×21	65×320	$459 + 682$

7. The table below shows how long Taoqi, Xiaoxiao, Qisi and Lele usually practise typing each day, and their results in one test (in the test, each chose their own article). Using this record, ask two maths questions and try to answer them.

		Taoqi	Xiaoxiao	Qisi	Lele
Daily practice time/min		20	60	30	35
Test record	Time/min	12	13	19	18
	Characters	384	728	931	846

8. For a speech contest, Miaoxiang wrote a speech of about 900 characters. Her speech lasts 4 minutes.

(1) If Miaoxiang types 60 characters a minute on average, about how long will it take her to type this speech?

(2) If Miaoxiang wants to give a 10-minute speech, about how many characters long should her speech be?

Unchanging Quotient Rule

Look at the two groups of number sentences below. Can you write another group like them? Say what you notice.

$8 \div 2 = 4$
$80 \div 20 = 4$
$800 \div 200 = 4$

$48 \div 24 = 2$
$24 \div 12 = 2$
$6 \div 3 = 2$

Make sure the quotients in each group are the same.

$100 \div 20 = 5$
$50 \div 10 = 5$
$10 \div 2 = 5$

Taoqi rewrote the three groups of number sentences. Do you agree? Try to state the rule in your own words.

$8 \div 2 = 4$
$(8 \times 10) \div (2 \times 10) = 4$
$(8 \times 100) \div (2 \times 100) = 4$

$48 \div 24 = 2$
$(48 \div 2) \div (24 \div 2) = 2$
$(48 \div 8) \div (24 \div 8) = 2$

$100 \div 20 = 5$
$(100 \div 2) \div (20 \div 2) = 5$
$(100 \div 10) \div (20 \div 10) = 5$

If the dividend and the divisor are both multiplied or divided by the same number (not zero), the quotient stays the same.

This is the unchanging quotient rule.

Can you explain why they calculate 350÷50 this way?

$350 \div 50 = 35 \div 5 = 7$。

Another way:

$$50\overline{)350}\quad 7,\ 35,\ 0$$

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Practice

1. Think about it, then tell your partner why the quotients are all the same.

$9 \div 3 = 3$	$9 \div 3 = 3$
$90 \div 30 = 3$	$27 \div 9 = 3$
$900 \div 300 = 3$	$81 \div 27 = 3$

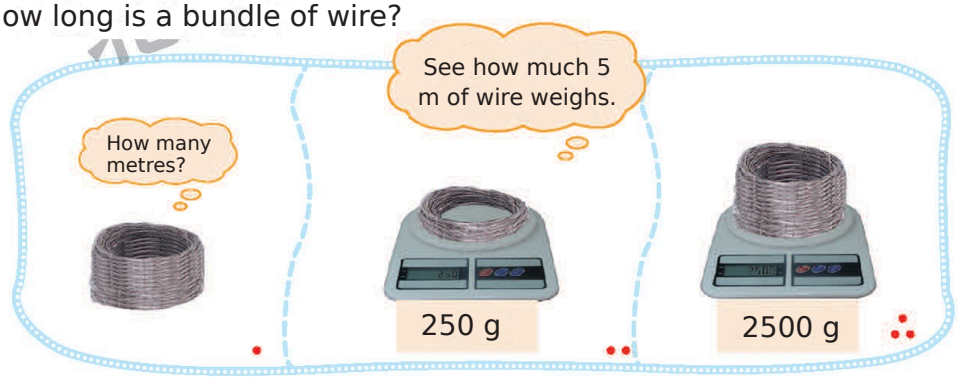
2. Are the calculations below correct? Discuss with your partner.

$$\begin{array}{r} 12 \\ 70\overline{)840} \\ 7 \\ \hline 14 \\ 14 \\ \hline 0 \end{array} \qquad \begin{array}{r} 310 \\ 210\overline{)6510} \\ 63 \\ \hline 21 \\ 21 \\ \hline 0 \end{array} \qquad \begin{array}{r} 12 \\ 800\overline{)9600} \\ 8 \\ \hline 16 \\ 16 \\ \hline 0 \end{array}$$

3. Calculate these, then discuss with your partner.

$240 \div 30$	$80 \div 20$	$360 \div 90$	$4800 \div 400$
$440 \div 20$	$9200 \div 400$	$120 \div 40$	$2400 \div 60$

4. How long is a bundle of wire?



5. Below is how Taoqi worked out $400 \div 25$. Look at each step. What ideas does it give you?

$$400 \div 25 = (400 \times 4) \div (25 \times 4) = 1600 \div 100 = 16$$

Can you calculate the following using this method?

$150 \div 25$ $\qquad$ $2000 \div 125$

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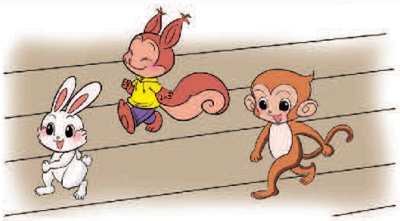
Distance, Time and Speed



Race walking results

	Time/min	Distance/m
Squirrel	4	280
Monkey	4	240
Rabbit	3	240

Guess: who walks the fastest?

The squirrel is faster than the monkey, because they took the same time and the squirrel went farther.

The rabbit is also faster than the monkey, because...

Who is faster, the rabbit or the squirrel?

How can we compare who is faster, the rabbit or the squirrel? Discuss with your partner.

We can work out how far the rabbit and the squirrel each walk in 1 min.

Rabbit per minute: 240 ÷ 3 = 80 (m)

Squirrel per minute: 280 ÷ 4 = 70 (m)

The rabbit is faster.

Let's learn.

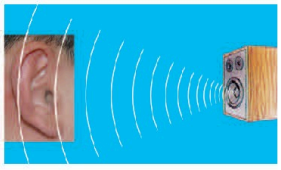
Speed = Distance ÷ Time

On average the rabbit walks 80 m in 1 min, so its speed is 80 m/min. Can you give the speeds of the squirrel and the monkey?

Look and say.

A person walks at about 4 km/h.

A plane flies at about 12 km/min.



Sound travels at about 340 m/s.

Light travels at about 300,000 km/s.

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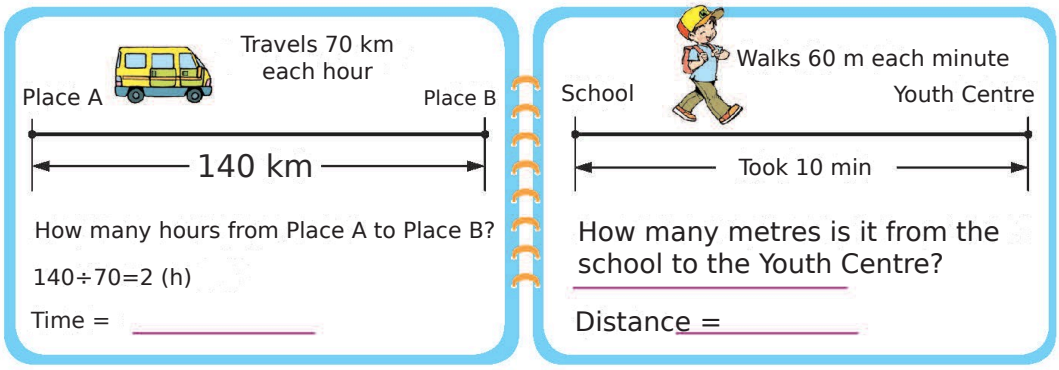
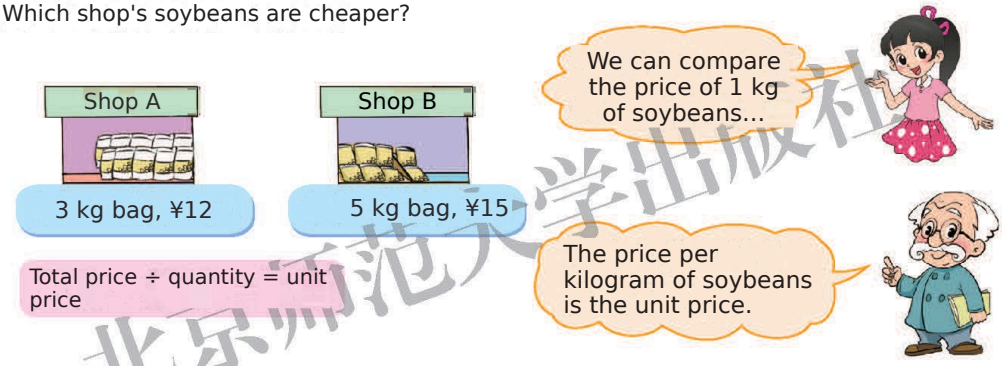
Try It

Think and fill in.



Which shop's soybeans are cheaper?



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Practice

1. Who travels faster?

(1) How do you compare who travels faster?

(2) Calculate who travels faster.

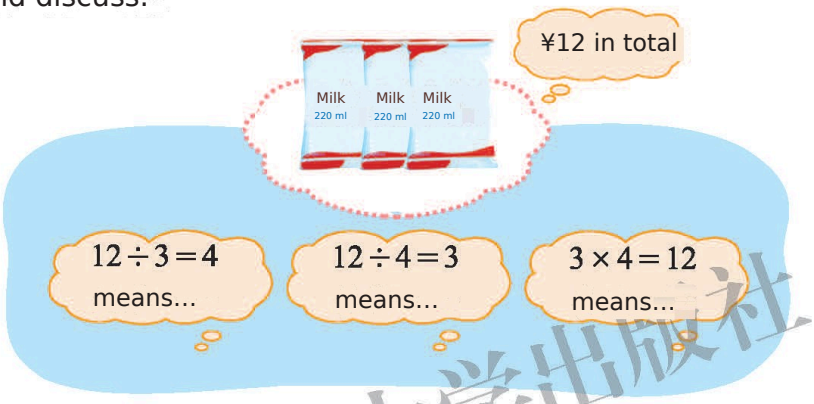
2. Do you know the first cosmic velocity? For a spacecraft to orbit the Earth, it must be launched at a speed of at least 7.9 km/s. This speed is called the first cosmic velocity. A person walks at about 4 km/h. Compare the two speeds. What do you notice?

3.

(1) The coach travels at 50 km/h. How much longer will it take to reach Beijing?

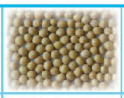
(2) The truck travels at 43 km/h. Can it reach Shijiazhuang in 8 h?

4. Look and discuss.

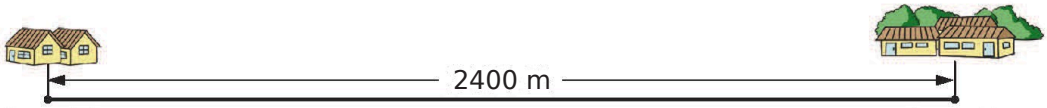


5. Calculate and fill in.

Distance/km	60		600
Time/h	4	9	
Speed/(km/h)		60	200

			
Total price/yuan	150	20	
Quantity/kg	50		6
Unit price/yuan		4	3

6.



Uncle Wang walks from Hexi Village to Hedong Village at 60 m per minute.

(1) About where is he 10 min after setting off? (Mark it with △ above.)

(2) Uncle Wang sets off at 9:15. At what time is he halfway?

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Exercise 6

1. Using the result of the first number sentence, write the other results directly.

$360 \div 30 = 12$ $\quad$ $3600 \div 300 =$ ☐

$36 \div 3 =$ ☐ $\quad$ $3600 \div 30 =$ ☐

2. Estimate, match and calculate.

1-digit quotient

$345 \div 31$

$312 \div 54$

$315 \div 35$

$564 \div 47$

2-digit quotient

3. The table below shows one day's children's book sales at Little Doctor Bookshop.

Title	Price (¥)	Sales (¥)
Children's Comics	12	192
Maths Stories	13	390
Cartoon Stories	28	504
Learning Science	19	494

(1) Which children's book sold best on this day?

(2) If the school spends ¥900 on Cartoon Stories, about how many copies can it buy?

(3) Based on this day's sales, estimate the monthly sales of Children's Comics.

(4) Ask another maths question and try to answer it.

4.

$966 \div 23$ $\quad$ $731 \div 79$ $\quad$ $980 \div 28$

$828 \div 36$ $\quad$ $689 \div 34$ $\quad$ $618 \div 88$

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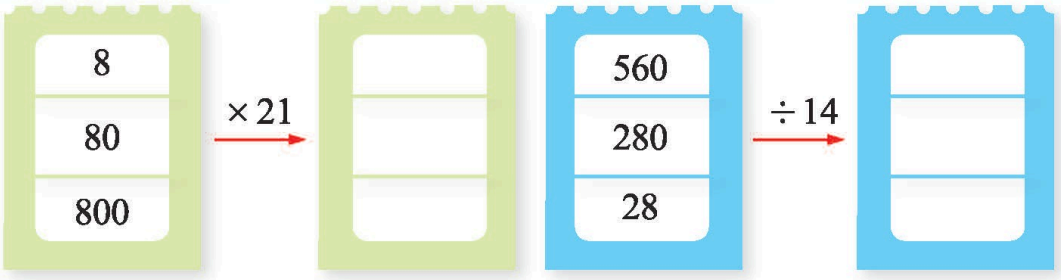
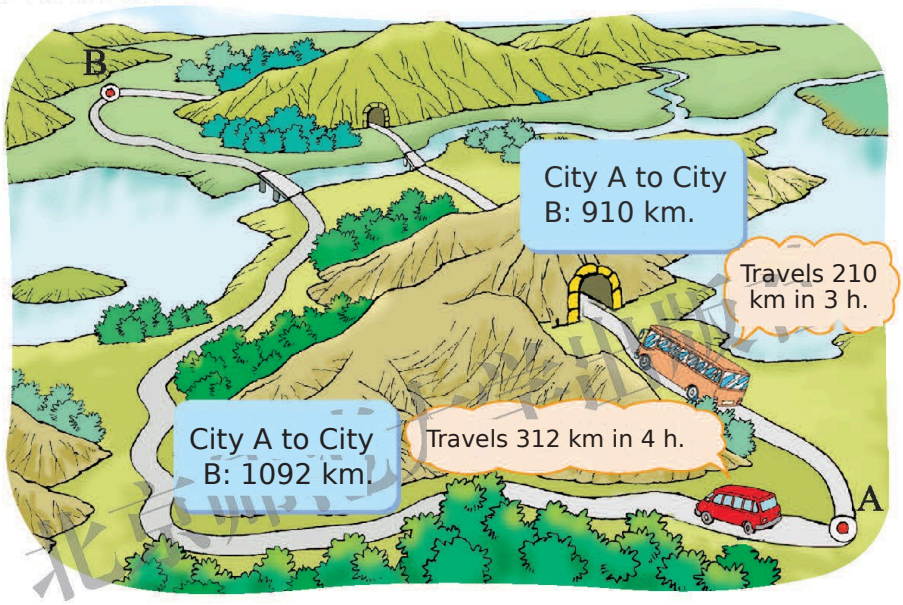
5. Fill in.



6. Who arrives first?



(1) Which travels faster, the coach or the van?

(2) Both leave City A for City B together. The van goes the long way round the mountain; the coach takes the short tunnel route. Which arrives first?

(3) On the actual journey, the coach stopped to rest for 65 min. Work out which vehicle arrives first.

7.

12	1	18
9	6	4
		3

In the square on the left, the product of the three numbers in each row, column and diagonal is equal. What are the two numbers in the blank squares?

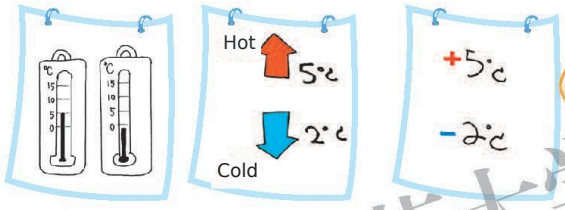
Unit 7 Negative Numbers in Life

Temperature

Beijing: −2℃ to 5℃; Mohe: −17℃ to −4℃; Xi'an: 0℃ to 6℃; Taipei: 17℃ to 24℃; Lhasa: −4℃ to 12℃;

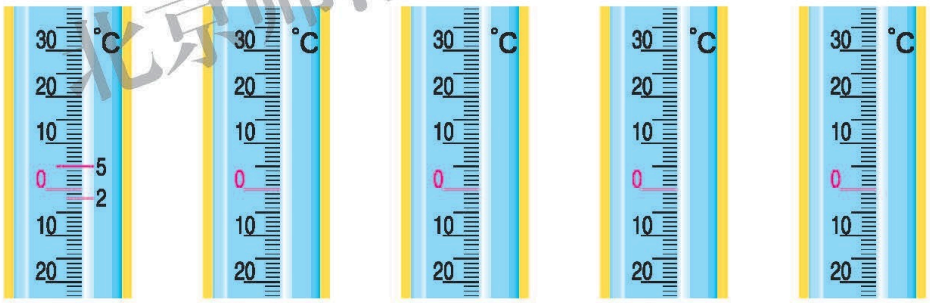
……

Find a way to show the highest and lowest temperatures in Beijing.



We usually write +5 ℃ for 5 ℃ above zero and −2 ℃ for 2 ℃ below zero.

Mark each city's highest and lowest temperatures on the thermometers below.



Beijing　Mohe　Xi'an　Taipei　Lhasa

Fill in and say which temperature is lowest.

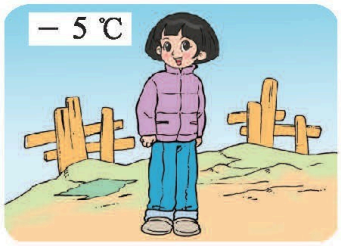
2℃ below zero	4℃ below zero	17℃ below zero
(　　)℃	(　　)℃	(　　)℃

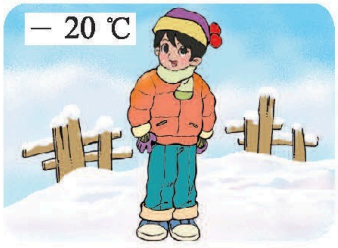
I'll look for them on the thermometer...

−2℃ is 2℃ lower than 0℃...

Practice

1. Discuss: which temperature is lower, −5℃ or −20℃?





2. The map shows the lowest temperatures of some cities on one day. Answer the questions.

(1) Which is warmer, Shanghai or Tianjin?

5 ℃ ◯ −2 ℃

(2) Which is warmer, Tianjin or Qingdao?

(3) Which is warmer, Changchun or Tianjin?

(4) Order the 4 cities' temperatures from highest to lowest. How did you compare them?

3. Find out the temperatures in some parts of China on one day.

_____ Date: _____

	Harbin	Xi'an	Hangzhou	Lhasa	Hong Kong	Taipei
Lowest/℃						
Highest/℃						

(1) Which city has the highest temperature, and which the lowest? What are they?

(2) Arrange the lowest temperatures of the cities from lowest to highest.

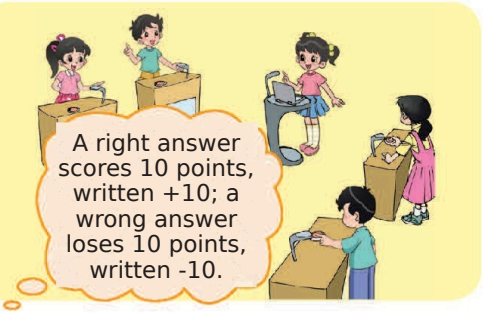
(3) Find these 6 cities on a map of China, and think about how a city's location is related to its temperature.

Positive and Negative Numbers

Look and say.

8844.43 m above sea level is written as +8844.43 m

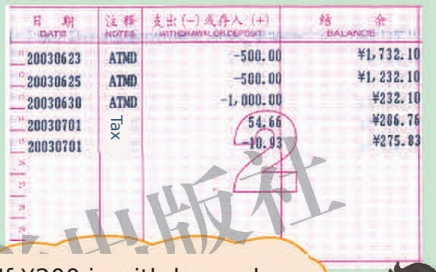
155 m below sea level is written as ( ).



A right answer scores 10 points, written +10; a wrong answer loses 10 points, written -10.

Charity shop results over 3 months

March	¥16900
April	−¥127
May	¥15200



The charity shop made a profit of ¥16900 in March, and in April...

If ¥200 is withdrawn, how will the passbook change?

I find that "+" "−" help us show quantities with opposite meanings.

Recognise and say.

Numbers like 10, 200, 8844.43, ⋯ are all positive numbers; a "+" sign can be added in front of a positive number, such as +10, +200, +8844.43.

Numbers like −1000, −500, −127, −100, ⋯ are all negative numbers. 0 is neither positive nor negative.

−10, 0, 200, ⋯ are all integers.

Find examples of positive and negative numbers in everyday life and discuss them with a partner.

For floors, 6 means floor 6 above ground and −2 means floor 2 below ground.

In a 100 m race, an athlete's result depends on the wind speed. A wind speed of +1.9 m/s means a tailwind of 1.9 m/s; −1.5 m/s means...

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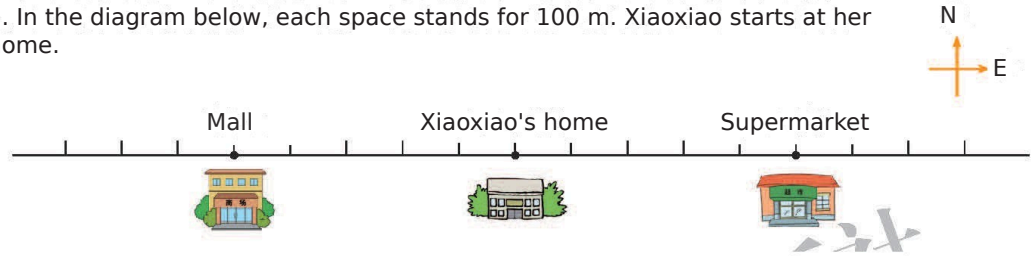
Practice

1. Record Lele's family's income and expenses in the table with positive and negative numbers.

4 May	Dad's wages received ¥1500
6 May	Water, electricity, gas paid ¥200
12 May	Phone bill paid ¥120
15 May	Mum's wages received ¥1400

Date	Amount (¥)
4 May	+1500
6 May	

2. In the diagram below, each space stands for 100 m. Xiaoxiao starts at her home.



(1) If walking 300 m east from home is written as +300 m, then walking 500 m west from home can be written as ____ m.

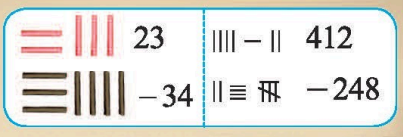
(2) If Xiaoxiao is now at the supermarket, she walked ____ m to the ____ of her home. This can be written as ____ m.

3. On average, pupils in Taoqi's class skip 80 times a minute. If Xiaoxiao's score is written as "−1", think and fill in the table.

	Xiaoxiao	Taoqi	Qisi	Miaoxiang
Skips	79	80	72	91
Written as	− 1			

Did You Know?

China was the first country in the world to recognise and use negative numbers. Over two thousand years ago, the Nine Chapters on the Mathematical Art already recorded positive and negative numbers.

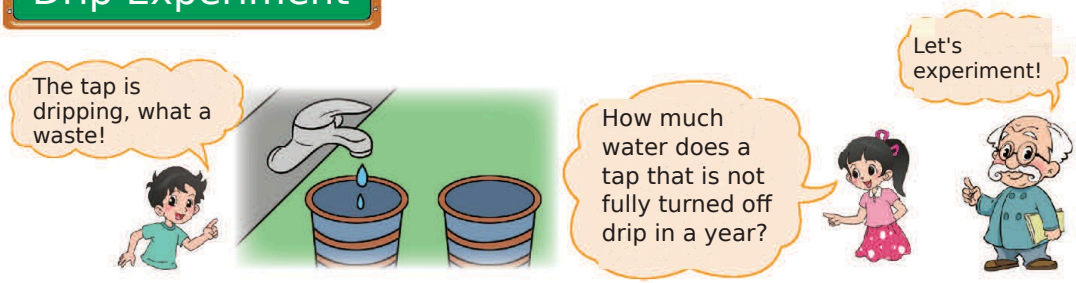


Long ago, negatives had their own notation: red counting rods meant positive, black ones negative; or upright rods positive, slanted rods negative. In writing, a slash on the last digit showed a negative number.

Write a negative number, then show it in this way.

Maths Is Fun

Drip Experiment

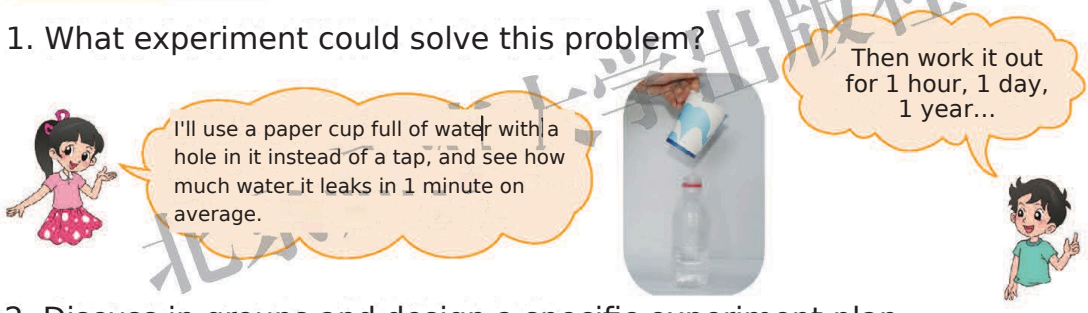


Task

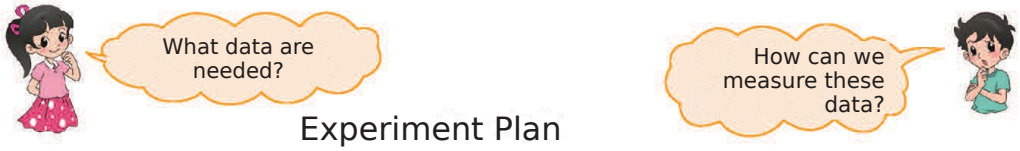
Design a drip experiment and work out roughly how much water a tap that is not fully turned off wastes in a year.

Design a Plan

1. What experiment could solve this problem?



2. Discuss in groups and design a specific experiment plan.



Experiment Plan

Experiment name	
Experimenters	
Measuring tools	
Experiment method and steps	
Division of work	

Experiment

Carry out the experiment according to the plan and fill in the experiment report.

Experiment Report

Experiment data	
Calculation process and conclusion	

Discussion

1. Share your experiment results with your partners.

2. Use your experiment data to answer these questions.

(1) Using everyday objects, describe roughly how much water leaks in 1 minute. If the water that leaks in 1 minute were put in a mineral water bottle, about where would the water level be? Draw it on your bottle.

(2) Using the data above and your everyday experience, describe roughly how much water a tap that is not fully turned off wastes in a year.

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3. Read and think

China has a serious water shortage and is one of the 13 countries with the least water per person in the world.

The water leaked in a month by a loose tap is enough for one person for a month.

Save

Waste

From the experiment data: if each person washes their hands 3 times a day for 30 seconds, then by washing the water-saving way, 3 people save enough water in a year for a family for a month.

(1) What are your thoughts after reading this?

(2) Look carefully around you. Where do you see water being wasted? What good ways of saving water do you know?

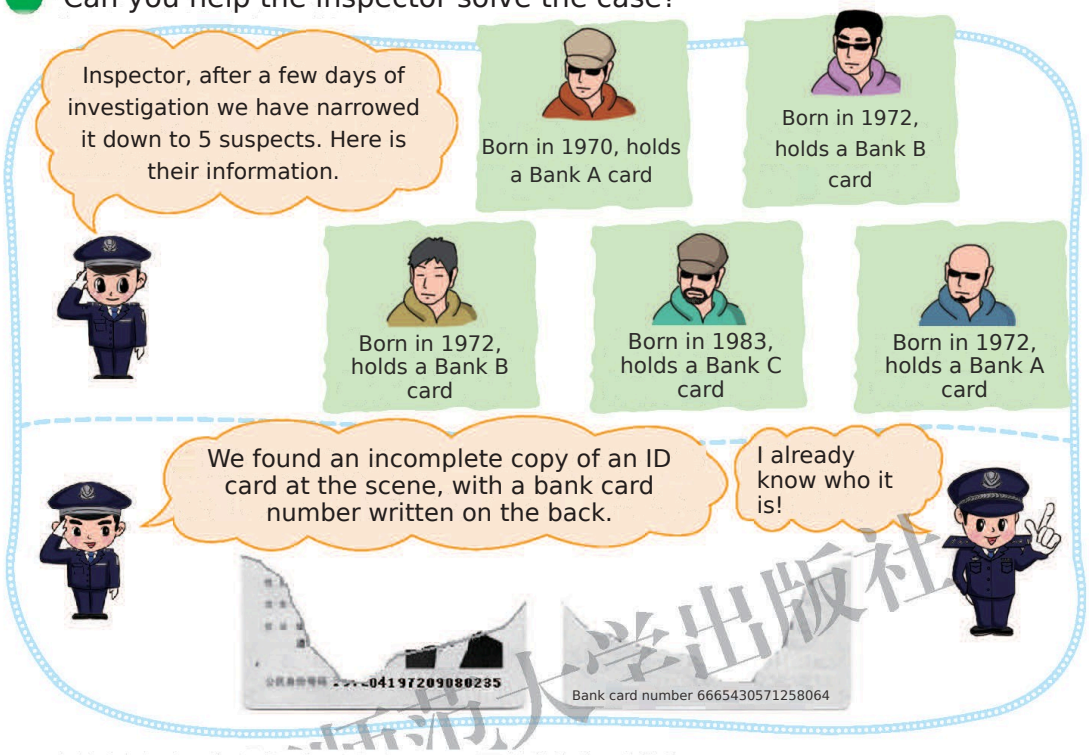
Self-Check

How I did in this activity (colour the ☆ after each item; colouring all 5 means the best):

The experiment plan is reasonable and feasible.	☆ ☆ ☆ ☆ ☆
The data I got is reasonable and reliable.	☆ ☆ ☆ ☆ ☆
Our group shared the work well and I joined in.	☆ ☆ ☆ ☆ ☆
I can use the data I got to solve problems.	☆ ☆ ☆ ☆ ☆
I can actively communicate with my partners.	☆ ☆ ☆ ☆ ☆

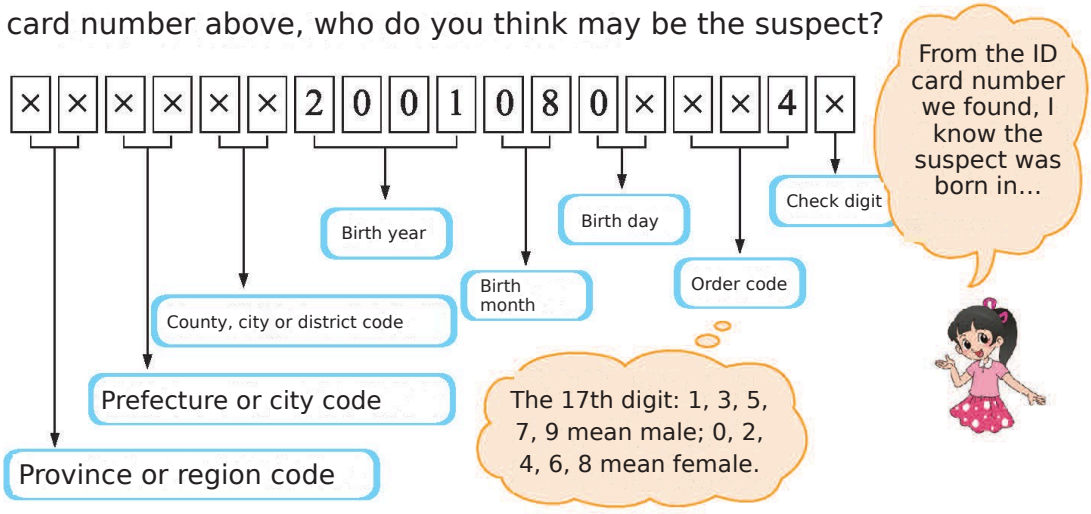
Codes

Can you help the inspector solve the case?

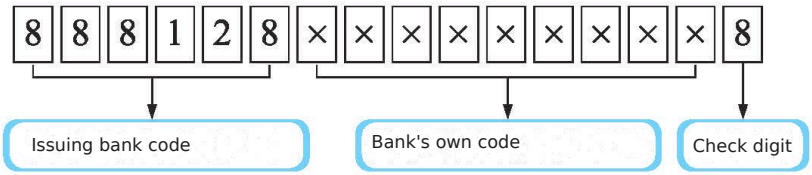


1. How did the inspector work out who the suspect is?

2. Do you know what the digits in an ID card number mean? Based on the ID card number above, who do you think may be the suspect?



3. This is how bank card numbers are coded. Use it to find the suspect.



These are the banks some of the codes stand for.

Issuing bank	Issuing bank code	Issuing bank	Issuing bank code
Bank A	666543	Bank B	888128
Bank A	666789	Bank B	888462
……	…	……	…

From the bank card number we found, we can tell... oh, I know who it is!

Make up a number for every student in the school.

1. What information do you think this code should show?

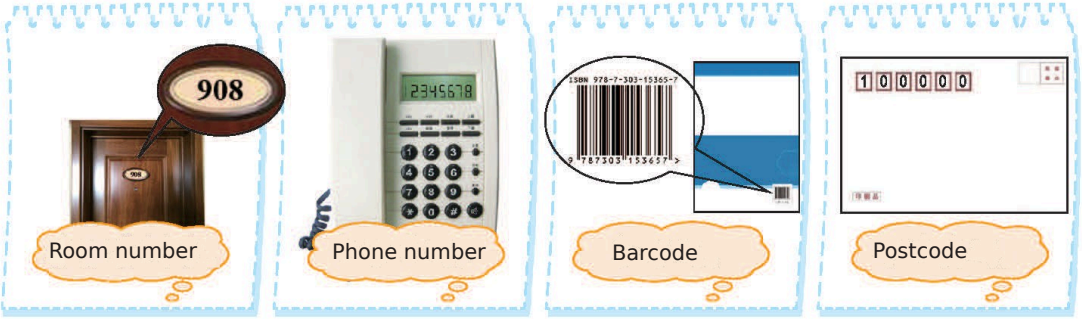
Shows the class.

Tells boys from girls.

No repeats across grades.

2. Taoqi made a number for every student in the school, where 201203321 means "a student in class (3) who entered in 2012, with student number 32, and this student is a boy". Can you see Taoqi's rule for the numbers? By this rule, which student does 200904172 mean?

Find examples of number codes used in everyday life and share them with the class.



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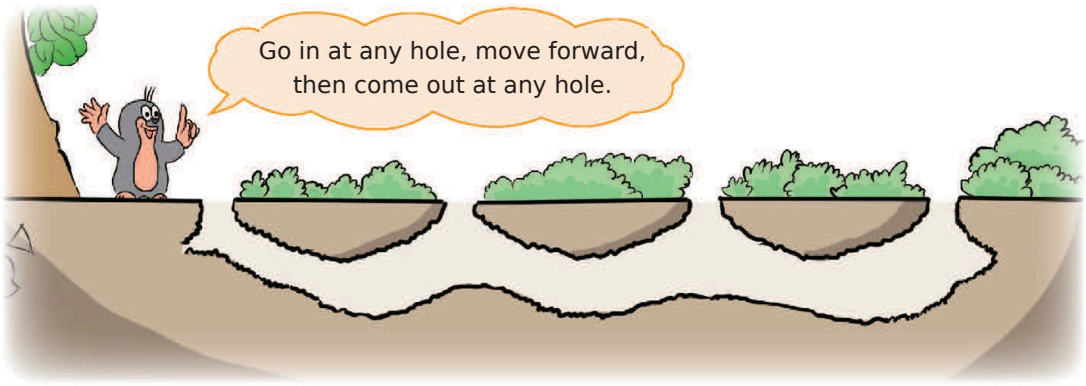
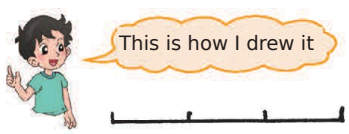
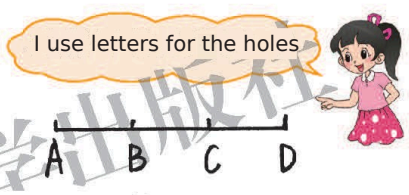
Counting Figures

Mole holes

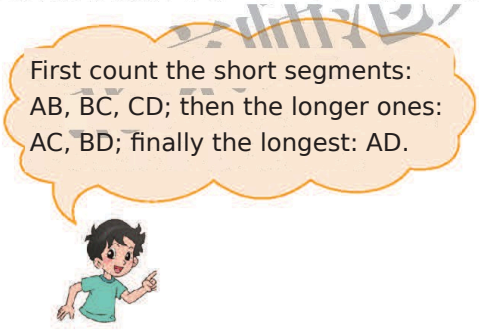


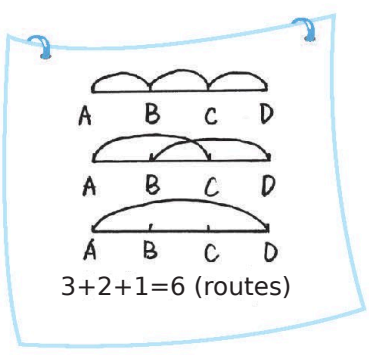
1. How many different routes are there? Draw a diagram.

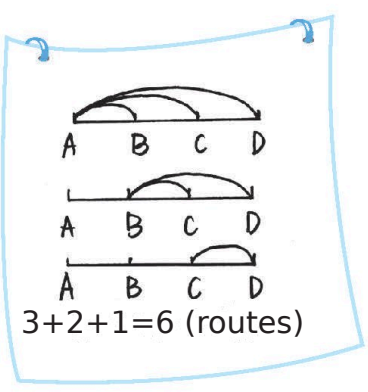


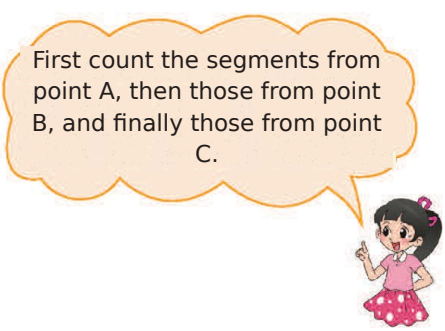


2. Count the different routes in order, without repeating or missing any.

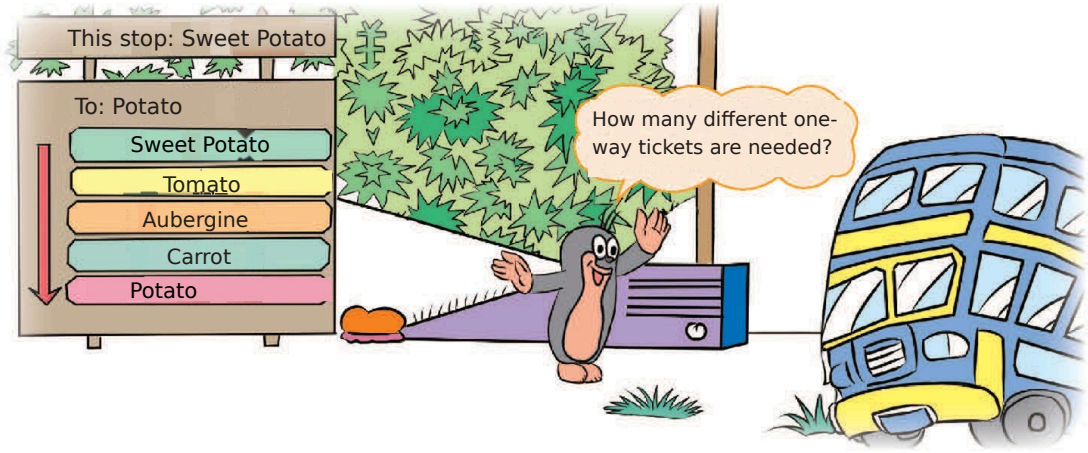




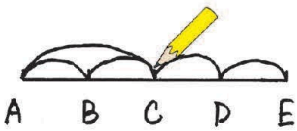




A trip to the vegetable patch.



1. Draw a diagram for the situation, count in order, and say how you counted.



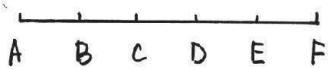
I counted like this, and there are...

I counted like this, and there are...

2. If there are 6 bus stops, how many different one-way tickets are needed?

After drawing the diagram, count again.

You can keep counting: starting from the result for 5 bus stops, add...



3. If there are 7 bus stops, how many different one-way tickets are needed? What about 8?

What do you notice?

When there are 7 bus stops...

5 stops: number of tickets = 4+3+2+1.

6 stops: number of tickets = 5+4+3+2+1.

7 stops: number of tickets = 6+5+4+3+2+1.

8 stops: number of tickets = 7+6+5+4+3+2+1.

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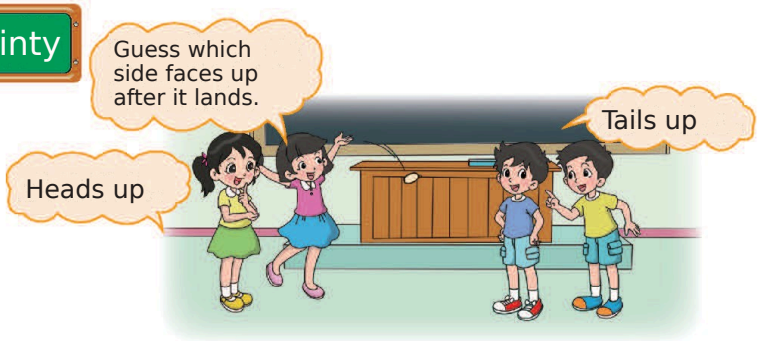
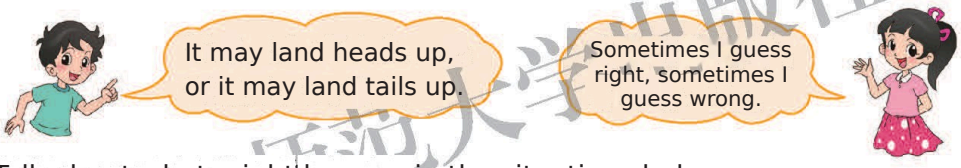
Unit 8 Probability

Uncertainty

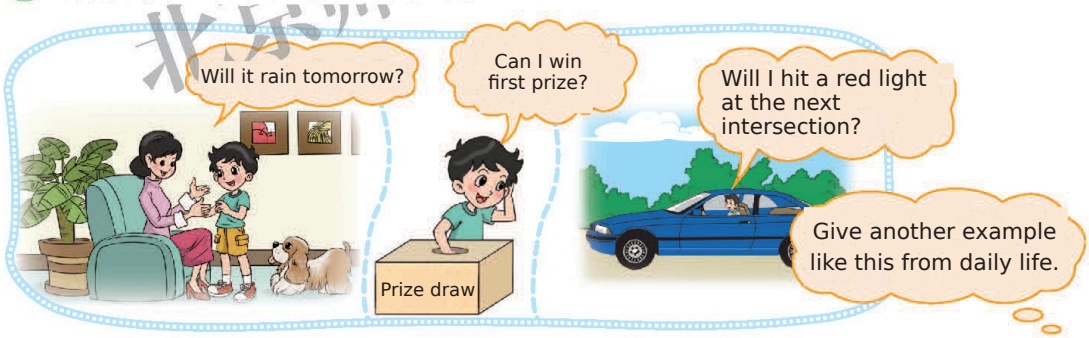


Take turns tossing a coin with your partner 10 times. First guess which side will face up, then record the actual result. Say what you notice.

Toss	1	2	3	4	5	6	7	8	9	10
My guess										
Result										



Talk about what might happen in the situations below.



Talk about it: what colour ball might come out of each box? How many possibilities are there? Then match.

Might be yellow | Cannot be yellow | Must be yellow

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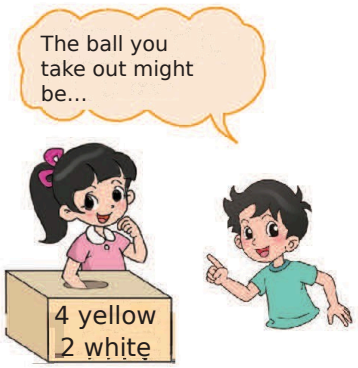
Practice

1. Draw 1 ball at a time, note its colour, then put it back and shake.

(1) Xiaoxiao draws once. What colour ball might she get?

(2) Take turns with your partner: first guess the colour of the ball, then draw one.

(3) If the first three balls are "yellow", "yellow", "yellow", what will the next ball be?



2. Will it snow in winter in the cities below? Answer with "must", "might" or "cannot".

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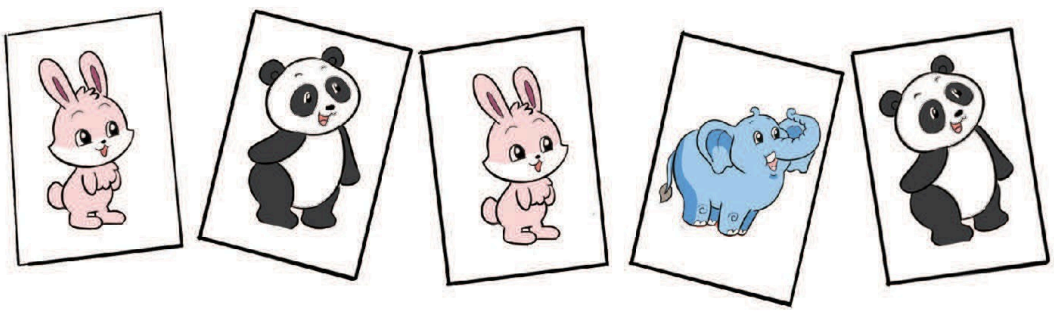
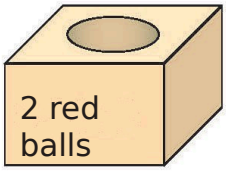
3. Mix the cards below together and pick one card at random. What could it be?

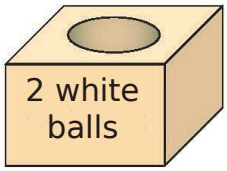


4. Use "must", "might" and "cannot" to talk about how likely some things in life are to happen.

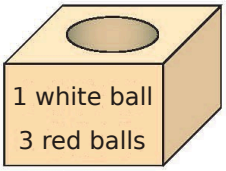
Pick a Ball



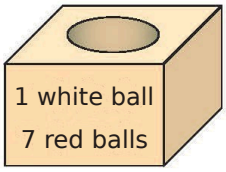
Box 1



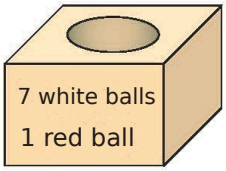
Box 2



Box 3



Box 4



Box 5

Draw 1 ball at random from each of these boxes. What ball might you draw?

From Box 1, you are sure to draw a red ball...

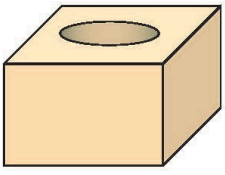
You can draw a white ball from Boxes 2, 3, 4 and 5.

Is the chance of drawing a white ball the same for Boxes 3, 4 and 5?

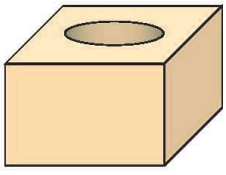
Chances can be big or small.

The chance of drawing a white ball from Box 5 is the greatest, because...

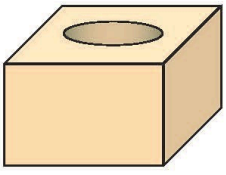
Put 6 balls in each box below so that each rule is met. Write what you put in.



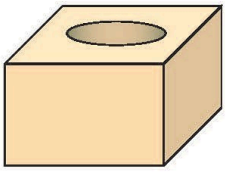
A white ball is more likely than a red ball



Sure to draw a red ball



Cannot draw a red ball



A red ball is more likely than a white ball

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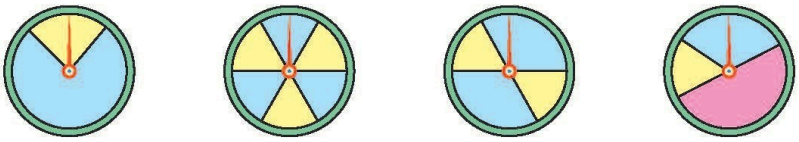
Practice

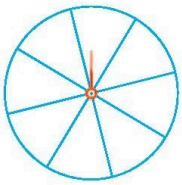
1. Spin the spinners below.



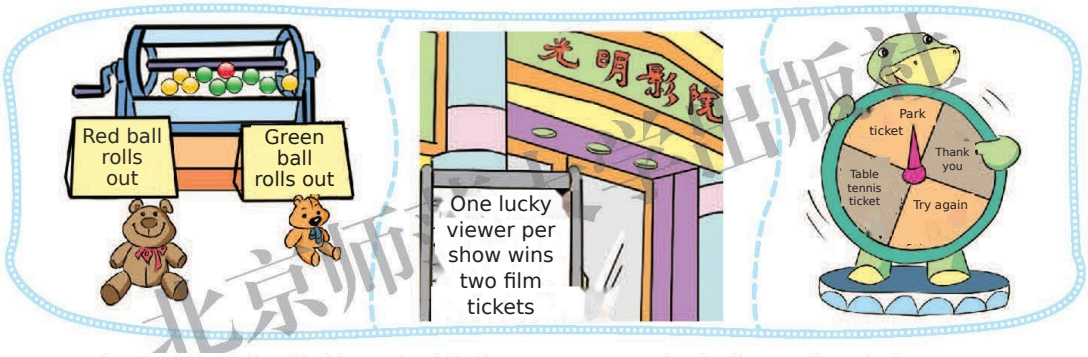
(1) When each spinner stops, which colours might the pointer point to?

(2) After the spinners stop, which pointer is most likely to point to blue?

2. The picture on the right shows a game spinner. Colour it red, yellow and blue so that the pointer is least likely to point to red and most likely to point to blue.



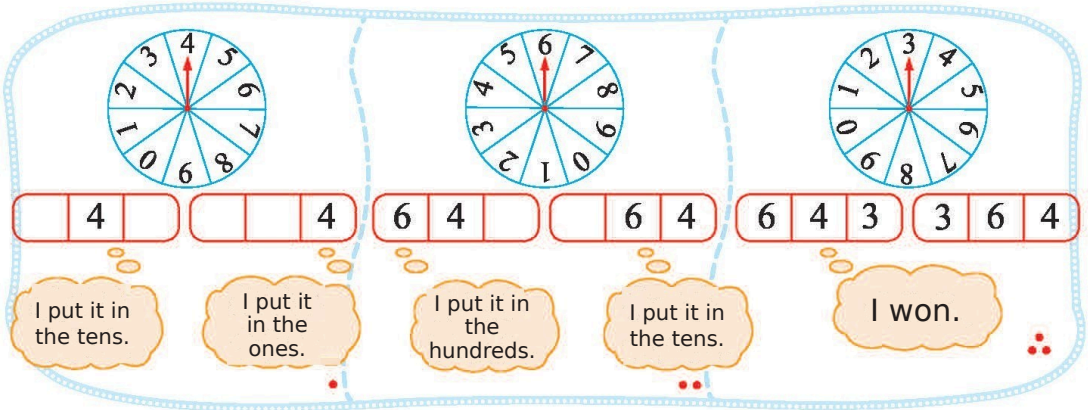
3. Taoqi and Xiaoxiao go to the park together. On the way they see these prize activities.



(1) Is the chance of winning high at each activity? Which one would you join?

(2) Can you design an attractive activity?

4. Maths game. Spin and compare: the bigger number wins. Tell your partner your tips for winning.



Final Review

Review and Share

Number and Algebra

On your own

1. Which numbers have we learned so far? Can you try to organise them?

2. Use examples to say how we use natural numbers, integers and negative numbers in life.

3. Look over your common mistakes. What must you watch for in multiplying and dividing integers?

4. Can you explain why they calculate like this?

89×5
$= (80+9) \times 5$
$= 80 \times 5 + 9 \times 5$
$= 400 + 45$
$= 445$

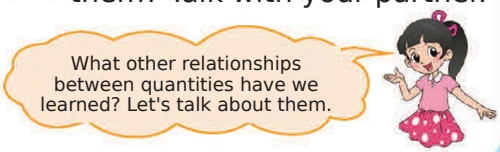
$225 \div 25$
$= (225 \times 4) \div (25 \times 4)$
$= 900 \div 100$
$= 9$

$$\begin{array}{r} 26 \\ 30\overline{)780} \\ 6 \\ \hline 18 \\ 18 \\ \hline 0 \end{array}$$

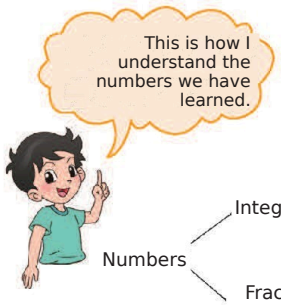
5. Tell a partner about the order of operations and the laws of arithmetic.

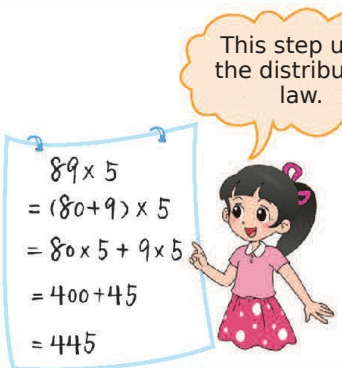
6. A train with an average speed of 98 km/h takes about 15 hours from Beijing to Shanghai. Since the Beijing–Shanghai high-speed railway opened, a high-speed train covers the 1318 km in only about 5 hours.

What maths questions can you ask? Which relationships between quantities did you use to solve them? Talk with your partner.



Share ideas





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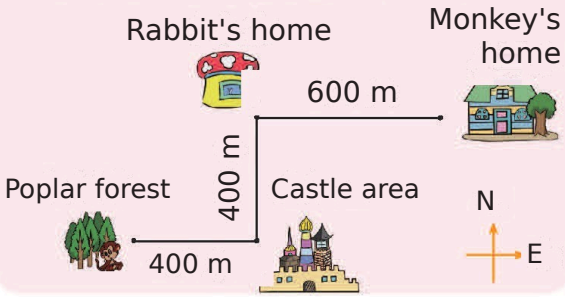
Shapes and Geometry

On your own

1. When are two lines parallel to each other? When are they perpendicular to each other? Say it in your own words.

2. What kinds of angles have you learned? How do you measure the size of an angle?

3. The little monkey is lost in the poplar forest. Can you tell it how to get back home?



Share ideas

Two lines are parallel if the number of squares between them is always the same.

When measuring an angle with a protractor, first see whether it is acute or obtuse, then decide whether to read the inner scale or...

The little monkey should first walk 400 m east to the castle area, then go north...

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Statistics and Probability

On your own

1. Can you give some examples of "uncertainty" from everyday life?

2. To make it more likely to get the highest score (or first prize), how should the two game rules below be changed?

Dart rules
- 5 points
- 3 points
- 2 points
- 1 point

Card rules
- ♠ 1st prize
- ♥ 2nd prize
- ♣ 3rd prize

Share ideas

Pick any classmate; it is uncertain which month their birthday is in.

In the dart game, if the 5-point bull's-eye is made bigger, the chance of getting the highest score is greater.

Exercises

Number and Algebra

1. Every day we come across lots of numbers. Have you noticed them?

Below are the results of China's sixth population census.

China's total population is 1370536875. Of these, the 31 mainland provinces, autonomous regions and municipalities and the armed forces counted in the census total 1339724852 people; Hong Kong SAR has 7097600 people; Macao SAR has 552300 people; and Taiwan has 23162123 people.

Population of the 31 mainland provinces, autonomous regions and municipalities, and active servicemen

By gender:

The male population is 686852572 people; the female population is 652872280 people.

Urban and rural:

The population living in urban areas is 665575306; the population living in rural areas is 674149546.

(1) Read out each population figure in the census results.

(2) 552300 has ( ) hundred thousands, ( ) ten thousands, ( ) thousands and ( ) hundreds.

(3) The rural population is about ( ) ten thousand people.

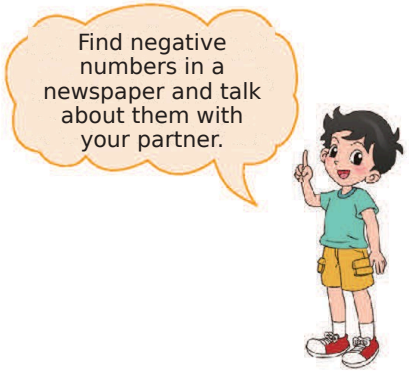
(4) Compare and fill in.

By gender: ( ) > ( )

2. Class 4 (1) had ¥60 in its class fund. It earned ¥20 selling waste paper, the teacher spent ¥15 on stickers for the class, it earned ¥30 recycling plastic bottles, then spent ¥33 on coloured paper.

If we write money received as positive numbers and money spent as negative numbers, can you help the teacher record the class fund income and expenses in the table below?

Item	Amount (¥)
Original class fund	+60
Selling waste paper	
Buying stickers	
Recycling plastic bottles	
Buying coloured paper	



3. The table below shows the length of highways newly built nationwide from 2006 to 2010. Round each number to the nearest ten-thousand km.

Year	2006	2007	2008	2009	2010
Length/km	62267	89687	75416	107675	72392
Rounded/10,000 km					

4. First estimate the product or quotient, then calculate.

253×56	503×32	45×240
$336 \div 21$	$858 \div 39$	$918 \div 27$

5.

$25 \times 37 \times 4$	$(125 \times 12) \times 8$	44×25
$27 \times 45 + 27 \times 55$	13×102	$800 \div 25$
$169 + 78 + 22$	$138 + 293 + 62 + 107$	

6. See who gets it right.

$(227 + 26) \div 11$	$459 \times (76 - 50)$
$(105 \times 12 - 635) \div 25$	$864 \div [(27 - 23) \times 12]$

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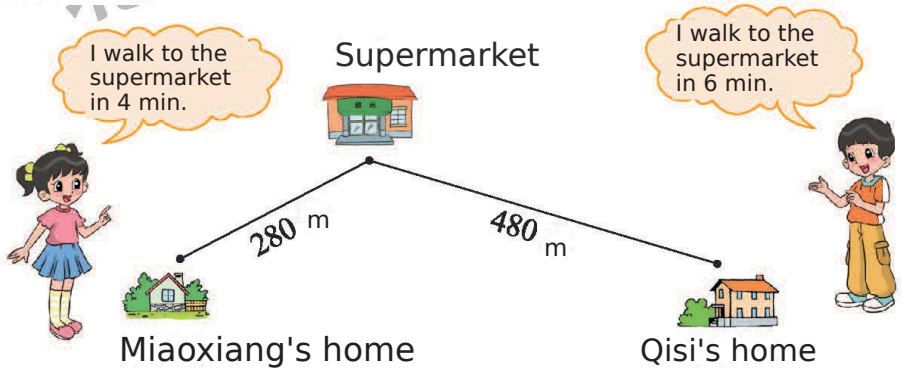
7. Who walks faster?

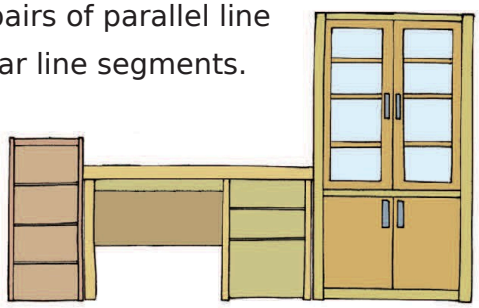


8. A sports shop sells footballs at ¥61 each. Mr Wang has ¥500. Is that enough to buy 9 footballs? What is the greatest number of footballs he can buy? How much money is left?

9. The New Year is coming. Xiaoxiao made 12 greeting cards by hand and plans to send them to relatives and friends in other towns. If each stamp costs 8 jiao and each envelope costs 2 jiao, how many yuan will it cost to send the 12 cards?

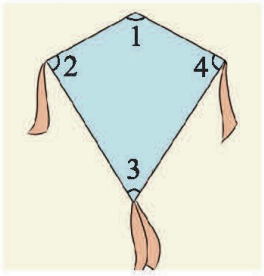
Shapes and Geometry

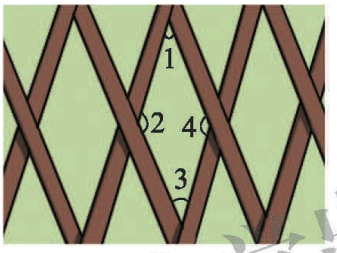
1. In the picture on the right, find two pairs of parallel line segments and two pairs of perpendicular line segments.



2. Draw a rectangle 3 cm long and 1 cm wide.

3. Measure the angles below in degrees.





4. Draw each angle below in a suitable way and say what kind of angle it is.

60° 90° 105° 120° 180°

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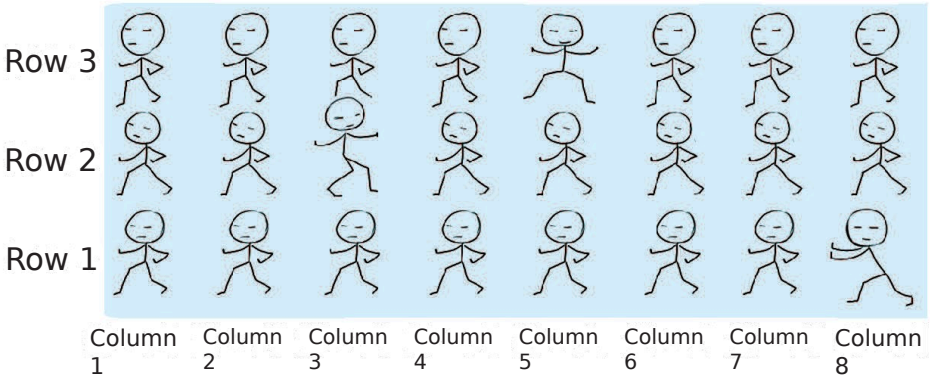
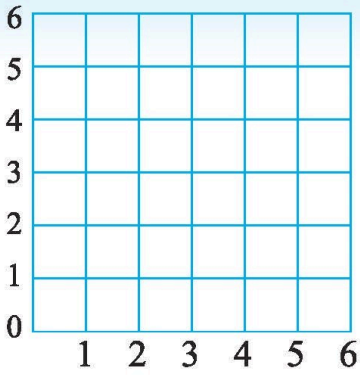
5. The children are doing tai chi in PE. Which children's poses are different from most of the others? Give their positions as number pairs.



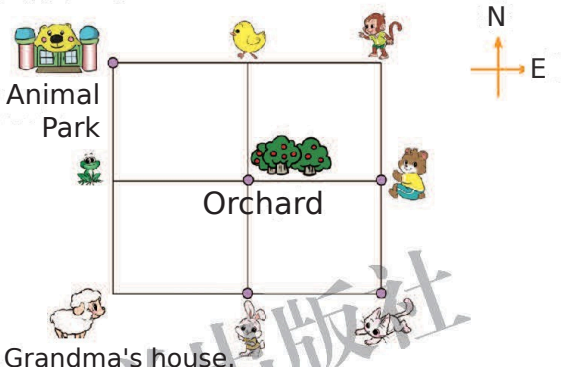
6. Mark points A, B, C, D on the grid. Their positions are (1, 1), (5, 1), (5, 4), (1, 4), and connect them in the order A→B→C→D→A.



(1) Say what shape this is.

(2) If the side of each small square stands for 1 cm, can you work out the perimeter of this shape?

7. The rabbit, the bear and the cat first go to the orchard to pick their favourite fruit, then go to the Animal Park for a meal together. Describe the routes they take.

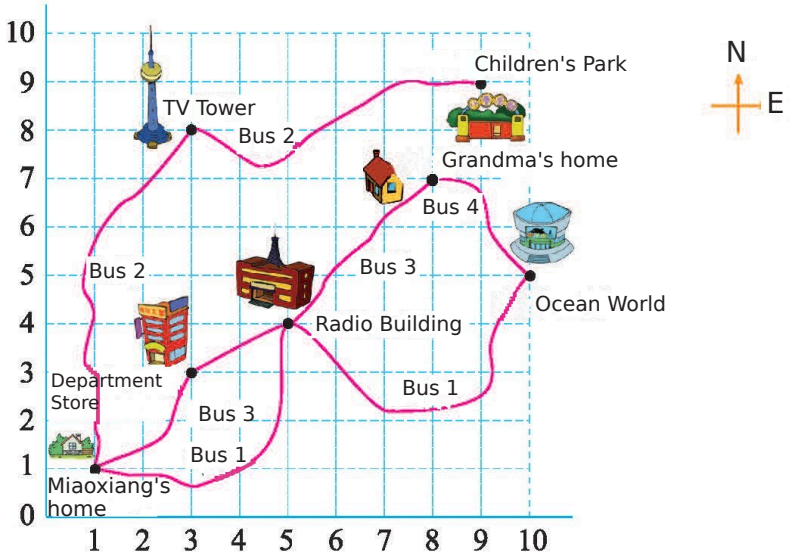


8. On Sunday, Mum took Miaoxiang to Grandma's house.

(1) Which buildings does their No. 3 bus pass? Where is each one?

(2) In the afternoon, they went to visit the attraction at (9, 9). Do you know where Miaoxiang went?

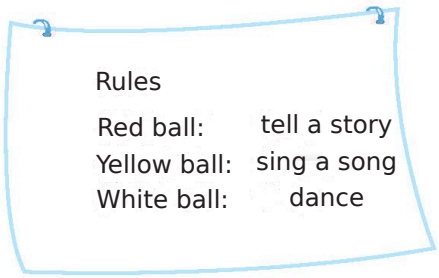
(3) After the visit, they took the bus home. Which buildings are along the way? Where is each one?

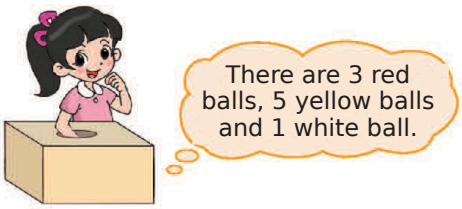


1. At the weekend, Xiaoxiao had a party with her classmates. They each drew a ball to decide what act to perform. It was Xiaoxiao's turn to draw.

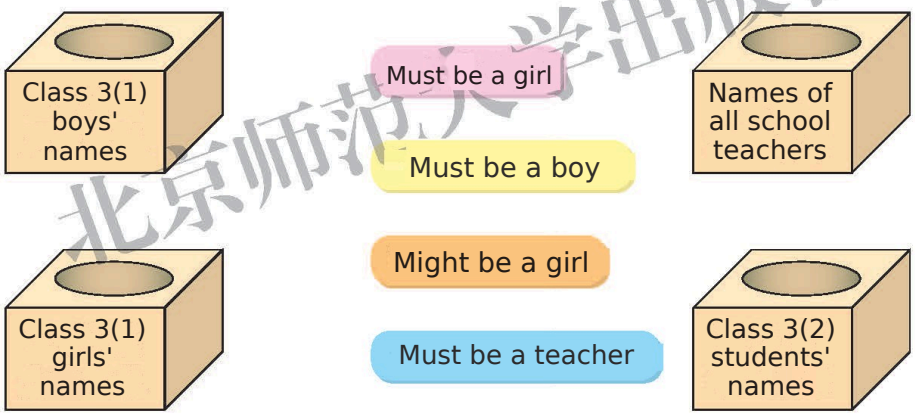
(1) What might Xiaoxiao perform?

(2) Which act is Xiaoxiao most likely to perform?





2. Draw one name from each box below. Which result do you get? Match them.



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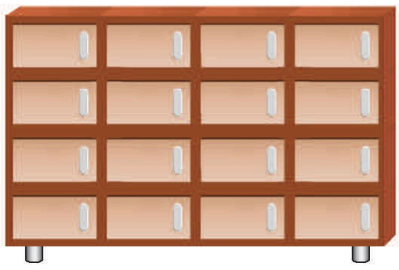
3. In the cupboard below, each compartment has 1 hat, with 2 red hats, 3 yellow hats, 8 white hats and 3 black hats in total. Open any compartment.

(1) Which colour hat is most likely?

(2) Which colour hat is least likely?

(3) Which two colours are equally likely?



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What Have You Learned This Term?

1. _____

2. _____

3. _____

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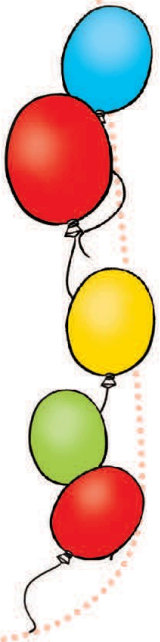

Question Bank

What maths problems have you found in everyday life and in your studies? Write them down. Can you solve them?

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Appendix

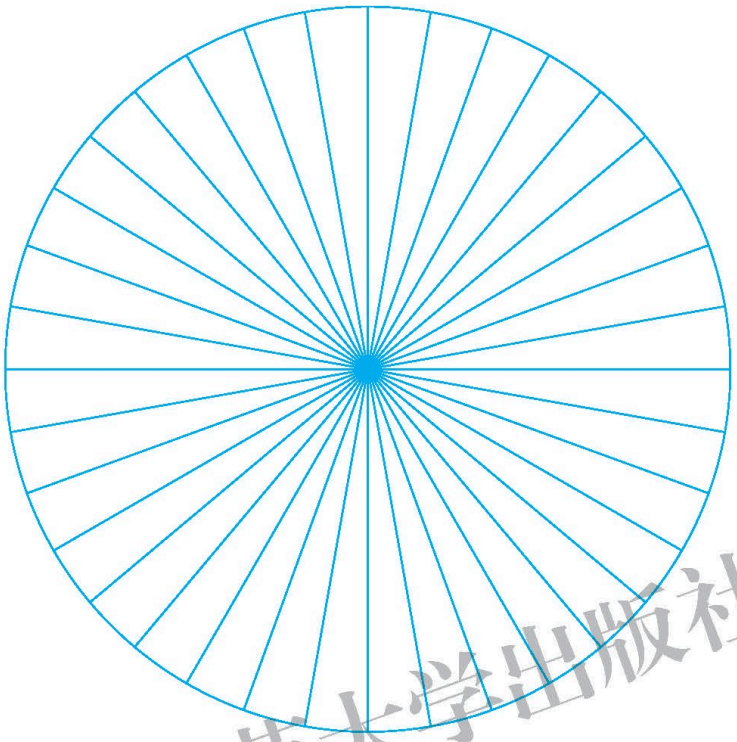

Figure 1

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Figure 2

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Afterword

The Compulsory Education Textbooks (Beijing Normal University Edition) were written by many heads and core members of the groups that drew up the national curriculum standards for basic education, together with subject experts, education experts, psychologists and master teachers. They rest on solid research and advanced educational ideas, and offer high-quality writing and professional service. The textbooks strive to reflect the spirit of the national curriculum standards for basic education, make use of many kinds of information resources, show the latest progress in each subject where appropriate, and stress the use of knowledge, skills and ways of thinking in real life. They stay close to students' lives, pay attention to how students learn, meet their different learning needs and promote the all-round development of every student.

The Compulsory Education Textbooks (Beijing Normal University Edition) · Mathematics (Grades 1–6), also known as the New Century Primary School Mathematics Textbooks, fully embody the basic ideas of the new century's reform of the mathematics curriculum, with the healthy growth of children as their fundamental aim. This is the 4th edition of the New Century Primary School Mathematics Textbooks. Building on the strengths of the first three editions, it has been fully revised in accordance with the Mathematics Curriculum Standards for Compulsory Education (2011 Edition).

The new edition has the following distinctive features. Its basic form of presentation is "a situation plus a series of questions", which seeks to make the unfolding of the content, the students' learning, the teachers' teaching and the reaching of the curriculum goals one whole, good for learning and for teaching, so that students keep going through the process of thinking a problem through "from beginning to end". It seeks to give students realistic, interesting and challenging learning materials, to give them time and space to explore and to share their ideas, to show how mathematical knowledge is formed and used, and to meet the needs of different students.

The aim is for students to: appreciate the close links between mathematics, nature and human society; gain the basic knowledge, basic skills, basic ideas and basic experience of activities that they need and that suit their age; develop the ability to find and pose problems and to analyse and solve them; understand the value of mathematics, become more interested in learning it and more confident of learning it well; and gain a first sense of innovation and a scientific attitude.

The chief editors of the New Century Primary School Mathematics Textbooks are Liu Jian, Kong Qiping and Zhang Dan. The other members of the writing group are Hu Guangti, Liu Keqin, Wang Yong, Tao Wenzhong, Wang Mingming, Zhu Dejiang, Chen Xiaomei, Zhang Hong, Qian Shouwang, He Fengbo, Hua Yinglong, Huang Lihua, Zhao Yanhui, Li Tie'an, Wei Huinü, Zhu Yuhong, Ren Jingye and Hou Huiying.

The chief editors of this volume are Zhao Yanhui and Huang Lihua. The main people who took part in revising and finalising it were Zhu Dejiang, Chen Xiaomei, Hou Huiying, Wang Mingming and Hu Qinzhu; the final text was compiled and approved by Liu Jian and Zhang Dan. In addition, Li Tie'an did a great deal of work in the early discussions and on the first draft of this volume and gave many constructive comments and suggestions. Many local teaching researchers and front-line teachers gave valuable advice on this new round of revision.

Special thanks go to Zhang Zhimeng and other children, and to Zhu Lei, who were invited to contribute their work.

Because time was short, errors in the textbook are hard to avoid; we sincerely ask our readers to point them out. You are welcome to contact us by phone or letter: Basic Education Branch, Beijing Normal University Press (100875), (010) 58802836, 58802832; Mathematics Studio, Basic Education Curriculum Research Centre, Beijing Normal University (100088), (010) 58435911, xsjmath@126.com.

Beijing Normal University Press